SMITHIAN FOLD THEORY - PHYSICS BRANCH PAPER 001

From Fold to Physics

An Exact, Parameter-Free and Machine-Closed Reconstruction of Physical Science from Smithian Fold Theory

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Third clean-room reconstruction - Physics inventory complete
132 required and 8 supplemental admitted derivations - 35,840 Physics candidates
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Abstract

This paper reports the third clean-room Smithian Fold Theory reconstruction of the Physics branch. It documents 132 required laws across measurement and metrology, mechanics, fields, waves, thermodynamics, physical quantum theory, matter and nuclei, spacetime and gravitation, collective matter, and the physics-to-cosmology boundary. Every required law passed the single admission engine with a complete 256-form grammar, exactly one survivor, minimality, named-shape uniqueness, four adverse controls, a source-bound derivation seal and an implementation-distinct reconstruction. Natural-law claims additionally carry capability-closed prediction, target custody, seal-before-open evaluation, complete external rows and falsification records. Eight supplemental claims backfill exact measured-value relations for earlier subbranches. Fourteen required or supplemental claims execute exact positive rational interval predictions against official external measurements without binary floating arithmetic. The paper distinguishes forced law from unit scale, measured initial state and finite census; it claims closure only at the declared generated and empirical boundaries.

1. Publication and authorship boundary

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2. Scope and result

The frozen inventory contains 132 obligations in ten ordered subbranches. All 132 are model-admitted. Their combined required grammar contains 33,792 candidates, 33,792 decisions, 132 survivors and 528 passing mandatory controls. Eight supplemental numerical-validation claims add 2,048 candidates, eight survivors and 32 controls. The complete repository census now contains 308 claims and 91,794 generated candidates across all closed branches.

Physics closure here means that every obligation in the frozen categorical inventory has one engine-admitted law at its exact boundary. It does not mean that every future physical observation, material, astronomical object or historical state has already been enumerated. Chemistry, materials science, astronomy and cosmology remain later branches.

3. Exact constitutional domain

The derivational domain admits the empty One as structural absence but never numerical zero. Proof magnitudes are positive generated counts and exact positive rational parts or ratios. Orientation, opposition and complement are held labels rather than negative quantities. Irrational, imaginary and binary floating values are barred from proof. Completed infinity, an ungenerated continuum, axioms, fitted coefficients and free parameters are also barred. External decimal measurements remain source-bound records. A finite decimal is converted to an exact rational interval only inside the empirical adapter and never gains authority to select the Fold law.

4. The single admission engine

Each claim enters one `SFTAdmissionEngine`. Registration rejects axioms, free parameters, missing root trace, unadmitted dependencies and source-identity failure before candidate execution. The engine then checks census cardinality and identity, one decision per candidate, exactly one survivor, minimality, named-shape uniqueness, the four mandatory controls, an implementation-distinct external reconstruction and, for empirical claims, prediction isolation, target custody, complete rows and falsification. A failure halts without model admission. Accepted receipts are immutable evidence identities; this paper does not replay or replace them.

5. Why superdeterminism permits uncertainty and quantum weights

No law in this branch installs ontic nondeterminism. The complete Fold state includes preparation, held labels, measurement setting, path and observation record. A probabilistic or quantum weight is an exact census ratio over unresolved support. Each branch execution remains deterministic. Measurement uncertainty records distinctions unavailable to an observation class; it does not assert causeless state selection. Bell correspondence therefore retains setting and preparation records in the complete state, identifies the factorization assumption that fails, and separately preserves the no-signalling marginal through complete remote-fibre enumeration.

6. Empirical constitution

External evidence follows one direction. First the Fold dependencies generate and eliminate candidate laws. Then a data-only Fold program receives only registered inputs and the sealed derivation identity. Its instruction set has no filesystem, network, subprocess, clock, environment, dynamic import or foreign-function capability. A distinct custodian commits target identity before execution and releases content only to the matching prediction seal. The evaluator preserves every registered row and must reject a deliberately altered or displaced control. BIPM, NIST/CODATA, NIST ASD, PDG, IAEA, IAPWS, GWOSC, CERN Open Data and NASA LAMBDA supply the external records.

7. Exact measured-value correspondence

Measured-value forcing means that a parameter-free Fold relation maps registered measured input carriers to a withheld measured target. It does not mean that the numerical size of the SI metre, the universe's initial temperature or a contingent particle mass is created from no physical record. Finite decimals and uncertainties are parsed as exact positive fractions. Multiplication and quotient propagate interval endpoints in the capability-closed interpreter; target overlap is evaluated only after release. Exact official decimal prefixes with ellipses are bounded by their next decimal place. Any non-overlap halts admission.

The value suite includes molar Planck composition, momentum, force, electric potential and field, wave frequency and energy, molar thermal energy, quantum action-energy time, mass-energy equivalence, limiting speed, molar mass, collective molar charge and COBE FIRAS background-temperature invariance. Every one passed its official uncertainty interval and rejected a displaced interval.

8. Reading the derivation ledger

Each claim section below is a self-contained prose projection of its machine evidence. The complete `candidate_census.json`, `elimination_receipt.json`, `controls.json`, `certificate.json`, external experiment registration and receipt remain authoritative. The paper includes every claim identity and admitted receipt hash so the evidence map can fail closed on omission or substitution.

9. Measurement Metrology

Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain.

1. Capability-closed exact Fold prediction form

Claim identity: `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Within the registered finite prediction grammar, a physical prediction can be target-inaccessible, exact, portable and completely traced in exactly one form: immutable registered Fold inputs pass through a data-only sequence of generated exact operations to one terminal emission, while every ambient host capability and unregistered operation is absent and any violation halts.

> *Registered exact inputs, a data-only closed opcode surface, no ambient capability, single-assignment state, one final emission, complete trace and no extra rule.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-COMP-FORM-STATE-TRANSITION-001`
- `SFT-COMP-FORM-OPERATIONAL-PROCESS-001`
- `SFT-COMP-SEM-EVALUATION-001`
- `SFT-COMP-SEM-TERMINATION-001`
- `SFT-COMP-SEM-VERIFICATION-001`

Generated grammar and closure boundary. Generate the complete product of input, value, operation, capability, state, termination, trace and extension forms. The declared exact boundary is: All finite data-only prediction interpreters assembled from admitted exact values, registered inputs, generated operations, immutable held states, a declared terminal emission and no host capability. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `registered-input-only__exact-fold-domain__data-only-generated-operations__no-ambient-capability__single-assignment-held-state__one-terminal-emission__complete-step-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`input`	`registered-input-only`	`ambient-input`	Ambient input can contain the withheld target or an answer-producing channel.
`value`	`exact-fold-domain`	`host-scalar-domain`	Host scalars import forbidden or platform-dependent proof values.
`operation`	`data-only-generated-operations`	`contributor-executable-source`	Arbitrary source can create operations outside the registered grammar.
`capability`	`no-ambient-capability`	`ambient-capability`	An ambient resource can reveal target, comparison or mutable authority state.
`state`	`single-assignment-held-state`	`mutable-overwrite`	Silent overwrite loses the exact predecessor and permits untraced replacement.
`termination`	`one-terminal-emission`	`implicit-or-open-ended`	Implicit or open-ended completion cannot certify a complete prediction artifact.
`trace`	`complete-step-trace`	`result-only`	A result alone hides the operational relation and its resource path.
`extension`	`no-extra-rule`	`extra-rule`	An extra instruction or capability is a free answer-selecting parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:d4c5b29b8c8c7fbf677579bb04cb996cdf6878a95ef678bac3998c373374bae6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form missing the first preservation; observed reject the generated form missing the first preservation; receipt sha256:101b735f39c0308acff4f1fbb4fc7c0845d09021fb0cada8326ad7a876d77f96.

- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:06e9ea349c585d18965d1f715617649d780dbd450ab59bce9090d38591920b1b.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1814b0879667d0341a6a7cf3b345a60550bcbc94d5ad113e00cc9e2db8dc00b6.
- `boundary`: passed; expected reject excluded capabilities, values or authority paths; observed reject excluded capabilities, values or authority paths; receipt sha256:0978722205b9e2a84a6ef2fb7e600cc50bc215a65ae8c025a2e96c53f0e6e3dd.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2ac5e37d95aea5263c2fcf026b93eb3549e21a04ce7e0e125ad91b62a956c0cd`. Independent certificate: `sha256:060dddafaba441cc92c20cb16ac1c5e960744ea2af940ffacb70c6bcce2d0d15`. Engine external-validation hash: `sha256:d6ad95de25a8dd5dd16045baead9da289366c3189e216cff4421a6308e4c7c2e`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- arbitrary Python or native contributor code
- filesystem, network, subprocess, clock, environment, import and foreign-function access
- floating, negative, irrational, imaginary and semantic-zero proof values
- unbounded or untraced execution

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:b11e9de76a85a54c63f86af81bcef25db27c06262293fb9236347dcb04fb9d88`; engine receipt `sha256:dc597e88dc8d1f725ef158e3e69a1bb6b4b402c4cc138372e40b1df21fe821d8` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001-dc597e88dc8d1f72.json`; empirical-validation hash `None`; measurement receipt `None`.

2. Portable target commitment and post-seal custody

Claim identity: `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Within the registered blind-exchange grammar, target content remains unavailable to prediction in exactly one form: a distinct custodian commits the complete named target package and prediction envelope before execution, then releases that unchanged package once to the matching experiment only after a matching prediction seal exists.

> *Pre-prediction complete commitment, distinct custodian, nonce-bound unchanged content, matching experiment and envelope, one post-seal release and no extra rule.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-COMP-SEC-HASHING-001`
- `SFT-COMP-SEC-COMMITMENT-001`
- `SFT-COMP-SEC-INTEGRITY-001`

Generated grammar and closure boundary. Generate the complete product of timing, holder, target support, content identity, experiment, envelope, release and extension forms. The declared exact boundary is: All finite portable target exchanges using admitted exact identities, pre-seal commitment, distinct custody and one post-seal release. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `pre-prediction-commitment__distinct-custodian__complete-named-support__nonce-bound-content-identity__matching-experiment-only__matching-envelope-only__one-postseal-release__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`timing`	`pre-prediction-commitment`	`post-prediction-choice`	A post-prediction choice can select favorable target content.
`holder`	`distinct-custodian`	`prediction-holder`	The predictor holding targets violates structural inaccessibility.
`support`	`complete-named-support`	`partial-or-open-support`	Partial or open support permits selective favorable rows.
`content`	`nonce-bound-content-identity`	`unbound-content`	Unbound content can be replaced after prediction.
`experiment`	`matching-experiment-only`	`cross-experiment-release`	Cross-experiment reuse breaks registration custody.
`envelope`	`matching-envelope-only`	`unbound-envelope`	An unbound seal can certify a different input or relation.
`release`	`one-postseal-release`	`preseal-or-repeat-release`	Pre-seal or repeat release leaks target information.
`extension`	`no-extra-rule`	`extra-rule`	Discretionary release is a free selection parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:d531095fb015f927f02c1f40fe9951b973bdf5f6a00934c768a162e60d3eb80c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form missing the first preservation; observed reject the generated form missing the first preservation; receipt `sha256:9d1ae2f40bb91451031c219678c477468054cc6bfecb398935d7356e3e0e059a`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:18e6f58bb38714edf42a2881d8bf8e19a8c4b4999a5132d7bde03a4a053f20c7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:53b93b6397aa18513fb73bcd48e2a739a26061f47428ae3790fa8d25f0248087.

- boundary: passed; expected reject excluded capabilities, values or authority paths; observed reject excluded capabilities, values or authority paths; receipt sha256:0f1c55cf30f7c60f78e3ff7f089b1c3a641fe34e00e84508ede332bab9d54b03.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:458f3324e7eb0127ec05aab491f9ddd6005718ba642b2d6ed3b67d0bc5adc1d9. Independent certificate: sha256:af9b5b3c7fcb2d140f04f3aa902aadd0392c9aec0bc027e7bb517042963abc13. Engine external-validation hash: sha256:d4f89d5039a9e646f187e2219dd0b0ee9354ef60010cf9277afb3226e49a0a9d.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- target access by the prediction executor
- release before a matching prediction seal
- selective target-row omission
- host-specific sandbox claims

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest sha256:638f04b6b15e15cffa178c9fb8bb8fe84b04a95aec12ff99edee1eae1b07a7ae; engine receipt sha256:f788edf20c3f0bb019b813cb63534724518765fccaa1afe7610fb736ae6b21a5 at receipts/engine/model_admitted/SFT-PHYS-MEAS-TARGET-CUSTODY-001-f788edf20c3f0bb0.json; empirical-validation hash None; measurement receipt None.

3. Hostile empirical-package and authority protection

Claim identity: SFT-PHYS-MEAS-HOSTILE-PACKAGE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Within the registered empirical-package grammar, contributor material can enter official prediction without gaining execution or scientific authority in exactly one form: strict data-only program fields are parsed by the admitted interpreter, the census and model-admitted receipts are identical before and after handling, and every extra field, executable source or protected-tree change halts admission.

> *Strict data-only Fold document, exact parser, no unknown fields or direct source execution, unchanged authority trees, separate comparison, fail-closed response and no extra route.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001

- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-COMP-SEC-INTEGRITY-001`
- `SFT-COMP-SEC-ADVERSARIAL-001`
- `SFT-COMP-SEM-VERIFICATION-001`

Generated grammar and closure boundary. Generate the complete product of package, parser, field, execution, authority, comparison, response and extension forms. The declared exact boundary is: All finite empirical prediction packages admitted through a strict data-only parser while census and model-admitted receipt authority remain protected. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `data-only-fold-document__exact-schema-parser__reject-extra-field__interpreter-only-execution__unchanged-protected-tree__postseal-separate-comparison__halt-without-admission__no-extra-route`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`package`	`data-only-fold-document`	`mixed-code-and-data`	Mixed packages can execute contributor-selected behavior.
`parser`	`exact-schema-parser`	`permissive-parser`	A permissive parser admits undeclared behavior.
`field`	`reject-extra-field`	`ignore-extra-field`	An ignored field creates divergent or hidden meaning.
`execution`	`interpreter-only-execution`	`execute-contributor-source`	Direct execution exposes every ambient host capability.
`authority`	`unchanged-protected-tree`	`mutable-authority-tree`	Changing census or accepted receipts can manufacture admission.
`comparison`	`postseal-separate-comparison`	`embedded-comparison`	Embedded comparison can expose or optimize against targets.
`response`	`halt-without-admission`	`warn-and-continue`	Continuation admits a known authority or capability breach.
`extension`	`no-extra-route`	`extra-route`	An extra route bypasses the protected admission path.

Base and successor. The closure proof is sealed by derivation hash `sha256:3b398ba6ad10467a48f4b6da1b13f21511b7decbb548c176ff7b069acf836de4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form missing the first preservation; observed reject the generated form missing the first preservation; receipt `sha256:155abd7902abed0cf2e85564b52bafb870f8f8565b09bc3b8c0847134a9399c5`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:9eadbe7d5481a418a910ff9ca6f16ef51c317220a3ac4b47900b0516ec7e419f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a319275abd552274cbf9c75121007066d2ae71142500f2fe884f53cd93276a61`.
- `boundary`: passed; expected reject excluded capabilities, values or authority paths; observed reject excluded capabilities, values or authority paths; receipt `sha256:119b65e69ed287f95df7763be66eb58b7997ef88302cae2a698c8651125a9034`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b5a6ba9afcebd3f858403bb8a48a8d8623ac2fde50ecb36700c68264453e7103`.

Independent certificate:
`sha256:2a2bc6e6aadd3b7c2753423f4cd5a8690935d26e4d58a44e1b65a601fef6d8f9`. Engine external-validation hash:
`sha256:8ab8a7d3c62932743b19764e0b71fd29d73f24f8b6556fa28d96d47eba360cef`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- contributor Python, native source, shell commands, modules or callables
- unknown package fields
- census or model-admitted receipt mutation
- target comparison within prediction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
`sha256:b6b6f54ba087d71f46fa57a106d83472d89b361c4ccc0bfa6ca6dd4ec914fc1d`; engine receipt
`sha256:20cd01fae8743456043006ac2f882cbe025b5c0bc5fd1062417ef44a5572b9b0` at
`receipts/engine/model_admitted/SFT-PHYS-MEAS-HOSTILE-PACKAGE-001-20cd01fae8743456.json`;
empirical-validation hash `None`; measurement receipt `None`.

4. Physical observation and retained distinction

Claim identity: `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical observation is forced to retain a distinguishable phenomenon, a held reference condition and the exact comparison record; an answer without those three coordinates is not an admitted physical observation.

> *The unique observation carrier is a source-bound phenomenon/reference pairing plus its held comparison record, executed without target access and with every row retained.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__distinction-reference-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`distinction-reference-record`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:daf450c3332bf1658e0d2f915c057a3832794c996ea290599a66571ec5979df1`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a34141df720d459de4b146698583cf8fdcde23b929ddcda6e2efd7685bf2a6cd`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:99dffccfc8546e03a99d83a3b7a5a24e5a6e352e2a77380cd7904173382c2d60`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f62791c52be506a073e57cb83a2a099e6fa76f0f8e0ecb4df7950d4ceafa44dd`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:eedf3b06cb5ccdc9b7ffad27fa80d7f553e977f01d14994793b7e2530c3cb9a2`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4a2fc44538091b1505d1da8b0424f73efbea60a315a3c253ca7307673dbabb08`. Independent certificate: `sha256:063d93d38c21349f348acef11572c524ffb601bf2e4b5eba7a05bba5b7e29b29`. Engine

external-validation hash:
`sha256:b5fdbc401d6760092c30caab2e52b9b20c44144bf03ada1a8fa3721e7ba45861`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- observation-carrier-1: predicted distinction-reference-comparison-record; observed distinction-reference-comparison-record; exact match True
- observation-carrier-2: predicted distinction-reference-comparison-record; observed distinction-reference-comparison-record; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: `sha256:f9fda2226a2fad7562b3a0dda4144bd637241a274a23a852eee1c9ad9c82e058`. Isolation certificate: `sha256:09f89a25ba5ccdc6ecee428dff4f3dae996c0d349a76a283f9c359a6141aafa5`. Custody certificate: `sha256:fc49213f497d434380833ef20ec477cd6a96efd7a2c99a5292ea53e2760ffa2b`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- observation-carrier-1 from BIPM-SI-BROCHURE-9-2026
- observation-carrier-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:99bdb19d7d84e7464f072fc44f536b7167af77971799ea02594035eb5b2632a0`; engine receipt `sha256:4c4ab1e9623d357d1fcf16813b4c172b508ede8a5216c78c6ad786950b657009` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-OBSERVATION-CARRIER-001-4c4ab1e9623d357d.json`; empirical-validation hash `sha256:5fc32330a8a436e034c5fb9f31a4cdb46b708bf99cdd29c1edaf118f2979b052`; measurement receipt `sha256:f9fda2226a2fad7562b3a0dda4144bd637241a274a23a852eee1c9ad9c82e058`.

5. Physical quantity as reference-equivalence carrier

Claim identity: `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical quantity is forced as the equivalence class of observations comparable through one generated kind-preserving reference relation; its measured value is a separate record, not the quantity itself.

> *The physical quantity carrier is the complete class of observations connected by a kind-preserving reference equivalence, with measured values retained separately.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__kind-preserving-reference-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`kind-preserving-reference-equivalence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f8b6e0a0cb283fec0c60c1d874db053051e2b09c5f425eccd52feee7b7c07ee9`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9888ea14f67d6c08a04a1cfc8d37a8522c8222c7855cd64c4a0414328a8486c2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1ad6611d696ab491614f73e53a56b3a3c1f91931a00215ebdb548ee37e121a47`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9a060ec9f8710478bdbf4ae6ff1056ad01e38498daa3fa8e712b5b97ec54d29c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:61dc4ec52ba6fb38968ea6c69f29e3b3164d7a336203b9af708cece27e69569a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8c3c165e5709b23eba3bd85ff8e52ef0582861f3da26491e87b9baf7932be60a`. Independent certificate: `sha256:e01dbd3531cd39593cd0477f80a7cd7dd4245304c6a3baec8a1a143c92de88c7`. Engine external-validation hash: `sha256:d46fd4e262c1f7e029cbdfee3f8334fd2495e66eb379acb8f63e8ae7775f6456`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- quantity-carrier-1: predicted property-kind-reference-equivalence; observed property-kind-reference-equivalence; exact match True
- quantity-carrier-2: predicted property-kind-reference-equivalence; observed property-kind-reference-equivalence; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: `sha256:bea784fbbfa39af43898c8730beb0d3255907a633c0731818d976945654310ae`. Isolation certificate: `sha256:651961f18efa758979d104f0ba373ce12383ee7e3f99c7e4e411fcb4dc1415cd`. Custody certificate: `sha256:d1d8736465a0e7dae1ba5cde42947671ee3fa25fdc5153388e07d3a426328e1b`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- quantity-carrier-1 from BIPM-SI-BROCHURE-9-2026
- quantity-carrier-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:48453cfd4c1b9afda889f62d48c83b3f01b885340ca28def33d9851ac067221d`; engine receipt `sha256:9966065b5a469cd330eca589f63e5d58cc86cb9815f6ff14dc26e620368efa6e` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-QUANTITY-CARRIER-001-9966065b5a469cd3.json`; empirical-validation hash `sha256:8ad865ff3aea5f0fec4331d26eff0d499e900f6531868df1a557eb6fc5d60047`; measurement receipt `sha256:bea784fbbfa39af43898c8730beb0d3255907a633c0731818d976945654310ae`.

6. Dimension composition and independence

Claim identity: `SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical dimension is forced as the canonical generated composition path of independent quantity kinds; two descriptions share a dimension exactly when their paths reduce to the same held form.

> *Dimensions are canonical finite products and exact quotients of independent quantity-kind carriers, compared by form identity rather than numerical value.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__canonical-dimension-path__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__com` plete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	canonical-dimension-path	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:12a26830ac90cd2e04871a7470fd081ea4243cfa12a67c99df9db8dbdfa7c78d. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:9eed98166f8ac9a14cdf047c4790e53789d749b4d06966057f64d641844c681a.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:8f143ffae762666331bd4572a420f8c08eb1eae0abc7edd002800541b3ce5abd.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:8ad77de29f4c2d52c5ebe06bb3ac7e97bd096be04e2fd47168eb43346cd9e3cf.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:77c4aa61b2b5b44faae27694a0736b89001989fd4d05a4aa02711ed84ae9f875.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:31540d9c686489dbdcd06cde3f9318e569bf13f73ff54208473367ca3ecca16b. Independent certificate: sha256:28ead99c9c104d5a6d65eb11f2e9c52ada94502a34c04b76ef4d6aea4a042043. Engine external-validation hash: sha256:536b553d70addec336fd9e506e1d4f304920a1b5218c02eb605993becec64a6f.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- dimension-composition-1: predicted canonical-generated-dimension-composition; observed canonical-generated-dimension-composition; exact match True
- dimension-composition-2: predicted canonical-generated-dimension-composition; observed canonical-generated-dimension-composition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: `sha256:db439e0a8276e051e917e86afc66345d767cc25a020d3402efc0097722c54b05`. Isolation certificate: `sha256:1fa9e925ed9d29be25c4b44d658e81c9f8f3a67e0d2661ad53717869130fd728`. Custody certificate: `sha256:920d6b69c7a10197b6f27dcb2b3b5c2aadba67d02327c3f8fbe08c631da762ff`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- dimension-composition-1 from BIPM-SI-BROCHURE-9-2026
- dimension-composition-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:bf601f63f8539b443eedf791852143e5a38ef3a2054f5c69a766c4a387a4ea7f`; engine receipt `sha256:88074780f2b97130169a864d0fe1e45073f0a0bc5544a0f71548c31739c12068` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001-88074780f2b97130.json`; empirical-validation hash `sha256:c5aff7dee978371151afc87e90611e3cbca553be2cf41f07afe32c8ec1fbcb29`; measurement receipt `sha256:db439e0a8276e051e917e86afc66345d767cc25a020d3402efc0097722c54b05`.

7. Unit comparison and exact scale transport

Claim identity: `SFT-PHYS-MEAS-UNIT-COMPARISON-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A unit is forced as one held reference quantity used to compare quantities of the same dimension; lawful scale transport is witnessed by an exact positive part ratio and cannot change dimension.

> *A unit comparison pairs a quantity with a held same-dimension reference and records an exact positive ratio; conversions compose those ratios and preserve dimension.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__same-dimension-reference-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`same-dimension-reference-ratio`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d8b385f06f47e3897a6326f3d2cc4bb8da82fed0b97984f823c25d6e8927e71d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:fcf38f4b41423e84d4a14af2d289e780faddd2a08dab621a12d3796b71ff65a6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5a03d4dfa95258d4fbc0a242c8209d3aaeb08e107094e45cd864e66d751d880d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e96864f5f86da2dd02aab6cbdfb056815ea19b5311500f036264d916ab479c18`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:cf9239eeb6851b1aea6c2989ab384f21ebfebab2c7a62a20ace5d8e4d77f9c52`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2da4fce7d904328003e628ffb9bc4da6ec55d6f5b30a191ffe73ec2efea7170d`. Independent certificate: `sha256:cd16c654f4ee555225b946883febceb54e755414770d71342cefdc47d57160ce`. Engine external-validation hash: `sha256:35361952b6e72219657f35ade169137ca3ccb3fd3455013b6c9d84b5f62cea74`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- unit-comparison-1: predicted same-dimension-reference-comparison; observed same-dimension-reference-comparison; exact match True
- unit-comparison-2: predicted same-dimension-reference-comparison; observed same-dimension-reference-comparison; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: `sha256:5db572226cdd4c53f814abfd8cb508de3da6cd8c70ed5de26f9e09638d9660af`. Isolation certificate: `sha256:ddd0194e1db1099bfbfb26e5df55d51222dfee70cc7500cc4731b1e417020592`. Custody certificate: `sha256:b400ef792a9d9a4f6a7c1cab9226f8f2ae05e3ef6229cfb7eae37ec0241341f2`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- unit-comparison-1 from BIPM-SI-BROCHURE-9-2026
- unit-comparison-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value

- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:e6729d5a5819169aba066bb7477481b2b5949f656944ee10eeafaf344fd0a9ef`; engine receipt `sha256:ddea20599918c381ea7bcf8172d46692ae36b76df2bbacbc168d54c1f3a2d5c0` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-UNIT-COMPARISON-001-ddea20599918c381.json`; empirical-validation hash `sha256:36ce553823b5b77257c5425c89595e3353e5a4151d3f29c0019e36b569d595a2`; measurement receipt `sha256:5db572226cdd4c53f814abfd8cb508de3da6cd8c70ed5de26f9e09638d9660af`.

8. Reference realization and traceability

Claim identity: `SFT-PHYS-MEAS-REFERENCE-REALIZATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical reference is forced to be operationally realized by a finite reproducible comparison process whose complete trace connects the definition to the observation; a name or assigned scalar alone is not a realization.

> *Reference realization is a complete finite operational trace from the held definition through each comparison to the observed record.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-UNIT-COMPARISON-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__definition-to-operation-trace__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	definition-to-operation-trace	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:fd441b4443a34e0d1efc0736e7f7f4981d651f31d27895887510fb09083ca339`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1f2dad9e88737b8f22f19d247b8bfb25c4447d0cac7c1e23c4c1407cf97387bb`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b233c37c78a4e4f34ac70474aceef70cc56604f1f671fc12d620f5ed00716faa`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:209f2b5c736cf23e4b94151005ca3062de1d396d0458739a460a514cd1d3d863`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:023389bb1e2e8f8d47b353faf6147bc91f36156ea0cc9c1f771b60c10e642473`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f78121815a382685c4dcaf3c638f980994cf29225e0e9471841814aad2fc4e12`. Independent certificate: `sha256:fc2c880293991e77dc198e7b11a4e81e14f5c0750c67b42ae86f0b95c6acb3a4`. Engine external-validation hash: `sha256:be2ab45528b95c573dda17a3f3b9eacad8d19a9e86dce508cf9bbef6c28c6b72`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- reference-realization-1: predicted definition-realized-by-reproducible-operation; observed definition-realized-by-reproducible-operation; exact match True
- reference-realization-2: predicted definition-realized-by-reproducible-operation; observed definition-realized-by-reproducible-operation; exact match True

- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: `sha256:53cc527e34d7f176884fc21bd5ab8ae6b52b3d18900539a69c83452cf8af3fa1`. Isolation certificate: `sha256:77acdb7059728686161907fa113088985fcd141162a873c11257d910db45534c`. Custody certificate: `sha256:47e6159570febdf9aa2cd8555a9109d7d654c764bc075a672e9fa151edf62ee9`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- reference-realization-1 from BIPM-SI-BROCHURE-9-2026
- reference-realization-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:47850de5d0f25c7228c1263b5d41235a42b70bdcea8cf7e57a6f4137fccdb248`; engine receipt `sha256:7d7698fb49b9b740e1e29560195980c7dc9d7249a03580bace583c378198588f` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-REFERENCE-REALIZATION-001-7d7698fb49b9b740.json`; empirical-validation hash `sha256:419dc28cdf09a88c37bc6d4795e65bcda17893ab9e102096558590adad730d5a`; measurement receipt `sha256:53cc527e34d7f176884fc21bd5ab8ae6b52b3d18900539a69c83452cf8af3fa1`.

9. Measured-value record separated from proof value

Claim identity: `SFT-PHYS-MEAS-VALUE-RECORD-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A measured value is forced as a held external record containing the exact reference identity, reported numerical inscription and observation provenance; the inscription never enters the proof as an SFT scalar.

> *The measured-value record holds the external number inscription, unit/reference identity and observation provenance while the derivation remains exact and separate.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`

- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-REFERENCE-REALIZATION-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__held-reference-number-provenance-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`held-reference-number-provenance-record`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ad437e5c37862af878495de4de7ed6dac5c60212dd1485b839c14653282baac0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8cbb8dd6f3a25af27290eb17336e1bb52b737779cc1e8c9fe26d29a3e92bb585`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:51109d21ed7e0a72b1655f6317744d65af2ed906d176ea92cc990b14ff8cbd2b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8dc482be052ba144450746d84f9dfa508643eabd770d8e96fdf38640b7ef8fed`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:bd48eb008e2e79f555cea3790deddd31fd01286d296245b56cdd01ffa97fe127`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2fd434613bc4467e473b86e1dd9b1720c04bfe2a0deaf8448a1aba855c765f5f`. Independent certificate: `sha256:8d58b9c85c1dcf9970b4a62cec01fa6ad66a07b2650d5aff08d6f66d6d7bc803`. Engine external-validation hash: `sha256:a6eb14b424a10de722279d3a83f80c9841bd8632aeec17339790dc44dfa099a5`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- value-record-1: predicted number-unit-provenance-measurement-record; observed number-unit-provenance-measurement-record; exact match True
- value-record-2: predicted number-unit-provenance-measurement-record; observed number-unit-provenance-measurement-record; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: `sha256:99159d4689f4d920e4a5b194667322505fbd2a174d38608008c73f0e089e6145`. Isolation certificate: `sha256:eeefbb16a836ef1a49a47e5482ad16a3ebab0bc16b97b95da7492ec354520d7d`. Custody certificate: `sha256:ab58f3f8c62e9e5bf2bc4dad61a5c14b9871a832c39f42890eafb89070a0ed42`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- value-record-1 from BIPM-SI-BROCHURE-9-2026
- value-record-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:9fda7c62c521e34a351fb363987d7ee44117239e9ffc558d3b505bb3d4f64b02`; engine receipt `sha256:ee71a58e3d4f2f8b3c646c64cb8d3f922a270cb45b751330a85d7b97e69b6d2c` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-VALUE-RECORD-001-ee71a58e3d4f2f8b.json`; empirical-validation hash `sha256:c4f2b4cba8722a979be0e2ced471cafd9f1c6bd9fc95f1111c8785467a49f63a`; measurement receipt `sha256:99159d4689f4d920e4a5b194667322505fbd2a174d38608008c73f0e089e6145`.

10. Measurement uncertainty as retained admissible alternatives

Claim identity: `SFT-PHYS-MEAS-UNCERTAINTY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Measurement uncertainty is forced as the explicitly retained support of observation/reference distinctions not closed by the measurement process; it is not an irrational proof value or permission to choose a favorable result.

> *Uncertainty is the source-bound set of still-admissible observation/reference relations, reported with the measured record and retained in full.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__retained-unresolved-observation-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`retained-unresolved-observation-support`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:89948e98f461aa717d8ac035f3f9604dbb9196252996540718f45bf6e989cd36`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7a0f4a18b8bb0dd172eab8a909284e061cbfbf6ef91fc8722df97f5662cec719`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0020b7b3b47807abfae0edd921d25970bd18f2ef64054c2ecdc4e902f8076653`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4bfeb3041e24c6d31b4aac408bd4e3b568816786a88191f65030d9d7b73f0b05`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:46a9d02402b8165db8a231977dd86732974422aa8a0d8e28ba871a18e228aeaa`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d194e478c4f9b3b77ffb8c7e737f86ddf25d01e64477403d65c8470c9a87413d`. Independent certificate: `sha256:3ed1ad98c0d3c0ec36d699270786a8ebf315001345b0cf2dd5cf9af2233d0b0c`. Engine external-validation hash: `sha256:1d61d49b663e4c34c4e77767f65c74ff94ce6a923d04ae6f36423a9158e2bf12`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- uncertainty-1: predicted retained-range-and-uncertainty-components; observed retained-range-and-uncertainty-components; exact match True
- uncertainty-2: predicted retained-range-and-uncertainty-components; observed retained-range-and-uncertainty-components; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: `sha256:25fac07fa5ab7c4124704c16ae7cb7b48d99c426206392b5caf3b118629c0ab4`. Isolation certificate: `sha256:900522af12ba110a676b283546d80ff986104da68eca768cbbd0a29f6c335dfe`. Custody certificate: `sha256:d53eb5b926907e899f77d2011528bbd53864fec059e5f5e860c8f767678b016e`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- uncertainty-1 from BIPM-SI-BROCHURE-9-2026
- uncertainty-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:9dd6c75ade4fb7cf74cc48c436826e9720dfbbe1fd572b4bd8e0bcc73404195c`; engine receipt `sha256:b65a1d7557983a7dd30c30fdcde54c55370d7c7ea8c22848abd1ebbcb7ced24c` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-UNCERTAINTY-001-b65a1d7557983a7d.json`; empirical-validation hash `sha256:fd74fdca51a498a4c37cad7dbd15d36e595ef6a6a9a0af99565dc836f51972ee`; measurement receipt `sha256:25fac07fa5ab7c4124704c16ae7cb7b48d99c426206392b5caf3b118629c0ab4`.

11. Calibration chain and comparison closure

Claim identity: `SFT-PHYS-MEAS-CALIBRATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Calibration is forced as the finite compositional chain of exact reference comparisons connecting an instrument record to the realized unit, with every intermediate identity and uncertainty support retained.

> *A calibration is the complete ordered composition of source-bound reference comparisons, closing at the realized unit and retaining every intermediate record.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`

- `SFT-PHYS-MEAS-UNCERTAINTY-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__complete-composed-reference-chain__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`complete-composed-reference-chain`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:582d610ed05178d5192eed4b4fdec549803f63d5301aac65277383bf73bc07fe`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:15f6f58bf9fe46ae235575ddc0319dcbb4eb91b12dbbefc8866777e205cc5f93`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:382c52e07b6cac7a858f688563fd4f4b9731dc6cabffa6739dfc50f3ac8a60c6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a1b9be492a77a444a1f989971a243fee651ffd731a8c7290a7dd879b05235af2`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a22c4401b242007fc6be19c41a51e85675303280663309b3fc00510adf2f0cfd`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9ab06c6a96ac0cb18688ea26ebb49f23022312c0bf37679c013f208842103c4c`. Independent certificate: `sha256:eebfa37af6a3e87822fd7a36b7e36c6e8945c251c837ed08a90f5b01e39cef39`. Engine

external-validation hash:
`sha256:7cb7a69daec5075021039bbe43cb787566c22df79088c9465d3b6b0142fc6b6d`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- calibration-1: predicted traceable-comparison-chain; observed traceable-comparison-chain; exact match True
- calibration-2: predicted traceable-comparison-chain; observed traceable-comparison-chain; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: `sha256:cc8f919e13fbfd2e36a5fc37868c528d5ecbad8e7fc7d236291b7b0a9b0ff5cd`. Isolation certificate: `sha256:d3fadf27ff637b82385e7ed2699ecf23603f71a2a739e8cd5ecd5c359ae0c2d0`. Custody certificate: `sha256:a60ead04453dca41c3030848ac7bdcb7d93e5ddae794edba0d60bbdab72cccaf`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- calibration-1 from BIPM-SI-BROCHURE-9-2026
- calibration-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
`sha256:f3389f0cb33655904ef8ab026b3be2ddb74241cb05f2a92a222c71954fad3f91`; engine receipt
`sha256:63c0243e02bdf6d5786ee2ed2ba0fcabfd6ecae5aeeb7d299923a1d2c6728b67` at
`receipts/engine/model_admitted/SFT-PHYS-MEAS-CALIBRATION-001-63c0243e02bdf6d5.json`;
empirical-validation hash
`sha256:0722575d8b7614615674d68833b5ba5d0a8d957cd8ac71919fac7ff54a2239a0`; measurement receipt
`sha256:cc8f919e13fbfd2e36a5fc37868c528d5ecbad8e7fc7d236291b7b0a9b0ff5cd`.

12. Dimensional consistency and lawful conversion

Claim identity: `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical relation is dimensionally lawful exactly when every compared or joined path has the same canonical dimension form and every conversion is a reversible exact same-dimension reference map.

> *Dimensional consistency is exact canonical path identity at every comparison, with conversions restricted to reversible same-dimension reference ratios.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-CALIBRATION-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__canonical-path-identity-and-reversible-conversion__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`canonical-path-identity-and-reversible-conversion`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9476757ccf62d26eefff26d852ba1adf9048da37c8927ba566c2036c1869529b`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:205006977cc520a84734c06e70ac637ab6c19a87ed851309b146b576a2f9849b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:fdd99f20271fea3b94c4d86056aa69692355c9e236302ae434dede45f04b99de`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5ebff673a39ef1ba8350015cf0f873f1ec3da528c4748d298b49eecb13fe0a81`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9baeb7fe6d1d1e66374d4e218194d3d58557c0e9dfa1859bfcafcdff895e73fc`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:1d4014ac9c8a6f913458caed124b7f4f6b16f8f2a0d478845ae3bc127e5c8f42`. Independent certificate: `sha256:72a1dbfa77d09de03ce057cf80b490ec72d08154ca662b2b452206c2ada8e128`. Engine external-validation hash: `sha256:3fab1249dba139a25bac4af63b21640b9e7990f01ab1d90ac6c75f2527bf5578`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- dimensional-consistency-1: predicted same-dimension-relation-and-conversion; observed same-dimension-relation-and-conversion; exact match True
- dimensional-consistency-2: predicted same-dimension-relation-and-conversion; observed same-dimension-relation-and-conversion; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: `sha256:376ba568e7bf5cf0374c42eab5a19efec99fe2a6fd075083d5c32e6d7dd5a770`. Isolation certificate: `sha256:0ba1bded32220f294aa19a94d90a7d48af17620e2b6433b46ca39c3c09c1687a`. Custody certificate: `sha256:98326d2f719846d08f7ed1689861ad62e03665eb01d595dea56039020dce685d`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- dimensional-consistency-1 from BIPM-SI-BROCHURE-9-2026
- dimensional-consistency-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:bc8de8498c6323c70772f812cc5fd8e1398a15eeb11b16c68cb9352fdcf94acd`; engine receipt `sha256:5310f3f1c40e05a2c14423b0d0d0d5b1775dc3ed45f585bd0971c05eab333c18` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001-5310f3f1c40e05a2.json`; empirical-validation hash `sha256:cd6b07b2c543de233c54c9eafa40dadcdb35a8c41c2797be0be4ae79e57f3e2e`; measurement receipt `sha256:376ba568e7bf5cf0374c42eab5a19efec99fe2a6fd075083d5c32e6d7dd5a770`.

10. Mechanics

Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace.

13. Physical event and generated change

Claim identity: `SFT-PHYS-MECH-EVENT-CHANGE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical event is a complete observed Fold state, and physical change is the source-bound transition between two such states; neither an unobserved background coordinate nor an answer-only scalar is required.

> *The minimal mechanical event/change law is a pair of complete observed states joined by one held, source-bound transition trace.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__state-bound-event-transition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`state-bound-event-transition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:34c6e0bc9be62ec68bc8493c18dcc5f25993cdabfef8ab370dfa43306f402cc3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e5732acdd20a744295dc44bc619a57918fbdc6192e1f90bc424ef81b4205556a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7602df232de970b2c8237b10af653597ff2791df45f2ce143274cb8982ec8c9b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e6d4253c7ea6e792d78e6130fd2e77d803da3e2aa1130caf377680e0067bcede`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d3bea2413cac687ffbde7295267a5c87d29d7607d2d9d6d3f0081d7ef424c1f8`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:6211d85755a0d738c265c26c686df476c122fb10f5751a8cfd2fab55c00e3458`. Independent certificate: `sha256:8362a480031dc0c7b73082a0266fca0f1d8ca29bf509f8f4c214b3b17c4964d1`. Engine external-validation hash: `sha256:bd83b1755b683b8add8ed2d67965299e07ec5590dcadc3cebf2744ae07d236d3`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- event-change-1: predicted observed-state-and-transition; observed observed-state-and-transition; exact match True
- event-change-2: predicted observed-state-and-transition; observed observed-state-and-transition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:aea2bb497ffa6a123b24025cdb4cc9e236077421681064a77ef96833985d0d38`. Isolation certificate: `sha256:ae46c4cd8c6c401ecb5c048f3de3b56837bea2f5ac84cb47496a5c25b7a0e490`. Custody certificate: `sha256:b077719713ea02b9030b4d92f0991c56dfc9e799128107fa8c6b366b96977893`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- event-change-1 from NIST-CODATA-2022-ALL-CONSTANTS
- event-change-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:8bf0896ea2c1944ef828c95af4fe562623ea5ce9b93e1eab92445e8627965a69`; engine receipt `sha256:04874257af857f7573d356ff8ad75f8f85ee0c441752d7f941dbd279bd766613` at `receipts/engine/model_admitted/SFT-PHYS-MECH-EVENT-CHANGE-001-04874257af857f75.json`; empirical-validation hash `sha256:4e8a5951b157520eab6931c979875d2face8326a3089484ecc67a1936225307e`; measurement receipt `sha256:aea2bb497ffa6a123b24025cdb4cc9e236077421681064a77ef96833985d0d38`.

14. Duration from ordered recurrence

Claim identity: `SFT-PHYS-MECH-DURATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Duration is forced as the exact positive ratio between generated transition count and a held reference recurrence; absence of a transition is structural Empty One, never numerical time zero.

> *Duration is the exact positive part ratio of ordered event transitions to a reproducible held recurrence.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-EVENT-CHANGE-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__transition-to-reference-recurrence-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	transition-to-reference-recurrence-ratio	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3f7e717f257f5dfa9844fd9a8c473b84f806abcd73fc517358ff23478fcac623`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b4f9404fd9b9ea4af729a789bef9b65234be6798a6b44e21e1a5243775bd15d2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:27378c512ce057e7da835630bcb0de56a2ebde6396583c756fd90916074215bc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ea92e1601fbcf473a19d93a380e15aa3d0c0a5380af04df94282e814cf88bfb5`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b6acf90f77e4d49ede0f0893e8ece79af4dc9c899fe02e00cff04557a9729da6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a6a15210ce7ca178cfc461603f430df76b1dac99bf6357c9ea0f581f4e1c2ea1`. Independent certificate: `sha256:ea03284dc046e848bd8ea95022c7babbffea01cc12ffdf2cdada7ae9e9d18b69`. Engine external-validation hash: `sha256:83cb0ea00cca669c4282da2a7bf29d1976f3c7fc0746218092545591b36f1889`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- duration-1: predicted ordered-recurrence-duration; observed ordered-recurrence-duration; exact match True
- duration-2: predicted ordered-recurrence-duration; observed ordered-recurrence-duration; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:0ba3539ce8eb8aae69f0b09c3ed790cc19fd75035dcdb15f9d5e5d132b1f2325`. Isolation certificate: `sha256:8b58c16645dc26778a971744c705f888211a21d4e3bb340b4204ca0dde56595b`. Custody certificate: `sha256:b3cf89f143f347b693518369a7fde863595ad083fd6f2a6996c3a1609a7a3a72`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- duration-1 from NIST-CODATA-2022-ALL-CONSTANTS
- duration-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:6685c1113883053673d628b4aa64960e026993967ee5a5cdf6a15918c75bfc18`; engine receipt `sha256:d7c78c834e95a2881236a24ff907323fd5974b1d19eef5b1f5233dcd1fe80bd2` at `receipts/engine/model_admitted/SFT-PHYS-MECH-DURATION-001-d7c78c834e95a288.json`; empirical-validation hash `sha256:eb96aaf54d6ff84876e888c812acde005eec2569729cde7a9ea80088f2007a01`; measurement receipt `sha256:0ba3539ce8eb8aae69f0b09c3ed790cc19fd75035dcdb15f9d5e5d132b1f2325`.

15. Location, separation and held displacement orientation

Claim identity: `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Mechanical location is a relation to generated reference support; displacement pairs two such relations and holds which endpoint retains the orientation rather than introducing negative length.

> *Location is reference-relative support identity; displacement is the exact paired separation plus its held endpoint orientation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-DURATION-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reference-separation-with-held-orientation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`reference-separation-with-held-orientation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:7f320fe3bfc2aa449ccb0dadfc09b746d5bedb845a698cd2a6c5bb57540283d8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:546b51f9b2e762bbd20eb2c8f6a9107e39d7be62c1269a1671ba0ada983d4e33`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:29579f288f2f858a4a2927b99128d64207896f55de5c193bc2c0b72278d69f26`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:00b0e58a2e79c26a3ead80fb8930d4fff51228d95a96ddd56183558069228f91`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7ffe334e6b45268ce30a65f8a96caa4cb7394c356ece583f159260a398aa7448`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ef7e7c152cab399048000402b884130d8b549982d2d4f4457b8f9d6079cc33b4`. Independent certificate: `sha256:74f4571e92dce781927dbbf696f483603323a18a90101594093db7831e6317ea`. Engine external-validation hash: `sha256:24411b1052f4246dd506912be909e6b09589365b094a5ebf0891fa99a31edaa8`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- location-displacement-1: predicted reference-relative-oriented-separation; observed reference-relative-oriented-separation; exact match True
- location-displacement-2: predicted reference-relative-oriented-separation; observed reference-relative-oriented-separation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:14243a9d2dbde5e2ebf8f210c8fd3888ac0e416b5697e32a164017a9846d02e9`. Isolation certificate: `sha256:5f0c2dea32097d940570fefbe86f2a3f8182ce1e95a64d622cdc2c933e81d0b4`. Custody certificate: `sha256:77790c1790f07e796bda0e5c81ed165ffda99c6166178b953e19716fecf62113`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- location-displacement-1 from NIST-CODATA-2022-ALL-CONSTANTS
- location-displacement-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:164cf71aa30f5139f0b486093b515375dcbc5b1f6354c6f72ab4fd8be2478a90`; engine receipt `sha256:651765eae5dc825c8f634ce43690116dbe0a0296306ed19a35ae5a7ff7c4e067` at `receipts/engine/model_admitted/SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001-651765eae5dc825c.json`; empirical-validation hash `sha256:896a6f188cb90aacd010827de8d8a0314217cd696bbe2d8610fef7455fe423af`; measurement receipt `sha256:14243a9d2dbde5e2ebf8f210c8fd3888ac0e416b5697e32a164017a9846d02e9`.

16. Speed and oriented velocity

Claim identity: `SFT-PHYS-MECH-SPEED-VELOCITY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Speed is forced as exact generated separation per reference recurrence, while velocity additionally retains displacement orientation; no signed magnitude enters the proof.

> *Speed is the exact positive separation/duration relation; velocity is that relation paired with held orientation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__separation-per-recurrence-with-orientation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__comple` te-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	separation-per-recurrence-with-orientation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:282820281c3cfa988386b97dc8e686b3d9265098cabdbb99e77bfe34739c7e42. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:857318859f939a1955cdc6e0ddb44e43ccbf563b0898011f7b1fb900a4599267.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:df735e60db928f0a532c8668596c46cb7368b0c6acb8af1cd32c744bbdc6116c.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:521586076d2e04c0f020770696dfb7737aba2718e4e5eb467d7d875639871c70.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:8a2dcbc1ee2621c044d84c1c0fd4a9d60bd6265d26eb03a28113d5b0bc332c6b.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:5f8459771e9c081691badd81c203fd3d4e2cf19ed97bfffe70892af1881fd68d. Independent certificate: sha256:4c406b62a112bbf68a3d9bd28663090c1890c69792279762418ab647e300b45d. Engine external-validation hash: sha256:16fd657af776151ecc4cb3d6a5d646936378ac5d2ac23ccb9ad467c79950f3a2.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- speed-velocity-1: predicted separation-duration-oriented-rate; observed separation-duration-oriented-rate; exact match True
- speed-velocity-2: predicted separation-duration-oriented-rate; observed separation-duration-oriented-rate; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:110d1d1936053ba41d1250d05ff8169e4f7fac90be528adf73a5d915fc128398`. Isolation certificate: `sha256:7514dd49531e595282745b3d7560a52e6fedb867cf332e7a35cbf08eac592066`. Custody certificate: `sha256:46fe4826afaf6528b7746829285b44956b7db6d581711cb53812c7f2a3d21b49`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- speed-velocity-1 from NIST-CODATA-2022-ALL-CONSTANTS
- speed-velocity-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:3ff73715d0314ee18de3427a6de114fb93411e3fb973fdd7bde5239912cde59d`; engine receipt `sha256:43e7e8b28583d51949d24a75fd8ca4024ba7c27e2dfa1d31fb39597b4fb2da80` at `receipts/engine/model_admitted/SFT-PHYS-MECH-SPEED-VELOCITY-001-43e7e8b28583d519.json`; empirical-validation hash `sha256:6e45b659df132991bda7ccca0d2cf1d009da3a80435d04306d0afae27e5f24c5`; measurement receipt `sha256:110d1d1936053ba41d1250d05ff8169e4f7fac90be528adf73a5d915fc128398`.

17. Change of velocity and acceleration

Claim identity: `SFT-PHYS-MECH-ACCELERATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Acceleration is forced as the exact oriented change between two velocity carriers per generated recurrence interval.

> *Acceleration is the held-oriented exact velocity difference per exact positive duration.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-SPEED-VELOCITY-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__velocity-change-per-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`velocity-change-per-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b71ee033260a9717ca14a4d1decdd18a48f40545af022e516d42061323f59df7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:eaa0e0d413902b77efec640ce3898f5916bf70effd6ea8b3ec11d4c59b3afe01`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b6f3db19eb18dbd89342011c6163d5cf7964513f4ed912afd464453f0fa87171`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7ef1899a7ad943756898dbdd1b19194c78cd65367b50c3d006cc0d9073dd043d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b3658d7236963e88665c6e90bc2f701eff616dd0df6cbc3f2de75eff96476e9b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a780de7d431097d31f0dfdd4e98b4f58f296084b8b5734e3fa3ff9ada31659ca`. Independent certificate: `sha256:6072fa5ba6bdf90653743a0fb99a02994860444b53ea4a2ab965068bdfed3ec6`. Engine external-validation hash: `sha256:aed7ceeb80d25df54062a697c7b5e71690380d422f3a3efb012fa6e75272ba00`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- acceleration-1: predicted oriented-velocity-change-rate; observed oriented-velocity-change-rate; exact match True
- acceleration-2: predicted oriented-velocity-change-rate; observed oriented-velocity-change-rate; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:687e98e78211b8fde686f05f2fd64922b33daeb45636e2beb6f3193779025249`. Isolation certificate: `sha256:8129185fb2fc469d365370a1ffb25169a2e57614094556417accbffe1cf37db9`. Custody certificate: `sha256:f9501e2b5c26e272e545fbe9aa34fa9a250d5f4a039dd6e7dcb3ffe03bdc3349`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- acceleration-1 from NIST-CODATA-2022-ALL-CONSTANTS
- acceleration-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:88ffe9c177e9655bc14605137d8db2884c7ffa8df0da7ba13014bf67de29b71f`; engine receipt `sha256:cef0caaf43f08f111540c5e96d75a0eac5bbb8c0c0a28d962742677f263e1352` at `receipts/engine/model_admitted/SFT-PHYS-MECH-ACCELERATION-001-cef0caaf43f08f11.json`; empirical-validation hash `sha256:d9db360cbdc2165cbd9fa54ff6a18a1ba5ce2d2a59ba927768ded5eb0988b730`; measurement receipt `sha256:687e98e78211b8fde686f05f2fd64922b33daeb45636e2beb6f3193779025249`.

18. Inertial persistence

Claim identity: `SFT-PHYS-MECH-INERTIA-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. When no external transfer trace enters a closed state path, the complete motion carrier is forced to recur unchanged; change without a source would erase provenance.

> *A mechanically closed path preserves its complete velocity carrier from one generated event to its successor.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-ACCELERATION-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-trace-motion-persistence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`closed-trace-motion-persistence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d3415a0f553f18c36c41819694dc800ad3e842d86c555ff5b0a32901e9edb746`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a9b00ca9f70199ff8a9224e7859716a9e3f719eb5a98bf3c81e3babbe0ea5ddb`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c304027d52c9d424f878fbcaed897116ce472297a60b57537bec1fdba21f2932`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:de0614da1cdb04608b176111e85876fe7c84ce21a8cda604ea6681bf5518271e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2edbb86bf293bad3273f3cb81facff99276ac4a6ba047c3b67f6035a9ea89464`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c75cc345554614e524cddf65683ede97bc16584472df14533e1bd20d8d035aa4`. Independent certificate: `sha256:c5aa0cab1a8a9aee2e9064bcdc05a566b64d977b6313cf2ae22f349ab41ad3aa`. Engine external-validation hash: `sha256:a31b2aab0c502670751555546b5e76be40368793ef13ce86a85cb49274f1f210`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- inertia-1: predicted closed-motion-carrier-persistence; observed closed-motion-carrier-persistence; exact match True
- inertia-2: predicted closed-motion-carrier-persistence; observed closed-motion-carrier-persistence; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:3453efd2545620b9860a79bdf108ff6a2f03e5f6c30543759843aef72a4495b3`. Isolation certificate: `sha256:8ffa8a8ce7156bdc01e028c232c35f485d7e98151c2b0d207cb49c0990e10c91`. Custody certificate: `sha256:274df124f0e49177f62ca43481ecf00935c6a0d4090eb243946357823981846a`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- inertia-1 from NIST-CODATA-2022-ALL-CONSTANTS
- inertia-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:d5b6704050d0ba9b99388045ce83e92338864f67bd1ab4dfd27ad33dec435cd1`; engine receipt `sha256:c3e7e2bccfecec9a476fc32379bcdb814aaef16ee29fa228ee0a3b4290e5f249` at `receipts/engine/model_admitted/SFT-PHYS-MECH-INERTIA-001-c3e7e2bccfecec9a.json`; empirical-validation hash `sha256:c8da1df56af1f3be2a30f9ecbceddfb70b2bd0de4d7c33bcd804e4157e61fcae`; measurement receipt `sha256:3453efd2545620b9860a79bdf108ff6a2f03e5f6c30543759843aef72a4495b3`.

19. Momentum carrier and transfer

Claim identity: `SFT-PHYS-MECH-MOMENTUM-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Momentum is forced as the complete pair-cell composition of inertia content with oriented velocity, because each inertial part must retain its motion label.

> *Momentum is the complete exact product of inertial content and oriented velocity, retaining both source coordinates.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-INERTIA-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__inertia-velocity-pair-cell-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`inertia-velocity-pair-cell-composition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:5443d18c952f458976a6b5f80fbd10efd11298525ea8cefe172722cd4a0203af`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e901ba60d3e54a104a96a0940b00151d67825f35111ab394275e3d7374237878`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ab52c54f3cc0dbede483e23d6b4e1eb6c27884892fc23836814945f13ae4daff`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f601ea2f1fc531758f7a20059bf9b4252cf59b4f3d03d16d5b122071afb39aaa`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:428e2d96ef2edbc2a6e12d1f215ef82164580a1670516a026448663b1b4e2cdf`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d11aff6c886dca731eb6adf5548b7a409bbb7b62d19340b7b6e43f37176d643f`. Independent certificate: `sha256:002be5aa14d8977be1ca48fd14c295a6980caca97a50b3d57aefcac50c3018e9`. Engine external-validation hash: `sha256:7710d8e84a3bb0b4d3b244cce82c56403e3ddba79405da66ab7bbef5ebe3607a`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- momentum-1: predicted inertia-times-oriented-velocity; observed inertia-times-oriented-velocity; exact match True
- momentum-2: predicted inertia-times-oriented-velocity; observed inertia-times-oriented-velocity; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:9d61a19063020ef65884d5fdf598a8cbe4b6db21303343f5cb561bafb7d0e1e9`. Isolation certificate: `sha256:544649d5b72f19f908dcacbba0aa1667b22998651a99cffa3a389d79fcc6fc83`. Custody certificate: `sha256:9b9c407e841b3adfe6741743eeee35fe240b79b4d80e7333c498367826bda7ca`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- momentum-1 from NIST-CODATA-2022-ALL-CONSTANTS
- momentum-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:443c1ce5c004f7f2bbcf8498a3dad47ff763a2df362a1eb20c3448a0dfdbacd5`; engine receipt `sha256:5b0049e80287dbca2cb5544338ad288ebc2727bbcb04473a90494926f8693615` at `receipts/engine/model_admitted/SFT-PHYS-MECH-MOMENTUM-001-5b0049e80287dbca.json`; empirical-validation hash `sha256:891b99aa46cbfdf4bdc5233e6f33a4a3434924b797ee894a37f250c8248b9315`; measurement receipt `sha256:9d61a19063020ef65884d5fdf598a8cbe4b6db21303343f5cb561bafb7d0e1e9`.

20. Force as constrained momentum transfer

Claim identity: `SFT-PHYS-MECH-FORCE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Force is forced as exact momentum transfer per generated duration; it is an interaction trace, not an independent primitive or fitted coefficient.

> *Force is the source-bound held momentum change divided by the exact positive duration of transfer.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-MOMENTUM-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__momentum-transfer-per-duration__source-bound-proof-of-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`momentum-transfer-per-duration`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:6c262ffceb08893750346dba3ee82ae915141f93ab55ad23bd6f5e557d76b7b2`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ca5ca4b9adef57f96c96e099214bb5382deca3bce3fe84c29d618e412193ed69`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:395f9b84abd524ce73fb49e305d8586bce8c48374b6c6700c29f7f7fac00579f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fb2cbd727c7e8484eb6e8c7e29b362cf9fcf2a70db9720fdf12b53c24dcab409`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0f7f4dbc467ffa45c5df7ff588e3db7985e174a71dea3cf8ec581d8604f9bab8`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:78246d7975345c0e149c90b0c9f340eab96f6257e25dca8be0149661fcfdbc78`. Independent certificate: `sha256:a8a6c056543be591933aeea56583421668f6b22d065b6f658424427424e2a125`. Engine external-validation hash: `sha256:f2105279f6dfd08f0cda6ee18a127b2c57558c41f5b0bf81272c81bb44bedaf3`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- force-1: predicted momentum-transfer-rate; observed momentum-transfer-rate; exact match True
- force-2: predicted momentum-transfer-rate; observed momentum-transfer-rate; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:6391b40a7fb5165066b750ff86ec7741197e6b8c32785a6692213beff372df15`. Isolation certificate: `sha256:3441927a71b44a759595e95d841563ea460b2057e36216426b48177b2780cacf`. Custody certificate: `sha256:e9a80a5fd55021be972f5df3389aba73f7ee7afa14d0e3568e18bfa704c0286a`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- force-1 from NIST-CODATA-2022-ALL-CONSTANTS
- force-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation

- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:cb39e08f950e946cd516455c2fc545cd9f22d3a8282e41fd8d8145488f5694db`; engine receipt `sha256:bd0576b7dcee5c595698b22128853aebe01eabf681d8e44d8b743c63e8d1f7ae` at `receipts/engine/model_admitted/SFT-PHYS-MECH-FORCE-001-bd0576b7dcee5c59.json`; empirical-validation hash `sha256:8f750398faddea1e4f53eddb60dde78b0d80dc69910521cb5a5545632dba0b1b`; measurement receipt `sha256:6391b40a7fb5165066b750ff86ec7741197e6b8c32785a6692213beff372df15`.

21. Work, kinetic change and energy carrier

Claim identity: `SFT-PHYS-MECH-WORK-ENERGY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Mechanical work is forced as the complete composition of transferred force with oriented displacement, and energy is the retained capacity/change record transported through that interaction.

> *Mechanical work is exact force/displacement composition; the corresponding closed change is retained as mechanical energy transfer.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-FORCE-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__force-displacement-transfer-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`force-displacement-transfer-composition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b4f01c166a34b4becb9a3e43730b6294c1330be023d709f59669ee6a04202f0c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3794d4cdc98beb8e04daa8a8cfd87e57160ee075d437dff47ce14644d7742d68`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cf747a2b9ba20a84e0203edbf6ae22b06374bda74bff411269e92f9e67d065d0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e51e82a4d323c5ddbdbfb852b651c858a7f99f04750c48968496df4131a56b5c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dc17d7767ecf8016267312e2843b86c503602c4f7e3bd803771af774bff6ac7c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a6e40d4f25794f17748b16e95cf9cb1d526c6edf9c54c6fb062d5c82ea7d28c9`. Independent certificate: `sha256:829fed84a305929b33019b69cd3a241bb984b134118246ea556eab8751ca507d`. Engine external-validation hash: `sha256:f97e670e49ef7954cc946fa4618376c595614659235e849dfee9a75ee842ee2b`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- work-energy-1: predicted force-displacement-energy-transfer; observed force-displacement-energy-transfer; exact match True
- work-energy-2: predicted force-displacement-energy-transfer; observed force-displacement-energy-transfer; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:7c4755e8f9a871aa70157b336b846dfb5f9bb76c858f7989b5a037e6e62953bf`.
Isolation certificate: `sha256:6b1a2aa2aecfbccd197cdd7dff07ba485d23cdb850eda97234d8aaf9cd81dd38`.
Custody certificate: `sha256:d881aeeed51852847d890c718bc2dec735fc4a1a8c8a47843bf0ee1e80f85f24`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- work-energy-1 from NIST-CODATA-2022-ALL-CONSTANTS
- work-energy-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:f58fd1918c003907e5f787681c7d21ec3689f7784cafb6b5a53d9f93550f2345`; engine receipt `sha256:117d45ba1a18c82dabffd028c5740f7dfb29924cd0502278ee0b2ec003d681b1` at `receipts/engine/model_admitted/SFT-PHYS-MECH-WORK-ENERGY-001-117d45ba1a18c82d.json`; empirical-validation hash `sha256:905220827ef6a08da36806667e80e88dee60a63e44e4941e99718ccbe311b012`; measurement receipt `sha256:7c4755e8f9a871aa70157b336b846dfb5f9bb76c858f7989b5a037e6e62953bf`.

22. Power as ordered energy transfer

Claim identity: `SFT-PHYS-MECH-POWER-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Power is forced as exact energy transfer per recurrence duration, retaining both the transferred energy record and its ordered interval.

> *Power is the exact positive energy-transfer/duration relation with held transfer orientation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`

- SFT-PHYS-MECH-WORK-ENERGY-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__energy-transfer-per-duration__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	energy-transfer-per-duration	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:39655ad1210f70812df6c2be217edb9e30fed6bfa40e21df534cabd5f336c45a. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:c59e356804c621da0ed401ab1c327333cd8cd8cf1547d3f2ff3a849bc5b34648.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:d38e55fd04c92b8ecef7ee144cde3160c67c590cf0cff9e56407c8cf62dd1699.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:480e4a1e1dd88d9395d3b145f2054cc0d1ba29234dfc29b55d72618b6c99c803.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:8cec23188e4bebe26dbd9eb0d22639e7ceecd2a87295c0d68d526e78ff1ee195.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:0196c246d57db36313a70ba28369b6216fc75ea295113c42d450748df61a0a39. Independent certificate: sha256:2aad9d67ad4a812c9bfb23eefae164480cc0385d5f466eb3618bbc5bbc161b7f. Engine

external-validation hash:
`sha256:f97953a256ed88a165edcd14c96972d782abc37855c4850ae1d63526909ae096`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- power-1: predicted energy-duration-transfer-rate; observed energy-duration-transfer-rate; exact match True
- power-2: predicted energy-duration-transfer-rate; observed energy-duration-transfer-rate; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:99998bde324272f38856889bb481160188fcf3ce04c736bd7e4fe5c9b513621f`. Isolation certificate: `sha256:3838d74d9ddf45f10bf58b4222bbcbcd2979a63ebc2e52163dddb34e45f7762e`. Custody certificate: `sha256:d84575e7628b1934b8ec6b93189e0c1cea0d01e4f9fe418ebfb1f29e9bf680e6`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- power-1 from NIST-CODATA-2022-ALL-CONSTANTS
- power-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:5f0a1752b0cd74fd8e92489b01155f19471eb892c5a17cacb6b3d2049d42f756`; engine receipt `sha256:08e14aded1d213c827ff0a0c870fa1cf8d7a2c1058b12237cc90bf34907aa59e` at `receipts/engine/model_admitted/SFT-PHYS-MECH-POWER-001-08e14aded1d213c8.json`; empirical-validation hash `sha256:30a30bdf860d1ee3b72357d8bfea69726dc79c96e845f26b83015ed4ebdd5154`; measurement receipt `sha256:99998bde324272f38856889bb481160188fcf3ce04c736bd7e4fe5c9b513621f`.

23. Angular motion and cyclic orientation

Claim identity: `SFT-PHYS-MECH-ANGULAR-MOTION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Angular motion is forced by recurrence on a closed ordered path; phase is the exact generated position within that recurrence and orientation is held by traversal order.

> *Angular position is an exact part of a closed recurrence; angular velocity is its held-oriented phase change per duration.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-POWER-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-path-recurrence-phase__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`closed-path-recurrence-phase`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f9f2e561d109d6e9de81751e7c653921f101f5845c9a78519db2240d4b34ce88`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:91c99573f2c68caa9dab7a90084374ea7f5c76a7cca7604ad7a525068c2fdb20`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8dd93a8a2f3ff99e337aa19195da236de39d77d20d48b13c82e563779cb863ed`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f84f49d5b56f4473cb212dea76ecf6be895a890c59545ba4bb21c54789b4283a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d8c5f976a54b6985e8ae93a096e8912a7d1284e0d83a52c8b90f3e143682ea3a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:debdc18905c1598597bd9c1c8fc989351f5eba652d479efdadba090aae9d66b0`. Independent certificate: `sha256:6e1a782602dd60976895fa4f344d672e0edb2fd9ea6a87cdcf3ac298ce39a6bd`. Engine external-validation hash: `sha256:55e4e9b1082ddb03a7645a8752eff3759c2cc8cc88924962b2f2f8bca26c8297`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- angular-motion-1: predicted cyclic-phase-and-oriented-rate; observed cyclic-phase-and-oriented-rate; exact match True
- angular-motion-2: predicted cyclic-phase-and-oriented-rate; observed cyclic-phase-and-oriented-rate; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:9d02a16ffde6975e36b764ef2f83642c8893a384b75c8058396cbd689f16ef12`. Isolation certificate: `sha256:64359ff87112c705b5e648f81d2c9c0bcdb9abfac10cc63d66d6823b2e2f4412`. Custody certificate: `sha256:f83b8936b522177942eb7779fde1b820da9ca8ac80c5eaeb67967a88f4a86476`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- angular-motion-1 from NIST-CODATA-2022-ALL-CONSTANTS
- angular-motion-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time

- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:d8888b5b9776b39c607788489bc8944c59e5d71c42e434e6a6bbdff890f43174`; engine receipt `sha256:b07d16797fa48c5cfbc499291558e44e8e02edea85b3730669acb58ce6730323` at `receipts/engine/model_admitted/SFT-PHYS-MECH-ANGULAR-MOTION-001-b07d16797fa48c5c.json`; empirical-validation hash `sha256:8438fd76e81ce7158897becd4f186c72d26cae296386f984f7ddaea9c87f0431`; measurement receipt `sha256:9d02a16ffde6975e36b764ef2f83642c8893a384b75c8058396cbd689f16ef12`.

24. Angular momentum and torque

Claim identity: `SFT-PHYS-MECH-ANGULAR-MOMENTUM-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Angular momentum is forced as oriented separation composed with linear momentum on a closed path; torque is its source-bound change per duration.

> *Angular momentum is exact oriented separation/momentum composition, and torque is its exact held change per duration.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-ANGULAR-MOTION-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__separation-momentum-cyclic-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`separation-momentum-cyclic-composition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f8442286386cb2ef239ac8703741fc47e3ec72a40ad5e894a58341bf2fe6e7f5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:232302a821809c4550eb83205445d4fe0816d8317ffa55d47fc0bfe0b8e931a1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3c2cbebde1987ac9cd48919d2c1eaa2540b6431c32946931533f660f03db1551`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6a9e927be1334cbfa1a2a3b9cb661f58799a598a850dd10c16a49c2a643d256b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:897299ac6dc97d35ba690f074a5d8ec1aa9272658f0f6f30e02bad9ea3ce39e5`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:7d49cd2f4c425b1cd3ec28f07c2fb50a5bb3a919834be024fa9dd88a2345cac0`. Independent certificate: `sha256:b3f560d8405276173d31892428ef9dee06ff2c19dc0ee2b925d60ed592cc8913`. Engine external-validation hash: `sha256:24e660a5bbee4afff8b5bab36d2012a84e1910a66d0e0e885b7ed380d506152c`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- angular-momentum-1: predicted oriented-angular-momentum-and-torque; observed oriented-angular-momentum-and-torque; exact match True
- angular-momentum-2: predicted oriented-angular-momentum-and-torque; observed oriented-angular-momentum-and-torque; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:01cfac6a131a3f9ac5459dfbba35c567836f69be942c01209c7843d316a28d8b`.
Isolation certificate: `sha256:3f05090edb4b41ca9175d271216469ec0d7c7b490589b77da9ebe8a8e8520ccc`.
Custody certificate: `sha256:9c73effc16a76cadee5e6089cca085746269013460e0796fc1d467e027adc36d`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- angular-momentum-1 from NIST-CODATA-2022-ALL-CONSTANTS
- angular-momentum-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:2d79b541a967090a256c94f8ed25e966d3c64e9ad764283c443b639997b5c91a`; engine receipt `sha256:bdb07ed36a0c97019a98b71ae63029df5455948ff7f61ca5f302d7a29d07f5f2` at `receipts/engine/model_admitted/SFT-PHYS-MECH-ANGULAR-MOMENTUM-001-bdb07ed36a0c9701.json`; empirical-validation hash `sha256:4c58f6964748badf6194d250a0260df9b904844b08f77eac8cac4394cca65467`; measurement receipt `sha256:01cfac6a131a3f9ac5459dfbba35c567836f69be942c01209c7843d316a28d8b`.

25. Composite motion and centre relation

Claim identity: `SFT-PHYS-MECH-COMPOSITE-CENTRE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The centre relation of a composite is forced by complete inertial-content weighting of every part's separation, divided by complete inertial support; no constituent may be omitted.

> *Composite centre is the exact complete inertia-weighted location relation and composite momentum is the disjoint junction of all part momenta.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-ANGULAR-MOMENTUM-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__complete-inertia-weighted-part-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`complete-inertia-weighted-part-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:797e257041d29d71bd8fb092cf3630f4b1ee06e17bcd1840555466e54a3a4981`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c511bfc1eab2cad1eea54aa1e729de50494692446e8d8ddc9a2b48351cbdf687`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8d19e87768f3fb35a39532c3a72e89d6cd2201521652f2d23d8f0e38a37f8cb0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:787aeda12671300fc6430e467f1113186dce377b5dbb79024aed55a65dd28b4c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1635fa44abf739cc12b3b0b18df0b1a5523b163829a538419f567f9f0264709a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:377e3ae1335d43420ae3d96342bc263578b30766dcf3fc926cb2dcd400536d22`. Independent certificate: `sha256:228be1a5037de6c20a8321e3a332e0ebe67c77d0e395bd9e6233c13878024dcd`. Engine external-validation hash: `sha256:531f607a293e50c7f12d25bade79499eab7c9afb2fc2d3d59b4d591332088fa3`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- composite-centre-1: predicted complete-weighted-composite-centre; observed complete-weighted-composite-centre; exact match True
- composite-centre-2: predicted complete-weighted-composite-centre; observed complete-weighted-composite-centre; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:57931fe0cc00ecf60c476441524aa57d318625ed5a295878643e6dd53b2b4dc1`. Isolation certificate: `sha256:897ae9e5366d65a399997cf6db6287fc023cad0713687d9b5ffd44774d7a73e3`. Custody certificate: `sha256:88392ad05e775f3a15b5611c286af38c935d9ac8500d8cf05bc7ab89943f7526`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- composite-centre-1 from NIST-CODATA-2022-ALL-CONSTANTS
- composite-centre-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:d89997a4c2ae4d17dc9639327e560bce38fd65602708dd485856b8f297f9fa61`; engine receipt `sha256:b565dc99daadf7225e7e9a817a731b8a9f7ab22359ea149bbbde3846dc5819c7` at `receipts/engin` `e/model_admitted/SFT-PHYS-MECH-COMPOSITE-CENTRE-001-b565dc99daadf722.json`; empirical-validation hash `sha256:ab5002712e1f825a0ca0bd12bbc74908248f3b01a15da25beef2157064d3190f`; measurement receipt `sha256:57931fe0cc00ecf60c476441524aa57d318625ed5a295878643e6dd53b2b4dc1`.

26. Mechanical conservation under closed transfer

Claim identity: `SFT-PHYS-MECH-CONSERVATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For a closed generated interaction network, every lost mechanical carrier at one node is forced to be held at another; complete internal transfer therefore preserves total momentum and energy support.

> *Closed mechanical transfer is a bijection on complete momentum and energy carriers, establishing exact conservation within the declared boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-COMPOSITE-CENTRE-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-network-transfer-bijection__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`closed-network-transfer-bijection`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:7b2d7365612ec55f3bacb18617644577330a71ee92fe7c17c3eafb14d7db3772`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:39dce981a6c1b9efa89ad292f1c4e993835d0c8c80e0a6e962373813517f9756`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f7a65471ba766cb9eaf7aad30613aa0c30042309ce32542d267a35a979d98972`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a3e760a6792fa1b929d65b81fa7a0783ce3d1f4239ee253cb6beeb11a987c5e2`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:87bf710c03122257d4572610f5ad7a93a11d22329112dfd09faebadb29c07d10`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4adb9788db5bc48fa2d40265858e1ffc9ba1e2076bdab833af6e35a6303a0c40`. Independent certificate: `sha256:b2b1a8e1de1c790784891d0fdf9fb10b142e7b8889317a4860d2d1013e0ea715`. Engine external-validation hash: `sha256:5703c8d77b0288455bcaafc21fcb52cf0dbb70e98f8dca2497c5264b41d8dd8c`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- conservation-1: predicted closed-transfer-conservation; observed closed-transfer-conservation; exact match True
- conservation-2: predicted closed-transfer-conservation; observed closed-transfer-conservation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:815762e2322cb47c3cf1ba62ba6bd075f141894a653579e247717e88a0e0d506`. Isolation certificate: `sha256:79bad415ec7d447d52728a8a38e0a853f64d5c3f5aa149c1cb1b6f805caa7a67`. Custody certificate: `sha256:a011ba04ebcca047f26d0eaab21eaa501601fccb6ff7efc0e870329b282ca966`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- conservation-1 from NIST-CODATA-2022-ALL-CONSTANTS
- conservation-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:27d63656f9e8b5b72f367ea2348f9d48c42ae7f133f78a3906c7c64fbbc06994`; engine receipt `sha256:01d57e618335bfc6fa3513336d0e1a30aa2fa8cf02bdd18420983b1a29a92093` at `receipts/engine/model_admitted/SFT-PHYS-MECH-CONSERVATION-001-01d57e618335bfc6.json`; empirical-validation hash `sha256:33093892c053c89f41bc7d34b5957be21dc7d8287e71fcd0f9129c86d21cd20d`; measurement receipt `sha256:815762e2322cb47c3cf1ba62ba6bd075f141894a653579e247717e88a0e0d506`.

27. Constrained motion, restoration and oscillation

Claim identity: `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A reversible constraint maps displacement orientation to a source-bound restoring transfer; finite repeated reversal forces a recurrence whose period is an exact transition count relative to the reference recurrence.

> *Constrained reversible restoration generates a closed oscillation with exact recurrence count, held phase and conserved carrier trace.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reversible-restoring-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`reversible-restoring-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4dd5fa2a86bb517aed073b8e5bf93a40107b23b1a419c638d67d140154b2e049`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a5d131fd9eb84ebdd960da36959498a70e1f2399d40f5c72f20e550924de89b3`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f3c229eb365b1fa9e7a68da36f338651cc373ef16a836b6eb2870c663af016f2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:69687c3dd3eb6008dd8e5e5029c70cf19fa4d6f867c34b19d96acb6fa4947ec4`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:475a40edbb23e6c9929bd30fd6c8f763a41a1d41b622238548f14a9241f93bea`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:716edae5dedaef06202415df671c379fdc51dff7d5e60afa73d9879daf953e06`. Independent certificate: `sha256:733c4ba617c22f1f664f6efb16868a19824da109570788e2d2caf08eda91e44a`. Engine external-validation hash: `sha256:b3c458d7a58a1cc360391e1b2c725a507d31cd3b00c4900519b87411d89157ef`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- constraint-oscillation-1: predicted reversible-restoration-oscillation; observed reversible-restoration-oscillation; exact match True
- constraint-oscillation-2: predicted reversible-restoration-oscillation; observed reversible-restoration-oscillation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:3bb6f0375bdece35468b92f96ca24d265ef1d765b0bcfa4d8be15776677dedf0`. Isolation certificate: `sha256:869c675b7fa2e7fdcf0a8492e20a6632dec690a4671767db909c1a31b548a0fc`. Custody certificate: `sha256:826366cd43f786fcd9e95b22d77c25aac7e89915e3e9a52fee1ee2e873296b0e`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- constraint-oscillation-1 from NIST-CODATA-2022-ALL-CONSTANTS
- constraint-oscillation-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics is reconstructed as exact state change, recurrence count, relational displacement and closed transfer. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:f4d4a396cfb3fc87f72258aa02ade12ef76331fd0ddd2e3924cdf0966031edfa`; engine receipt `sha256:269bf50548ac623910ae13c882311660ecf1ff58a23a1af3340fc9c6e7c80dab` at `receipts/engine/model_admitted/SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001-269bf50548ac6239.json`; empirical-validation hash `sha256:766fbe6bfd29d3d36d43619a67424be1ae5a2a33ca9f50346f35fb81854fbb29`; measurement receipt `sha256:3bb6f0375bdece35468b92f96ca24d265ef1d765b0bcfa4d8be15776677dedf0`.

11. Interactions Fields

Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace.

28. Source, response and field carrier

Claim identity: `SFT-PHYS-FIELD-SOURCE-RESPONSE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical field is forced as the complete source-bound map from a localized distinction to every generated accessible response cell; a value detached from source and support is not a field.

> *A field is the complete finite map of a held source distinction through generated accessible support to source-bound response records.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__source-bound-support-map__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`source-bound-support-map`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ab689d82a4bdef284f4320b91502eab091951f6214e0e2f6f14f46bf220aa998`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:041f78c0bd302fd7b80d81d704c363231a1d9f969d5afa0bb5616744aeeee376`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:744369e625fc3e621706b51f234a09906a2d26b6548cc4fa29f9d5610a8e1b41`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c6481452651e95b15b4bd1d40d946b6b1b6da41e0ce684ca8ccf6b65a4b31b95`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:33b67fad6c33cc5a25ce4f9fd87d6234a1f7c5112d3a4ecdd2f6ac1fb376d3bd`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:1eb0754b3023b1f21e490d51ab8ccea125ba60c9247fc197cbd4fa99ede542df`. Independent certificate: `sha256:4557b6f886c13ea1c0718f82f936530fad652700a247c5c61936cdbcec75d8d2`. Engine external-validation hash: `sha256:829fa28c07fdea5f2d37d66e599608a93e54260f49c880d43cf64f2f1fd6003e`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- source-response-1: predicted source-support-response-field; observed source-support-response-field; exact match True
- source-response-2: predicted source-support-response-field; observed source-support-response-field; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:ebc58b140ebc9bb3b1a12b505dff0de9ef6fadb8e9adbdccfe53045a8203daaf`. Isolation certificate: `sha256:b422bc0135ec06f820f1fa2b84bd367455372cc06b4ba71b194d8c1ead580528`. Custody certificate: `sha256:770b5eec779fcecf8674e1aef286d851d1821f49bafa4f6a7871be7af944849b`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- source-response-1 from NIST-CODATA-2022-ALL-CONSTANTS
- source-response-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:6a8409307ffe7115cc2e0eedc4f99557d8bb07df360cb88cceb8ec040bb390ff`; engine receipt `sha256:203231945e022e79b7ffae5fc692aa84ec9e0cf0acea135fe22ac2bc6838ebc9` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-SOURCE-RESPONSE-001-203231945e022e79.json`; empirical-validation hash `sha256:69a79da801e537f1ac5d55c7c785a98140d4a4b89b6f2c30c47a31cd43822e96`; measurement receipt `sha256:ebc58b140ebc9bb3b1a12b505dff0de9ef6fadb8e9adbdccfe53045a8203daaf`.

29. Generated support and geometric dilution

Claim identity: `SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. When one conserved source trace is distributed across a complete generated boundary, response per cell is forced as the exact source support divided by boundary-cell count.

> *Field dilution is the exact conserved source carrier divided across every generated equivalent boundary cell.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-SOURCE-RESPONSE-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__conserved-source-over-complete-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`conserved-source-over-complete-boundary`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a5ad5c02f73a2f73028454f9df4ce617a86055220c5025e2945f6240c8da7bfb`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8d76878d3e63df5947b4e677c368dcbc9f06995db3539663069867031b6026e7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:17870c6c9111f8027c5a04318a92cb02613787d79a5dd91ff6e809bff8af9604`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ea33d8af27399c9e8e03ca0d37ed92667744e367f6daf1870754adb4a7a6496e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:52211b87aa710adc74dd296af6b6cb3f011a2ff8ee05bb75a61c1483607b29c4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ff7dc6ce86a1c911c501f6f7dd6fa3d786b3daf1fcd34650b2f0fe9e9b83de57`. Independent certificate: `sha256:31100409c4d7ddd477f1b87fd73fc90377f5755a34b9ecf3c00a30dffdfd3fa0`. Engine external-validation hash: `sha256:43689921203884f3fa5d4e49909b5f1993a5044ae43aeb130a97af448f5cf1cf`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- geometric-dilution-1: predicted conserved-geometric-support-dilution; observed conserved-geometric-support-dilution; exact match True
- geometric-dilution-2: predicted conserved-geometric-support-dilution; observed conserved-geometric-support-dilution; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:7c949ef7a741f0d750e380e39695e428563e895f4056966995f73fa721d4bd3e`. Isolation certificate: `sha256:d7b278138765a0a1629cf81e84303985a0b9c8c0d9cdff39b0e5b463f5cf2fb7`. Custody certificate: `sha256:92d3436db300e05ebffad7816837749b98ad6b481cbf56d5c8a1cae914c84ef2`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- geometric-dilution-1 from NIST-CODATA-2022-ALL-CONSTANTS
- geometric-dilution-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:a1d9d6acc9d4794e6ca8f81d22c2caf35ff18a9d4876197736779c206b294cfc`; engine receipt `sha256:afa50140059d7f2d81931e46b6761c0053f4e77eeee0538c1127843dfa690ceb` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001-afa50140059d7f2d.json`; empirical-validation hash `sha256:1984c29055fd7e0858da0bdad73676ee0b11df022da20fef9c51036c1f6cb3c9`; measurement receipt `sha256:7c949ef7a741f0d750e380e39695e428563e895f4056966995f73fa721d4bd3e`.

30. Gravitational interaction at the weak relational boundary

Claim identity: `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A gravitational response is forced as a universally source-linked relational change of free motion, with geometric support dilution and no held charge-family alternative.

> *Weak relational gravitation is universal inertial-source response transported over generated support and observed as a free-motion change.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__universal-inertial-source-response__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`universal-inertial-source-response`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:98df42a85cb0c69e1cd0d12ceab9721d463fc60730527b17e5eb9bfcdc677167`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c8f956d03349e201afeea9b8b1d15f3416a3d0728a92431d69d7af3c6cbf8c52`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4c07f5b01b4f9dd95aa95edadd518b8e04541fb9492c9dc9f6d8fb1e42a6a722`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6606681b51c0a9a8c9e7e3af1497013f2f0339e449f349a6ad24519836bebd28`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4bf70ab1a1ab6aa0165875951085c651fda9fdfef6628f26556865be009095ee`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:93e83ead15508eef1a445383097eb16e7061d3dbc4eb5cf3812f52e9070193fb`. Independent certificate: `sha256:6b77d9e624382a50e26aef406e60763d677fd957d4ed398ed5202f15308a76bd`. Engine external-validation hash: `sha256:442209281cc34af2cde089152fb07d5c4d80efcebdd7627edf5ff30777a1ebdf`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravitational-interaction-1: predicted universal-inertial-gravitational-response; observed universal-inertial-gravitational-response; exact match True
- gravitational-interaction-2: predicted universal-inertial-gravitational-response; observed universal-inertial-gravitational-response; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:927062d762861272cdd1cbceacf880e129da86595b727c481e7c1c23c1ea5924`. Isolation certificate: `sha256:a230a0bba51fb5940c2035dfd7628114f864906291072450629 7f4627960e785`. Custody certificate: `sha256:a3ced0e0bd48469617d4df7b261050ba56cd256bbc3b93a020887649c25f8b20`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravitational-interaction-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravitational-interaction-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:959b4df999632bda004902de2ef28119033c172f7551c25e097d3e8b949fbd3b`; engine receipt `sha256:82395b33bc2287af04a87f7dead671259e3f3cbcb24d0cf507b64379649d7f23` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001-82395b33bc2287af.json`; empirical-validation hash `sha256:f45f6c9954f64dbbddfbf33e554d27c607df83a6c89480ca0ce6442292a4d804`; measurement receipt `sha256:927062d762861272cdd1cbceacf880e129da86595b727c481e7c1c23c1ea5924`.

31. Electric distinction and held charge orientation

Claim identity: `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Electric charge is forced as a conserved two-fibre source label whose opposite orientations are held distinctions rather than positive and negative proof magnitudes.

> *Electric distinction is a conserved two-fibre held source label; its orientation changes interaction class without introducing negative quantity.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__conserved-two-fibre-source-label__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`conserved-two-fibre-source-label`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2ce0dc22064dab08ad3fffba1c19bc0ee823b4f9b7eca2e4e6839561390c5233`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:662bd8009ab209cfa1bf46d5a554e8d4b79eb53d9f4f82c88719857fff9fe78b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:be9c50d45008014f4fdd20aa8bfa18f5e91f9a120a755427781b5ccb608d5d7b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d467869edf735c08251cb166a4868024d2acb9cdeae472dfe01f267faff6511a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a8d75ee0c59080e854470b31c6db28bb990b07783468a97e7c971f921ebac4b1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9a2814a2eb56fc4b08cd99d8fd678f66d1f3b4ad98b593730e5d5229030fd0da`. Independent certificate: `sha256:1dc68a09b17229216ebee4249420c082e95bcfc78502b5c053714467bad10b29`. Engine external-validation hash: `sha256:a7bc39123649080929c475be867911646fee104456cbba883ae8fc16c46c640b`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- electric-distinction-1: predicted conserved-held-electric-orientation; observed conserved-held-electric-orientation; exact match True
- electric-distinction-2: predicted conserved-held-electric-orientation; observed conserved-held-electric-orientation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:22c918d208c1b4bc38aeb319dc9ba5aeed21cac8d558392edeb6c341a85e2325`. Isolation certificate: `sha256:6d626766c8a56217c2f794556dfd746c7ac879b9ef41e6a5ab2839758785b8d1`. Custody certificate: `sha256:90da359acd9cee371bf96463f8116ad13eff990c664b0074301a772c797d1558`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- electric-distinction-1 from NIST-CODATA-2022-ALL-CONSTANTS
- electric-distinction-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:3b19a83d707e022cee35bb81cd57554e60965da8cd660562699270b1aaa43212`; engine receipt `sha256:479be6249f070c662d307aeec59a6ceeeceb4981de428f3200a78575f05e0cf3` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001-479be6249f070c66.json`; empirical-validation hash `sha256:fee8a5e70fbcfa86d433625149b2927d948bde0ae51c92ac12783236bdf75863`; measurement receipt `sha256:22c918d208c1b4bc38aeb319dc9ba5aeed21cac8d558392edeb6c341a85e2325`.

32. Electric field, potential and work correspondence

Claim identity: `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Electric field response is force transfer per held charge carrier; potential difference is the exact work transfer per charge between two generated support relations.

> *Electric field is charge-normalized force response and electric potential difference is charge-normalized work transfer between held endpoints.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__charge-normalized-force-and-work__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`charge-normalized-force-and-work`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ececeb9261cf699b09d96214d6465e648b3ae8179afb60f5c4d5b78771f98817`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:822036fc67f0501f0487767675b89306c3fd296509762afc01915cdf4786acea`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4c3a8711ed4de88b4c8578e65a3f32681900e1c7d93b94a6a46f8b78ecfb38ca`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8c5e7fbf3054bddf4f5a969d20418adfbdd03f419c1f058e8d14e048244a9257`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4da112bff0e21eb5e42d2bd033cf0da2ce75ca226c5d371ecae8c9e562e422fe`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:29a14819bf54c8423f2fbdbd136d4462c6698875ac2ae78bb63b70a7aa605f8d`. Independent certificate: `sha256:ba419390f0d39e89fc7bbe511178cb83411759ebdc52d44d4ccd1cdd02c32bee`. Engine external-validation hash: `sha256:d4b8eb4595f2c2c31999a1746c93080d18bfbf03741304e20ef8ee9e12725f07`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- electric-potential-1: predicted charge-normalized-field-and-potential; observed charge-normalized-field-and-potential; exact match True
- electric-potential-2: predicted charge-normalized-field-and-potential; observed charge-normalized-field-and-potential; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:07aaa4990023f31405ca92090ea2aa9e9ddad31f4cb58413884f1a42cdaead89`.
Isolation certificate: `sha256:9bc4092f84a6c2a1b24da8eb1ea5466f0c6b7b2f85320d14544ffc2e62351d49`.
Custody certificate: `sha256:54097d62ec821195f76cd07134ef73de0060c9ad252a1b3370bbb14cc4e699bd`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- electric-potential-1 from NIST-CODATA-2022-ALL-CONSTANTS
- electric-potential-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:974168706fd3887f0fd2fb69728d2b38b064d2fa5fcb00130cf5be4220f942d0`; engine receipt `sha256:404307768a30ea39fce5e7ab47589d6a5e103841f6c7c532a63328bed4a4feca` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001-404307768a30ea39.json`; empirical-validation hash `sha256:6af026f60911b1f1a0642a8ac4949daea63e99b0859ba57eaa9876439a5046fe`; measurement receipt `sha256:07aaa4990023f31405ca92090ea2aa9e9ddad31f4cb58413884f1a42cdaead89`.

33. Magnetic response to oriented change

Claim identity: `SFT-PHYS-FIELD-MAGNETIC-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Magnetic response is forced when an electric source label is transported with held orientation: the response is transverse to the source-motion path and reverses with either held orientation.

> *Magnetic field response is the transverse held relation generated by oriented electric-source transport.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__oriented-charge-motion-transverse-response__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	oriented-charge-motion-transverse-response	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:46fb7deadb9ab1e5d70004af859171d67cae1ecb18b29c6a4fda07104eb42872. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:30c315164b77365902d50426dc07b08e9aa4c4641eed78a15394675cbed09921.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:56f4d2c79bb6eeb8f1113d3eca4ef88cf61b28c469e91d55d85c2d6cc55ace57.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:ebc25039336454dbcf6bca3ac5bd76621720cc1ad5b7dd049184db31ba6c6239.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:e1c50337da9ca2040e691d3aa093e4ae1e50247e0c03fcb77f13512767ba34eb.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:92c14267164b0df4326dc155fac09d5a7ebf9b6469774beeaac2999709d34fbc.

Independent certificate:
`sha256:63f9580fa4c97bde378def33968315d4581aafcbcfe25c81a9bedc902106a2fa`. Engine external-validation hash:
`sha256:c916d84bc22d7175aa68f7d0576f93db52786e5a556780546dbcdf054df39ee1`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- magnetic-1: predicted oriented-electric-transport-magnetic-response; observed oriented-electric-transport-magnetic-response; exact match True
- magnetic-2: predicted oriented-electric-transport-magnetic-response; observed oriented-electric-transport-magnetic-response; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:7cc2da823d648c756e5c806a61c7dc0b4d5e00d24c847c4b9b6b742962595c52`. Isolation certificate: `sha256:c0c7428cdbd6d2a40e2cd5bf2bf774b135424bc54e0951056b7264ba322031ad`. Custody certificate: `sha256:abd25cb8c5572b0bf375789e2b66872da1c254ac08ae95f3d8e9209e544a7602`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- magnetic-1 from NIST-CODATA-2022-ALL-CONSTANTS
- magnetic-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
`sha256:26b1b3c56afb0e115418eb8afb799524ff9b54ca8b2084f02dbedce5e165192e`; engine receipt
`sha256:4a212db2fc1113d2502f514d023caf210814886a20253226b2b4eeae0328f238` at
`receipts/engine/model_admitted/SFT-PHYS-FIELD-MAGNETIC-001-4a212db2fc1113d2.json`;
empirical-validation hash
`sha256:c064915bfc94f33a0d756ee592e8b77849b75320ea296143a33cdc28380dfda6`; measurement receipt

sha256:7cc2da823d648c756e5c806a61c7dc0b4d5e00d24c847c4b9b6b742962595c52.

34. Induction from changing linked support

Claim identity: `SFT-PHYS-FIELD-INDUCTION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A change in magnetic support linked through a closed generated path forces an oriented electric work-transfer trace around that path, preserving complete source-change provenance.

> *Induction maps exact change of linked magnetic support to held cyclic electric work transfer on the complete boundary path.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-MAGNETIC-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__linked-support-change-to-cyclic-transfer__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`linked-support-change-to-cyclic-transfer`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9a4b3fa00363266d9583a011a13e0da7c501edfbe373eec19dca2ab8cd1792ce`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:effda451435526b806ea9a3e3a39865f35fdd7448e7f0a4af4b255e828947cfe`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:aac1ccdceaf7af651fc29f7cad1a33665c2beabf5356f8529b6dcd4b284becb9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e69bf6759c052cd10deef515b60131f7cb67a3860c4cb71f8fa7e9f5d045f5b5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dfbbafb8db5bfa6776ac91b2499e152d8979e5ee70eed6cc80cd3b22bdb345d1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d967a5095815ae755cd2805bf7772dbee85d90a62d8f7da10cdd67a2eacc7056`. Independent certificate: `sha256:b16e812eb78b1e7d7dad39616b999b6e02ae2356070dea3fb3e9ce3ae490862b`. Engine external-validation hash: `sha256:007b6be4a433d5f4afd85c2540fbf8eeb3e90f27a4af627b30273915f4de9ee8`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- induction-1: predicted changing-linked-support-induction; observed changing-linked-support-induction; exact match True
- induction-2: predicted changing-linked-support-induction; observed changing-linked-support-induction; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:82ddf178dd9b2f895e85ccfeed9b438ba2c20abbab8d6872acdd92ade0f54328`. Isolation certificate: `sha256:bc9122549dfdda1d4dd33778f963a8b6d1eefd1171390a00c9e1311e55eb3438`. Custody certificate: `sha256:1cd9be2be5393c86c7b6a4e3f7b11212d3e1bc1971d91703ed7bd8ca5310c4ca`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- induction-1 from NIST-CODATA-2022-ALL-CONSTANTS
- induction-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:91048d8bdf43be16b8fe48671f5b7bd714b7ea78acf2903749347e3049beadfe`; engine receipt `sha256:65da863a9a1b716d62b7541040863b59ee56c0055a972eaa5ac8c2b5e82959c8` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-INDUCTION-001-65da863a9a1b716d.json`; empirical-validation hash `sha256:a2fc5738662e6375822daa99fca34c25b0efcc44a4cac79f29529db6dd10e3df`; measurement receipt `sha256:82ddf178dd9b2f895e85ccfeed9b438ba2c20abbab8d6872acdd92ade0f54328`.

35. Electric-magnetic compositional closure

Claim identity: `SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Electric and magnetic responses are forced as two observation projections of one conserved oriented source/propagation structure, because each changing projection generates the other on a closed trace.

> *Electromagnetism is the closed joint carrier of electric and magnetic response projections with common source, orientation and propagation trace.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INDUCTION-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__joint-electric-magnetic-propagation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`joint-electric-magnetic-propagation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0103b751c089c7d16638e42187f95f5e6adff96bff0b0560e0a2996664df99b9`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:baa78f1d8cedfcd5821c132f2d797357cd64b2b029c3826f66f0c65fcf38617a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b0dcc38e2cd08fe6ce5242e906ae9baba21037229d30005d0b0edb1d2d6a282a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:559638b588ba3218eed5a3d470e7e6d6357b9e13f08f851b8ca2495d2385ac2d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0fdb3b10e60f99983b701490c84f3a76a9ca115ad9ce5af6806024f3401f0fb7`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:82b79da2d7b5c6902bbda52aca1ba49b9602ce9bd1c667d9329cacbc77a7cb05`. Independent certificate: `sha256:0c9579c39d3c0045693c0c1ed912f75c31f422581621ea0dc0d9c7d25de531c9`. Engine external-validation hash: `sha256:bcfa813af18126aa970ec5ab55ff4a3272ff0a3ac60232fb485fdbfc0a5a9cf3`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- electromagnetic-composition-1: predicted joint-electromagnetic-field-composition; observed joint-electromagnetic-field-composition; exact match True

- electromagnetic-composition-2: predicted joint-electromagnetic-field-composition; observed joint-electromagnetic-field-composition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:84f2644f6405541a10ec990c1840254adc51d1930d8adeee9ebbbe359f0d20b0`. Isolation certificate: `sha256:b31f387c1c96ed708d6d43743be58cdcec496efab02aaa77b5b688943cb3bf17`. Custody certificate: `sha256:bb9967ccb10f141c024eea8f8c0456c3ffaa193c0439001c47d97404de2f60b4`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- electromagnetic-composition-1 from NIST-CODATA-2022-ALL-CONSTANTS
- electromagnetic-composition-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:dba269304955eec2c12619e9043329c08e61e0ae65151f17c09951511441c2e4`; engine receipt `sha256:470397ea06dec2c8b33de5b77e4f6d0ba6681999d445b82eec45b93ab9233080` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001-470397ea06dec2c8.json`; empirical-validation hash `sha256:315ce6e7344ac779f48d29f8127718531c10815e571ec6c46a7fa8e4e368742a`; measurement receipt `sha256:84f2644f6405541a10ec990c1840254adc51d1930d8adeee9ebbbe359f0d20b0`.

36. Gauge-equivalent descriptions and retained observables

Claim identity: `SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Two field descriptions are physically equivalent when their complete observation and transfer traces are identical; unobserved internal label-path differences remain held descriptive redundancy.

> *Gauge equivalence is canonical equality of complete observable field traces under changes confined to held unobserved description labels.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__observable-trace-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	observable-trace-equivalence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:05abd37a18d2cf1e1193aaa56575c12b04dc0245e1d47c30e785a96dd5b95e2c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f145c14ad85295d09e564e20bdeeb87ee7d7536b03cc9e636ec2232c0cf1306e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3baeec4b82b342a0307585a4f83a80fc186153a8092922a7879bf09be0d9b070`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:2b4b3589ac32501ee81c6df1c75b4a840c876e0917f8198373476cef0d4b7d29`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:34fb113f6a06ec0fcbe160ede6077e08f7693735c946d1a0254368c79c94bd63`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:85c93bc7f68ae39b5bae7150435500555cd25db22bb3e4c327ef6a0c5f381644`. Independent certificate: `sha256:fa780de77bbc520df4440a87525dad6b4dee4903aac27d5548bc8087781609eb`. Engine external-validation hash: `sha256:b61301442986bf32d32ff580c85e81da5d8fbfa52bfab2397cab5c70b2e773a7`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- gauge-equivalence-1: predicted observable-equivalence-with-held-gauge-redundancy; observed observable-equivalence-with-held-gauge-redundancy; exact match True
- gauge-equivalence-2: predicted observable-equivalence-with-held-gauge-redundancy; observed observable-equivalence-with-held-gauge-redundancy; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:ee9425b090876521cf61a35a68cf4cf8f8c82707c88083cbd5f841525eb42c27`. Isolation certificate: `sha256:cb240904ec157cde290f7f8bcf9ce498ef871efc85231c8487631844e824097a`. Custody certificate: `sha256:7f7d9343777dcde781f49a2337a3624f1929d1d33379f07498d1c8305434115b`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- gauge-equivalence-1 from PDG-2025-SUMMARY-TABLES
- gauge-equivalence-2 from PDG-2025-SUMMARY-TABLES

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:6ec0cd95c2db07b0f5aed6500a49bbaf23e243af48fa165a86885b944c29678f`; engine receipt `sha256:7c9c2d1366bf84479e96fb53603fa5a94312f5f7606d35b495f6ec03320b5d70` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001-7c9c2d1366bf8447.json`; empirical-validation hash `sha256:8a968022783129910d6e052c8a995e855f03394964a31d93f7fbd92ae1979651`; measurement receipt `sha256:ee9425b090876521cf61a35a68cf4cf8f8c82707c88083cbd5f841525eb42c27`.

37. Local propagation and causal accessibility

Claim identity: `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A field response can change only through a generated adjacent transition path from its source; responses outside the completed propagation trace are inaccessible at that event.

> *Physical locality is source-bound propagation through generated adjacent support; causal accessibility is exactly the completed reachable path set.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__adjacent-source-bound-propagation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`adjacent-source-bound-propagation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:dfa113a0f4fa67a4c725e960962eadb784caa5080a85eafe8123e8becd658a6a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f20e2e9440d80fd6bcae3687d0e494568f05d309366eb8696458443b6d246d9a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:20a8fa5d1d76dfc37bb8cde320c1a4c7d0bfbaed15178472d88f54a43d619f7e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:faca424ddf42910e2626c8ce34cce75ccb38f8bc320ba7575f90e44bb9fec166`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e78bbac06cdc159ee09ba8105fb36f3a4d70f60be94c674b07fc110c88834373`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e431b510e133b534898a11b27facedb9b7a4bd7f31a177c767a41caa1ad9d93c`. Independent certificate: `sha256:7dc28b76c3b7fcb33d0f298b40abaa561d8afc3379211f37d9d45b7aad4ded11`. Engine external-validation hash: `sha256:9bdc983558b526ca79d5ad83c58dab49c7f6d7ed2ab722b68ae214342a389206`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- locality-causality-1: predicted local-causal-propagation-support; observed local-causal-propagation-support; exact match True
- locality-causality-2: predicted local-causal-propagation-support; observed local-causal-propagation-support; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:d7471789b05174ea0fb30ba3ab266534b351b2a406ab1a69b97a4cdc77d0adb6`. Isolation certificate: `sha256:3b05dd2f162d9bad7bce9f9842b6d098c80279fabc0309e22eb73103638ce2dd`. Custody certificate: `sha256:6ca63d0bc4075baa504a47497241814a4ef2f89befb82405781a1802540908f1`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- locality-causality-1 from GWOSC-V2-CATALOGS-2026-07-23

- locality-causality-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:377aad9ec09b8b2fc819f404397cc879a95cb81abb380af3a4a88970a6c3c2c2`; engine receipt `sha256:53eddf9f05386d8a26a3de9279a68d1a36451f8dfb290681a1121bfbbd70a6ac` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001-53eddf9f05386d8a.json`; empirical-validation hash `sha256:4c0da852185c5b19f157ed8b3f7f3a2c2a7abb6a74efba679b195834fab5d18d`; measurement receipt `sha256:d7471789b05174ea0fb30ba3ab266534b351b2a406ab1a69b97a4cdc77d0adb6`.

38. Radiative transport and source detachment

Claim identity: `SFT-PHYS-FIELD-RADIATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Radiation is forced when a changing field carrier closes into a self-propagating joint electric-magnetic recurrence that transports energy after local source contact ends.

> *Radiation is detached self-propagating field recurrence carrying source-bound energy, momentum, frequency and polarization records.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__self-propagating-field-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`self-propagating-field-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1de63aec7335d5500c16812437defd664f48d36f81dad6c6f32fc2ab8980adef`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1f6b8207b6701110505f7659295bfe8a2ec2e235b8c808e563443cc5247128cd`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0386c56bd1470024b84aebef849f3212bc179c348936e9d956f0be64faf63d43`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c7bc8f0e89263a161dfc2cd37fadf82e444cbf923916f175b457a998652ed914`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:37e033b8d370e9a37b1fc41f42c70c50360fa36303ae51c724540ffb76e430d0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:653788cbfd634f0f02f407fe78b430ef7621899f2566f0f9f331d61de391f899`. Independent certificate: `sha256:37813a5b21b4330e7fbd979b8c0b4362c6bae623debd74a927f726652b8d0649`. Engine external-validation hash: `sha256:e4a17f72fdaf9e94fd7f9ead60850d3d06914a3ed9c8a0237de30c00c8cc2705`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- radiation-1: predicted self-propagating-radiative-field; observed self-propagating-radiative-field; exact match True
- radiation-2: predicted self-propagating-radiative-field; observed self-propagating-radiative-field; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:f67a662a2e3f152a9d529c835ad00b1044fd652fcac0664c17aae5a7314e9866`. Isolation certificate: `sha256:b69678067f8b258f8647fe43418f389c40d3a1b601f7f3f0d733bdc072026011`. Custody certificate: `sha256:023f9d4a0ff2081af2fc363af4ceca078b7576fe52ee0ce526dc0212751287b9`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- radiation-1 from NIST-ASD-5.12
- radiation-2 from NIST-ASD-5.12

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:ab2660f76840eb85ad5503c84c0e2ed5c340caaac7da09d66532d4236964f5c3`; engine receipt `sha256:0d61c283fb2b593db4705f03525522b29ed15179ed4b36519d281ca9b34e2fcd` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-RADIATION-001-0d61c283fb2b593d.json`; empirical-validation hash `sha256:87d41f9553bcb7449b6c0d58978b6d3c0677da708d44f703e56c56f337813742`; measurement receipt `sha256:f67a662a2e3f152a9d529c835ad00b1044fd652fcac0664c17aae5a7314e9866`.

39. Empirical closure of physical interaction classes

Claim identity: `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Physical interaction classes are distinguished only by non-equivalent conserved source labels, propagation carriers and response traces; classes with identical complete traces merge.

> *Interaction classes are the canonical equivalence classes of conserved source labels, propagation carriers and observable response traces at the registered PDG boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__source-carrier-response-equivalence-classes__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`source-carrier-response-equivalence-classes`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:cea7252af8ef32564d8f0a514734455cdae5a4e6d929166fd65add448517c8a5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e8989ed9853bb54628f49d4906cddab80fbfe215b97e578c08fd1704fb36aebc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:144c6ff57254d0b52293019574441dcaa66442fc0b6ad1bf6777c8762413ee23`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:78bcc6733218517beeaf151d8b9344b635050b59c70ec5e6198e1720225ce333`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1c60ec0ba523f3de5dc16d6243568dc364cdea5198008b018e0b968fc396a39a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ba2bc950bb6b33ec3c1c0977df1ade1852e9ddab8565b8bc35a812d32017dddc`. Independent certificate: `sha256:4f5d728bf1d7d1d3fb6652019cf0d89a5341c7670ad1b6089960d65e1ada77be`. Engine external-validation hash: `sha256:dc9c45a31eb720f3d0a10515cbfa0749d937fdc4410d223af2f62ca3b63b583d`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- interaction-classes-1: predicted four-observed-interaction-trace-classes; observed four-observed-interaction-trace-classes; exact match True
- interaction-classes-2: predicted four-observed-interaction-trace-classes; observed four-observed-interaction-trace-classes; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:dd3a368b3edf145c3b387532f7c14b2d2607dfd08a62ca953b3c4c338c13e624`. Isolation certificate: `sha256:75bc8bc0f83cc04e94ffb22c49e21a3f1e9cb68432fb86fd38e73875926fdae0`. Custody certificate: `sha256:c9e3741bcb4889ce945c4524238bbad97b0253a63d1975e91743f2071523a8f6`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- interaction-classes-1 from PDG-2025-SUMMARY-TABLES
- interaction-classes-2 from PDG-2025-SUMMARY-TABLES

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:01c6393d317faab6655b107e5dc2bae30615a8d7c2c0a5046b257e1b14cf5a7d`; engine receipt `sha256:ac4ce2d6711e53b55ee78939d1ab2717b841dc92a84a5e0c2af31db12218d322` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-INTERACTION-CLASSES-001-ac4ce2d6711e53b5.json`; empirical-validation hash `sha256:8800c5a991882decab96e3717aa028b3a2a12c13bf52b9c3bc88882655e77f1a`; measurement receipt `sha256:dd3a368b3edf145c3b387532f7c14b2d2607dfd08a62ca953b3c4c338c13e624`.

40. Source continuity and field conservation

Claim identity: `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every closed spacetime support region, a conserved source label can leave only through a named boundary transfer; the interior change and boundary flow are exact paired records.

> *Source continuity is exact pairing of interior carrier change with oriented boundary flow over complete generated support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__interior-change-boundary-flow-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`interior-change-boundary-flow-pairing`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4bf4acd80b7c78f40c6da6dd796c6ad616bd5587156d37a22c9ef6453015c454`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6879367734b5c7f2286583b5c8739c36b436ccd219dd57c532e16c9774c34680`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e8f40f2e01ff4eb0c12eb93c01a8242f37d8c629af632fb58e5338bf93c54a9b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:01913905e0c22c6005ce422f4d205530bf82a361f6f38e9892691c85f9b9ae2a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0bd5fd4d412afec3911663c85265a74a95472f66fe019bd969830de967cecb13`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8dc81feb7603dc204bf2e45dbf55e742474325f372bf5cfb173a989d8023133f`. Independent certificate: `sha256:ba915b374fda4dfcddf48182128a3f396ec2f0009d99257fc36f6255c7d37d49`. Engine external-validation hash: `sha256:191b254d9733f0440bf8be1f35fbe5010107246c5155300490675d3e34f834e6`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- conserved-source-1: predicted source-continuity-boundary-flow; observed source-continuity-boundary-flow; exact match True
- conserved-source-2: predicted source-continuity-boundary-flow; observed source-continuity-boundary-flow; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:07029ac4d3d3fd288e1146ece1493bbabcf5bb46dd0b829d26dfd6a208db6e75`. Isolation certificate: `sha256:1ea5949027540ca2c7b70961e88852bbc50e58e20aa15c8c45e79abc24882caa`. Custody certificate: `sha256:9a1dc3a31fa8085db6a80b142d713494366aaf55391203747b7dc22832501c36`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- conserved-source-1 from PDG-2025-SUMMARY-TABLES

- conserved-source-2 from PDG-2025-SUMMARY-TABLES

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:e9267d967e4bbffebe4ad0239802582a994936b5e949d563c55533c0839d54ac`; engine receipt `sha256:48f15b7c73add25dba9c828842d81560ccc94d5733518a79f4d96de248334f53` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-CONSERVED-SOURCE-001-48f15b7c73add25d.json`; empirical-validation hash `sha256:183057ccc690f97a75b6782893c5236ebd3c122936080aac8c7eff8d4f9120d3`; measurement receipt `sha256:07029ac4d3d3fd288e1146ece1493bbabcf5bb46dd0b829d26dfd6a208db6e75`.

41. Reciprocal interaction and transferred distinction

Claim identity: `SFT-PHYS-FIELD-ACTION-REACTION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A closed two-body interaction trace forces each transferred momentum carrier to be held once at the source loss and once at the recipient gain with opposite held orientations.

> *Action and reaction are the two held orientations of one exact transferred carrier on a closed interaction trace.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reciprocal-transfer-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`reciprocal-transfer-pairing`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:7351627b625df2d9146aba382796b02791d2d6fd2546a6a81d0a23e76c6bba66`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b874366c3800f4054158f9713a7cde61f1f13dc9a0c6706bf1e49f109b3538ec`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d0ef8c30d8dda7e6d8f0f9488112990148001c81a80e05bda93387c9ab8cf98a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ed8d40efc11b42ee302db2490e861dfb97b376eb07abf21d491efef140de596b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6024740d1e081816ec3b01895ca3341fda0d213857a57c7ce3983ef90c375311`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:297dbef62cbb8d0564d3ab582ef9d773fb2060dc95e4b1253ab052a336e62fbd`. Independent certificate: `sha256:40404e78785faacae2f881f4a7d8abe1930ee3a3c3219182d8fa5f8ecd683d63`. Engine external-validation hash: `sha256:d5ad4d4b37c295529ccc834f33d1ffeef1ce619e7581ba3d853dd59a925c8636`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- action-reaction-1: predicted reciprocal-closed-transfer; observed reciprocal-closed-transfer; exact match True
- action-reaction-2: predicted reciprocal-closed-transfer; observed reciprocal-closed-transfer; exact match True

- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:dfd339e24fdf144d8825e7ba69c9990478519cc4d372a6c0b2fcaea5f4f99b73`. Isolation certificate: `sha256:83b22c70622bef70e54c78af39cfeeff20133a559cbb2ab94244c949f502c4f0`. Custody certificate: `sha256:516bfab4f70a7d4cfcac7d25a4aff2c346c61a5fe68856ec2fa484bda82d84c0`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- action-reaction-1 from NIST-CODATA-2022-ALL-CONSTANTS
- action-reaction-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Fields are source-bound response maps over generated support. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:6fd0b816fe52b6f475f046aceb099bea7065e9fcbd8f24befa3eb181c5ac35dd`; engine receipt `sha256:78ed10fe165871136905b7cde51a30df16627b13339643b9b340f69c368fb343` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-ACTION-REACTION-001-78ed10fe16587113.json`; empirical-validation hash `sha256:be0b4ef2b80815d848e946c18f5c5fe98d11974a1296949fff5f83e5b7dcc7f4`; measurement receipt `sha256:dfd339e24fdf144d8825e7ba69c9990478519cc4d372a6c0b2fcaea5f4f99b73`.

12. Waves

Waves are finite recurrence and propagation structures. Superposition, interference, diffraction, polarization, dispersion and resonance arise from complete path support and held recurrence labels rather than continuum amplitudes.

42. Period, recurrence and frequency

Claim identity: `SFT-PHYS-WAVE-PERIOD-FREQUENCY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A wave period is the exact positive transition count closing one recurrence relative to a held clock recurrence; frequency is the exact count of closed recurrences per duration.

> *Period and frequency are reciprocal exact positive recurrence ratios with complete phase and reference traces.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-recurrence-count-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`closed-recurrence-count-ratio`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:601b00b48bd3b97f643a7590795cbcf3f2f51ff535526a7fa531c4651f27f6a4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2484c7ec3efe10bd74c654b8c4bbafab282445c5022e2ec5122bcc189c67beda`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9e9a291e31cd136797288998c7d57a7734b99f165857c38894b07da5b609103c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6c93a003263282930c974a19e7419ebc1d785be762ba17f2067b9daa2599c8a6`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d28ae4833fb702e2b45d4b5815b2f139b60e7e315c6ef5dda145286584fa3b9e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:818304b482a4fcf476f693504599edfc466792c241c25186ce113f92a0d6947a`. Independent certificate: `sha256:c281bae88a0df7680ab1638b39dcc30c3badd302974d8dcb05edc9a995ff95f9`. Engine external-validation hash: `sha256:5d4ed5f7998755133cd0b29ef3ad75b9ee0fe331b80168bdb8a8e0d67c243017`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- period-frequency-1: predicted periodic-recurrence-frequency; observed periodic-recurrence-frequency; exact match True
- period-frequency-2: predicted periodic-recurrence-frequency; observed periodic-recurrence-frequency; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:00cf3fbd760d4912e92c38bce1e58c22b3b5abf1a644966e44f430085a9a0cba`. Isolation certificate: `sha256:81b4bfa06cf26fb3e492949bd7caba3d3a7ea2951358c131409849d2b92333ee`. Custody certificate: `sha256:248bfad8bcbd6458b8a6e19a43909aabdba5795241d266edf87aaf25c63f4919`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- period-frequency-1 from NIST-ASD-5.12
- period-frequency-2 from NIST-ASD-5.12

Meaning. Waves are finite recurrence and propagation structures. Superposition, interference, diffraction, polarization, dispersion and resonance arise from complete path support and held recurrence labels rather than continuum amplitudes. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:8c089d92e70f95b50a09fe3ccc510250303d72ce0d7297d31030ac46a0733f87`; engine receipt `sha256:55aecf75af90158e67ccf21446f06dc6d604ca0420482afe08637e465da99f93` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-PERIOD-FREQUENCY-001-55aecf75af90158e.json`;

empirical-validation hash `sha256:9c2404e18eb33aa4056099c5c3dbb9fbc21c0b8fcddbd59cdf2cd4dfc4df951b`; measurement receipt `sha256:00cf3fbd760d4912e92c38bce1e58c22b3b5abf1a644966e44f430085a9a0cba`.

43. Generated wave propagation

Claim identity: `SFT-PHYS-WAVE-PROPAGATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Wave propagation is forced as repeated transfer of one recurrence form through adjacent generated support while retaining source frequency, phase relation and carrier identity.

> *A wave propagates by source-bound adjacent transport of a recurrent state relation across generated support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-PERIOD-FREQUENCY-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__adjacent-recurrence-transport__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`adjacent-recurrence-transport`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d6ac3a6995197c0391122fe470977d800b160fc7a1e54abffe23648d1a188d9f`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3af7c76f5029ff2a83b9b23a8d8d1288514912fce6f4735d002320d632f77091`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:026a75a6671c260841c751c47ee49f7c0ffef947edab1117e95a6f68c78cb99d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:51c5ebc2f063f0349e8df80b91de3ba1e3f140bb831bf587b2831a6cb8a74558`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1f14778ed9b928d9abc8031d0a05a2d11cb2923e5a2586118d92c0a8d8ad4be6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0b3cc4a2294b47b1b090f55adc9b3fcc5ff6d6a46be5a52f0559462b58bfb242`. Independent certificate: `sha256:e9589e5e804c9d00fa36110b5c966679b9c10b80471a1db19aa91ba08120e2d8`. Engine external-validation hash: `sha256:b302a3e3777ef9c97a1914adb076610527d3e23b43e81bbd3533e63a38051ba1`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- propagation-1: predicted adjacent-wave-propagation; observed adjacent-wave-propagation; exact match True
- propagation-2: predicted adjacent-wave-propagation; observed adjacent-wave-propagation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:ecbdf53b4616a61a417ab3a833b22defe0381ea259b331f5795cf2c77d663f00`. Isolation certificate: `sha256:66632297b37d6eb268dde1190e813653f9b1b5bd6c23b7901e4dffbedc221fc1`. Custody certificate: `sha256:01b129f8343c2016bd0a36e1f6b06d2063af396859775dca2451e6e4d3a81368`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- propagation-1 from NIST-ASD-5.12
- propagation-2 from NIST-ASD-5.12

Meaning. Waves are finite recurrence and propagation structures. Superposition, interference, diffraction, polarization, dispersion and resonance arise from complete path support and held recurrence labels rather than continuum amplitudes. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:583d105d16ec8e339b799ea1bf11abfb70f5a129ed8f38091429bda7f81f7db7`; engine receipt `sha256:50a4686ca478198084edb8fed8ddd16cf51d791da3aee98fde34e393b23d0998` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-PROPAGATION-001-50a4686ca4781980.json`; empirical-validation hash `sha256:c69dd8f7d2a4f89cc7b94d6bb085107d89e6fb483a87bbec02aaa683dea47791`; measurement receipt `sha256:ecbdf53b4616a61a417ab3a833b22defe0381ea259b331f5795cf2c77d663f00`.

44. Propagation speed, wavelength and frequency relation

Claim identity: `SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. One recurrence transported through one spatial repeat forces propagation speed to equal exact wavelength support per period, equivalently wavelength composed with frequency.

> *Wave speed is exact wavelength/period, equivalently the exact product of wavelength and frequency.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__spatial-repeat-times-recurrence-rate__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`spatial-repeat-times-recurrence-rate`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:84bc6f297507e8ed257a73faea2c6b1251e8c2770fe66ca8154e05f3981593a6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:721206e1b8d38e686b94457bbb13dba587d49e5f497e5c5c8a57a530960f79ce`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6d6384f19549de2fc0a722fabb0821203feecee12ec799e493473eeabf067477`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1a0ef5a811b9a9e91b38d86e6cf20175aa034745139bc71502307dd236ef5256`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f687c6c853d8362126831975c0bfde0602fe37fae27ca089e39f4c1e5f5da554`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:eebd9cc41ac9e81564b19dc8bac9a4fac725278a96973dd497ce9dcf779bb0cf`. Independent certificate: `sha256:185d968975a383a7b3674347375ff79eb1b104c817038c86f9c0185783888192`. Engine external-validation hash: `sha256:5ec317567459c5d9366b5dfee20a9cca42e9abeb67132e2322a520e25afdd22a`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- speed-length-frequency-1: predicted speed-wavelength-frequency-closure; observed speed-wavelength-frequency-closure; exact match True
- speed-length-frequency-2: predicted speed-wavelength-frequency-closure; observed speed-wavelength-frequency-closure; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:e7fa7a2d678af00aa3058d405464161c3862a771103dec03a4e8390ef7d8ffd3`. Isolation certificate: `sha256:dede6f8886a6d3ca7742e007a62f8d411f29b215abd7fcc10359c51c7a2939e8`. Custody certificate: `sha256:d6b78e7dfb0a31f06bd9e7f65112db6aed1bbec4834edf70fb5c255733519ffb`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- speed-length-frequency-1 from NIST-ASD-5.12
- speed-length-frequency-2 from NIST-ASD-5.12

Meaning. Waves are finite recurrence and propagation structures. Superposition, interference, diffraction, polarization, dispersion and resonance arise from complete path support and held recurrence labels rather than continuum amplitudes. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:ef50b7f656f86bbb15cd4cb0d3c8172cf7736f419eda1817621fb875f7031a87`; engine receipt `sha256:0b1798571aa501d2d2f115ac1863142881c886214b0d3cbada2cc1cc03f47fc5` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001-0b1798571aa501d2.json`; empirical-validation hash `sha256:abd7e043ca2448e874c0bb8b5cf68cb1acef82d8ec018a52d35d065705e51571`; measurement receipt `sha256:e7fa7a2d678af00aa3058d405464161c3862a771103dec03a4e8390ef7d8ffd3`.

45. Wave superposition

Claim identity: `SFT-PHYS-WAVE-SUPERPOSITION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. When distinguishable wave traces share support without interaction, the complete observed carrier is their disjoint labelled junction; no constituent trace may be erased or double counted.

> *Independent waves superpose as the complete labelled junction of their source-bound traces on shared support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__labelled-disjoint-wave-junction__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`labelled-disjoint-wave-junction`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2287a8073cd6c4f58938d70412d176ec7b4d334e98bd0682e33e49944f99e627`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f1f498bfe822763d05d981ff2cc13c82e3f3097dffef33b20b93d9d8a5822624`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:99f36ca76c44e49a26f7202b22981998295b57f995d844b37caebafd3224db03`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6593ae7f0e26039c15d45cbfb55fd71c1b7c22fde364c32e43e5c2725db35932`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f0f1d5f47f0661d51353f05b0af7d28f6e7468e898d8da0d01f07db78038b7fb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:706b5acee19878334d33cfaa3a90accacc00bfef8af9b97731d640cff64a0257`. Independent certificate: `sha256:8759b77eeb48a7cb010732754817c7d3b36337011dfe2c6f1c13fd6e56140010`. Engine external-validation hash: `sha256:aaf37dc4120601e7703e8b255ef4e44249d75be6fce160661265a859e7296413`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- superposition-1: predicted complete-wave-superposition; observed complete-wave-superposition; exact match True
- superposition-2: predicted complete-wave-superposition; observed complete-wave-superposition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:6179415fbcec744b0d304d9e41c8a46334d46232548acf05dcd34bab9dfba677`. Isolation certificate: `sha256:400b79d945a54433caac58e8f8f37854c16180a125423c205b6b34f46ac82c73`. Custody certificate: `sha256:2fc0372700c99a06f8e135e0113cc733b63888e12f8bbecbf65da99cab4f9d71`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- superposition-1 from NIST-ASD-5.12
- superposition-2 from NIST-ASD-5.12

Meaning. Waves are finite recurrence and propagation structures. Superposition, interference, diffraction, polarization, dispersion and resonance arise from complete path support and held recurrence labels rather than continuum amplitudes. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:1e084c75cd32f819fb2724505d0d805269b650e12b6cd78784cb2d5aff46c356`; engine receipt `sha256:f8421b45a2cf9ca516fca24b3a33fc71e3f9b11e3435c2eaf6a6ccfe471f45e3` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-SUPERPOSITION-001-f8421b45a2cf9ca5.json`; empirical-validation hash `sha256:c8364b6cc5f873f40e52a5c5daf2f57d6deaba3727a12ed8ba4256bda4970cf7`; measurement receipt `sha256:6179415fbcec744b0d304d9e41c8a46334d46232548acf05dcd34bab9dfba677`.

46. Wave interference and predecessor merging

Claim identity: `SFT-PHYS-WAVE-INTERFERENCE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Interference is forced where superposed phase-labelled predecessors map to the same observation cell: matching labels retain support and opposed labels close distinctions while the predecessor record remains held.

> *Interference is exact phase-labelled predecessor merging into common observation support with all source traces retained.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-SUPERPOSITION-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__phase-labelled-predecessor-merging__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`phase-labelled-predecessor-merging`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:46db1b341d3a10315674f6a5e61afa5514f4424ce741e02d9ef56b726d999c95`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:61f255d3b9a55856cf0f48b1e67feba27a0381851562b7f11689609142ee024c`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:fd5497bd6c92c00934809764fb5fb06e0b0b89053f4b9178917c650676f37a56`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4a0f57ead6c024248f0960760259083e4d58d81e304758eae4dcfa2963683216`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6488f7f50675f6119b15bfe58e27a1093eebd210f4168384e7fbb19376b1ca40`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d60ddac9afa052b1c537ac69eff482d0be3eba6c7802fac752852e5c6d9a9ace`. Independent certificate: `sha256:0ccaf4891c616c47996b7881a390fa5561f34a1d1ba36511022dc21a8e4df570`. Engine external-validation hash: `sha256:361bcefd540ebc670a80b6c89e4728a464d21be6916f12ca19c47726b1160cb1`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- interference-1: predicted phase-interference-merging; observed phase-interference-merging; exact match True
- interference-2: predicted phase-interference-merging; observed phase-interference-merging; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:27ee402e11aa853c8e3f3907f43db528e593512879dac3c0bfb19d0e00baa1e0`. Isolation certificate: `sha256:78c598145c936624eb2bc701fc6a6f5838e16432aaa1faacee1a75c806745857`. Custody certificate: `sha256:ad7a666c859e2f17ccf25cf643e533cf1138ea226ac5cdad93472dd0e51df6eb`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- interference-1 from NIST-ASD-5.12
- interference-2 from NIST-ASD-5.12

Meaning. Waves are finite recurrence and propagation structures. Superposition, interference, diffraction, polarization, dispersion and resonance arise from complete path support and held recurrence labels rather than continuum amplitudes. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:be7b9c926641bd8cabda9e0e20899b92c7e630147cf731a62da44fede8cd2f66`; engine receipt `sha256:fecacb47e9a61a33c7df01e0cd569f3306ca1d298ed47232817357fbf247afee` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-INTERFERENCE-001-fecacb47e9a61a33.json`; empirical-validation hash `sha256:c77d3e82f8dfd7a43ac79cc5a07ff9d95e0e396dfa0230cc753b4866f51fdb7c`; measurement receipt `sha256:27ee402e11aa853c8e3f3907f43db528e593512879dac3c0bfb19d0e00baa1e0`.

47. Diffraction at constrained support

Claim identity: `SFT-PHYS-WAVE-DIFFRACTION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Constraining a wavefront removes propagation paths; complete successor generation from the retained aperture paths forces redistribution across every reachable output cell.

> *Diffraction is complete reachable-support regeneration from a constrained set of incident propagation paths.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__aperture-constrained-path-regeneration__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`aperture-constrained-path-regeneration`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.

Axis	Forced form	Eliminated form	Elimination reason
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b00f0250f1d752d0d3213d9197a24d07cc28a664ef07afd90f726a7f9c7f14e6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bba07f5cb14a9f143451e64b7e4ed0521d51a4361bea4867c0cff3c17cc88532`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:004b18f3ae136055c5cc72e2ff8713666ae74bed318be4a7b3e774c9636363ed`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a8c972847fbb73f4a71f95a8b5345ea9606159ce6f9d80b66103bce3da98142b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:12ddb094a78e44398cc42ca75027c12655d74c384edec117c098e8dae4a59684`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8221e65a8a3ec1dc7622846aec9d52a0a6e961eb4d198c14ba09de5e9faf1b69`. Independent certificate: `sha256:6f251dc4cda533fb92ec13b262a73815ea0591429ec15e92f200aed0ad8dc027`. Engine external-validation hash: `sha256:5cfbc19af638d2549d0cc0dff5b3e66c0f487669c5223dee2441ee098c2e9250`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- diffraction-1: predicted constrained-support-diffraction; observed constrained-support-diffraction; exact match True
- diffraction-2: predicted constrained-support-diffraction; observed constrained-support-diffraction; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:fddfbfb736f2506f418f6988471945f19870456bd066697ca2e5154e2ed5e891`. Isolation certificate: `sha256:f6e0757b310fca56631225d0a65abd4a8ec3c21f7b34bab9117caf846b14a7fe`. Custody certificate: `sha256:30b61e49ddae2bc061146a367ebd3ab6c70c79eb4681ed642a42d5cba370d53e`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- diffraction-1 from NIST-ASD-5.12
- diffraction-2 from NIST-ASD-5.12

Meaning. Waves are finite recurrence and propagation structures. Superposition, interference, diffraction, polarization, dispersion and resonance arise from complete path support and held recurrence labels rather than continuum amplitudes. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:7161343d80b1f1dfcca31709a1932fc0c38ac92cc61923966f7f6819137a1c93`; engine receipt `sha256:4c9922f000b07cfacacfbae4f820e76af89cc3e83f0e211e5396c5625b61902a` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-DIFFRACTION-001-4c9922f000b07cfa.json`; empirical-validation hash `sha256:2e8d10315460c7a7da499c47da3fd3ece4ecf21dd3f19edb6381bd040fcd3b84`; measurement receipt `sha256:fddfbfb736f2506f418f6988471945f19870456bd066697ca2e5154e2ed5e891`.

48. Polarization as held transverse orientation

Claim identity: `SFT-PHYS-WAVE-POLARIZATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Polarization is forced as the held cyclic orientation class of a transverse field recurrence relative to its propagation path.

> *Polarization is the held orientation of transverse recurrence support relative to wave propagation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-DIFFRACTION-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__held-transverse-recurrence-orientation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	held-transverse-recurrence-orientation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a4ad10b48dcea194c75b167d7f7389598c0436845ccb5a7df89a465cbb1eef04`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:41cdd4f74dea4fb6becc4f167cc10573d48f3d6f9d5ade29432ef3ea78cb1c93`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ce6994b049e729471b20447aef4c60376ee5b8704c069dacdb456903a20c7e34`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1f733f085dd0a916578c1b1caae7c363800f0c258ec6aea122c73732dc60d512`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a9e2a076ae00e455be645d7c404747acee5880f3f5bb7674a828ebb80e500453`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:bde18730b7e73c8a93fa972423db6a074505a121a3c8b0a929f8a1d343650232`. Independent certificate: `sha256:81f3f282e0c3917b2bfbda79bd3bb7ce8644efc67b6c8a920e624b2994b4f702`. Engine external-validation hash: `sha256:5027cfbc5f3d5eef797069fca17a1d5d392c63fd53ea967f47872e6ec56682ae`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- polarization-1: predicted transverse-held-polarization; observed transverse-held-polarization; exact match True
- polarization-2: predicted transverse-held-polarization; observed transverse-held-polarization; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:a43d059197b55a8f4b0ec6822538a49fa62203e937ac3c1083dd3cb87af201e9`. Isolation certificate: `sha256:c14780d43a13378b84cf7a70620b262e5890233beefd1c8af26f75452f2b0e35`. Custody certificate: `sha256:390344ae0725188e5f16dba6d73bcd2470369002f84b4da729f66bbd9900f9a8`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- polarization-1 from NIST-ASD-5.12
- polarization-2 from NIST-ASD-5.12

Meaning. Waves are finite recurrence and propagation structures. Superposition, interference, diffraction, polarization, dispersion and resonance arise from complete path support and held recurrence labels rather than continuum amplitudes. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:ff32660b2dd6d39c4658c6320647c05df6c2f984de9ebf2883d67a9628371925`; engine receipt `sha256:ed7c412e559c3488049747a95ba94327326d631f7896cdd18ddc33db557ec81e` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-POLARIZATION-001-ed7c412e559c3488.json`; empirical-validation hash `sha256:9191952e3689a1574ad3c1119eabba9143b8df9766c151c1fcfd33e6b816c324`; measurement receipt `sha256:a43d059197b55a8f4b0ec6822538a49fa62203e937ac3c1083dd3cb87af201e9`.

49. Dispersion and support-dependent propagation

Claim identity: `SFT-PHYS-WAVE-DISPERSION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. When support transition rules distinguish recurrence labels, each frequency class is forced to carry its own exact propagation ratio; their phase relations separate with distance.

> *Dispersion is label-dependent exact propagation induced by the registered support transition structure.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`

- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-POLARIZATION-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__frequency-labelled-support-response__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`frequency-labelled-support-response`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:c817a32960319800251cba6ae5313a77bb8e7067abd9c5a1992a92ddb2797006`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2eba8297442d282c1c34126487ae4d5c2ac8f27b318b97212e55bf0fc9e203cf`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:50935063db3acaf93337341250dd63aef37f4dfb434e90242c843de5ca391af0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:cda9f78a03b2c8254ced238a1568c73e687eb09e119d3ae133d8849d704054cc`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c3a78ce30373f0e5c85cc27b15c81a1804d07b4db02289d2de65f52acf0c2e46`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c2ce37fc1e3f802f9683fc17d1faf3dbcbcbe30b6db49cd878ca1ccd5a5d9186`. Independent certificate: `sha256:8ad4fb98343d0ebfa422aefd8e8c1ca0b4bb0b4c729e98daf0abdd1d3f4e9fd3`. Engine external-validation hash: `sha256:8aa4e151bef9045e9399b16114078d10e1e72479806e4fc3ed9277b09abbdcef`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- dispersion-1: predicted frequency-dependent-dispersion; observed frequency-dependent-dispersion; exact match True
- dispersion-2: predicted frequency-dependent-dispersion; observed frequency-dependent-dispersion; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:faf14b33a5493c9d3f99a9920b8f33f44d5ba2b2ddb1a04bdf29c3d0f8f7420f`. Isolation certificate: `sha256:62d5c149716fb3b2bcea271c557a24eca7414d34d1f1a1f24b85870a70c5a0ff`. Custody certificate: `sha256:5fd09bde126cebce88449193bb4ab517fffb880d6cab844e1e67f335dcde5093`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- dispersion-1 from NIST-ASD-5.12
- dispersion-2 from NIST-ASD-5.12

Meaning. Waves are finite recurrence and propagation structures. Superposition, interference, diffraction, polarization, dispersion and resonance arise from complete path support and held recurrence labels rather than continuum amplitudes. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:99dfb753c9092b1a1d9e26c0d60986514bd3b990a3c69ce166e2a40c5802da83`; engine receipt `sha256:72749162a46a877c7d54d3cde6f09848b2d383360ed687b201e6c94475a29fe8` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-DISPERSION-001-72749162a46a877c.json`; empirical-validation hash `sha256:31b6281252888e47b64ef536c44289125704bed608fdc95a2f077fcbeeb7841f`; measurement receipt `sha256:faf14b33a5493c9d3f99a9920b8f33f44d5ba2b2ddb1a04bdf29c3d0f8f7420f`.

50. Resonance and recurrence matching

Claim identity: `SFT-PHYS-WAVE-RESONANCE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A driven recurrence accumulates coherently only when driver and supported recurrence close on a common exact refinement; unmatched phases merge without persistent accumulation.

> *Resonance is exact common-refinement closure between driving and supported recurrences, producing repeated phase-aligned transfer.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-DISPERSION-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__common-refinement-recurrence-matching__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`common-refinement-recurrence-matching`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:bdf5ac36ea7790d3b829fef41996ebe76cc9dfbd4530456db3e643fbf741cb56`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated

by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d2df52b8708ce52d02c5f9c8950d5ea9b5b2f9b272fa5f7be3f6a9c9200a7c4e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:fc794b574799a1058ffb08f667dc5a42ee0523191d34ae3fcec6bbb143f4e083`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:cc80fe73f2b44d6c50179e804afae5bd6155a2698bee3689f5982707b729a1ec`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4f8e5411052b53acc7dc47aa78d57b2aa010994dd757abfc6b3ddc9e0bc6440a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9c5906e5e5607427d0a3ea0a15716c7686f7a6b02ae1e2baa369e857a81c88e8`. Independent certificate: `sha256:e89d71e94981cf1e8eadbbae1d46929ddc31bb7a5ef70cbfa926c3f173ed58eb`. Engine external-validation hash: `sha256:8ccefae22d2afded9378f274eceb0876bdeca43fa4ce14b2a9b0830bc57e08ad`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- resonance-1: predicted recurrence-matched-resonance; observed recurrence-matched-resonance; exact match True
- resonance-2: predicted recurrence-matched-resonance; observed recurrence-matched-resonance; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:5587beb44326be1e025b8a6a32d5769ea98ff4821fae2f0a40cf5de33a9ad33a`. Isolation certificate: `sha256:481d971c9b9bca81cb43c15ef93383756349195ff48f36be010284f866e43a47`. Custody certificate: `sha256:a46b5db8532e537f99537ae6aa82dd634c492b4b76260f535eb232b0fb440bf6`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- resonance-1 from NIST-ASD-5.12
- resonance-2 from NIST-ASD-5.12

Meaning. Waves are finite recurrence and propagation structures. Superposition, interference, diffraction, polarization, dispersion and resonance arise from complete path support and held recurrence labels rather than continuum amplitudes. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:4530e5320e83f41278376ce35c1394e1134d846d07515c70aa7c78fbf5b24e35`; engine receipt `sha256:cac57d85a74fcf5b8f323aa8ea98203d242e8e40084465de77b89d33b6db55e3` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-RESONANCE-001-cac57d85a74fcf5b.json`; empirical-validation hash `sha256:76602285c6259cca45dc32916d234702b3b607e013728be5e2e94f616a0891ee`; measurement receipt `sha256:5587beb44326be1e025b8a6a32d5769ea98ff4821fae2f0a40cf5de33a9ad33a`.

51. Wave energy and momentum transport

Claim identity: `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A propagating recurrence that performs work at a remote response cell necessarily transports the corresponding energy carrier and held propagation momentum along its complete causal path.

> *Wave propagation transports exact energy and oriented momentum carriers, paired at source, path and response.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-RESONANCE-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__causal-wave-transfer-carrier__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`causal-wave-transfer-carrier`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a2abc6592f4d4db9f1221685ee18b30ec9cd7db7ff109a260d50cb87ea83c9d5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:703a05f7937d3054ac10d8e2b66119064b82bf4a2891940671546eb6949d7fb4`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6f905acb05ac57cb4fd664d461b08ac448933f7adf404e432f6eaa911be54d65`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ba67bebcc0dfce85a8cc736d4cfd96f48b98a5617d1fe807d34091e25168e7cd`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:16137d5af11336ef1b1fdacf28e27fe6f63731cb2900c72fbe062bc6d96cea17`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:de27387e232a634fa0a663a09bcaf09a64290408013be504fd5727c5e36537d2`. Independent certificate: `sha256:e852817a61bb2659ee2d9b24e0d60b77e91ce8f0b8470271e302ab187c1d448e`. Engine external-validation hash: `sha256:f723a1659fe39ba6ec5f0e2479fd37e1dec3fd195820725fba702a5c04cceafe`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- energy-momentum-1: predicted wave-energy-momentum-transport; observed wave-energy-momentum-transport; exact match True
- energy-momentum-2: predicted wave-energy-momentum-transport; observed wave-energy-momentum-transport; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:eb45a52316baae6215822fdfb276a428e8883cad50b45dcab87087a8fb9fc7af`.
Isolation certificate: `sha256:0359905dc37a5254d9ee57565fd80b3e660d5a76079decaae9ac6ff8b3f7da41`.
Custody certificate: `sha256:72fee7f977609282e4a6f3102db753b3f19ef7de65429a8a70eebe15ea308622`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- energy-momentum-1 from NIST-ASD-5.12
- energy-momentum-2 from NIST-ASD-5.12

Meaning. Waves are finite recurrence and propagation structures. Superposition, interference, diffraction, polarization, dispersion and resonance arise from complete path support and held recurrence labels rather than continuum amplitudes. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:f58817363c14545510ba7152e7f925e69303e204f13e93e236d43a392980330a`; engine receipt `sha256:1a25f4c7e2b95a50fd6b37e122372952e6563575c716a6cda083ce1d0755e6de` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-ENERGY-MOMENTUM-001-1a25f4c7e2b95a50.json`; empirical-validation hash `sha256:337ba7284de709779a228b8d01faeba71e03195569fadbabfe66d18071ba31ab`; measurement receipt `sha256:eb45a52316baae6215822fdfb276a428e8883cad50b45dcab87087a8fb9fc7af`.

13. Thermodynamics

Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced.

52. Microstate-to-macrostate observation

Claim identity: `SFT-PHYS-THERMO-MICRO-MACRO-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A thermodynamic macrostate is forced as one exact observation class of a completely generated finite microstate support; its unresolved multiplicity is retained as the class fibre.

> *A macrostate is an observation fibre over complete finite microstate support, with exact multiplicity and retained provenance.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-microstate-observation-partition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`finite-microstate-observation-partition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a0bf480b1fb3ed3410aa2c6f693be952019578ecb0c8d769576671e830a5aa74`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c65afd05be1ff96daaf309433e10c3c85d371e4ea512f710f549859f68a4e929`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:214504c9124b489399cc4a308931c0f822a1bbd5a24e492d83ffab44306bccde`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a78d1727daba665726753bb0b20c97d621b23f07d07deabb555620c2535cd011`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:773249981ef18b98d2128f5e7ce4e000e212289760660ea99eeddab18dc06bf1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ff94c98d730d930a00e95a403b5d023e29d8cf70e5e516a7e5f2d03ebe47b6a5`. Independent certificate: `sha256:c245399cfd8062c8a754212fb79e5e6f01f976b0bc62de9ff139118d008eeb03`. Engine external-validation hash: `sha256:94302a7d7d0ef8e2194a635506e0940f8e66c3b2121d881c7b3f7e9ea00a2c98`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- micro-macro-1: predicted finite-micro-macro-observation; observed finite-micro-macro-observation; exact match True
- micro-macro-2: predicted finite-micro-macro-observation; observed finite-micro-macro-observation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:4a564a747c5f3959532436ca15634960455791ad2dffc83a600d2cbebf3a6926`. Isolation certificate: `sha256:1822e678b5f3bd0fd61c6fb633afe806abf85b8e9ae2b40b85d43b4991837226`. Custody certificate: `sha256:5910b505b5bbe5b3906cf20324252a234879ac3133fcdab7971b0081b0b795c5`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- micro-macro-1 from IAPWS-RELEASES-2026-07-23
- micro-macro-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:cafff80fffa91147655094797dc206b2b2a726fff4aafc70cd55cb66789c48bc`; engine receipt `sha256:11b63417e8a9d91dc1ceb4ecc356fe1083c36cf84cccb0522b42faae15a7e067` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-MICRO-MACRO-001-11b63417e8a9d91d.json`; empirical-validation hash `sha256:adce150365d521a8ea0d96844c6c91803c18a1e6be04b8d7781dc87522e88ce5`; measurement receipt `sha256:4a564a747c5f3959532436ca15634960455791ad2dffc83a600d2cbebf3a6926`.

53. Temperature carrier and thermal ordering

Claim identity: `SFT-PHYS-THERMO-TEMPERATURE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Temperature is forced as the exact ordering carrier comparing energy-support change with accessible-state multiplicity change relative to one held thermal reference.

> *Temperature is an exact reference-relative order of energy support per accessible-state change, not an imported continuum coordinate.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-MICRO-MACRO-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__energy-support-to-accessible-state-order__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`energy-support-to-accessible-state-order`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3740214d264d8442af6b2e386567eaa01e64fad19b15ae07c7984aac0cac2c33`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:35041c1fc11ce723d4560fe772069cea9216e1ef156e73cdbb47a18788d13330`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2956553dd7226e7cceeceec10db3a4024e4baf3ca57ee01ae13d7df370aadb3a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d51c89c2466ecb996b33557ce6855bab5fc11ff8b4192e2bdc31e2aa64e9cb78`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:862d8898fc88b7eaad0379763249ab5281a3c34b9a5884ded00a786707a0ba27`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:47726e82008630482d9781ed4e574ab970284d5e510faaa8965ef1a8d2e85d25`. Independent certificate: `sha256:ec2560a3be4715f87c1b8831a4f9103a04ee602088163e87d10b20fbb79208f2`. Engine external-validation hash: `sha256:26edbb1e3a330662b802e7bae193acf93c95500274f46ce17792801c9ca6be8f`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- temperature-1: predicted exact-thermal-order-carrier; observed exact-thermal-order-carrier; exact match True
- temperature-2: predicted exact-thermal-order-carrier; observed exact-thermal-order-carrier; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:4d1eb98a218d275eaf351ed5538c5981c465ed13a8b3305a46873ef7f85ae315`. Isolation certificate: `sha256:e7f04de4252891257ea2232f95b54ee8a44af79f7d9bca63acf13abf530e1209`. Custody certificate: `sha256:4de3c72f269ec8861d9837098926011741d57af35c2ed25db40ef5ab0fde6f87`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- temperature-1 from IAPWS-RELEASES-2026-07-23
- temperature-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:fa5a2f29b5b494f06a5addbb0c869a18bff1bf098bdaa2d286a07abe0f05ea23`; engine receipt `sha256:d68a2030d9e3e914068a84fb738d109c84f7d1b7be48c88c97d37d26646c09f8` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-TEMPERATURE-001-d68a2030d9e3e914.json`; empirical-validation hash `sha256:ab9efc043970126329e33dc9b07174c9dcdf0277d8e1a347a6ed4a8d0ccc0558`; measurement receipt `sha256:4d1eb98a218d275eaf351ed5538c5981c465ed13a8b3305a46873ef7f85ae315`.

54. Thermal equilibrium equivalence

Claim identity: `SFT-PHYS-THERMO-EQUILIBRIUM-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Two coupled supports are in thermal equilibrium exactly when reciprocal energy exchanges preserve their separate macro-observation classes and common temperature ordering.

> *Thermal equilibrium is exact equivalence under reciprocal energy exchange at a shared thermal order.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-TEMPERATURE-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reciprocal-exchange-observation-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`reciprocal-exchange-observation-equivalence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:00b0fda7c9c51d799d39cdef0c10cddf1ac3e40d4ef95460cd11551403d23ebd`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1b728c43b61ae39d3405b5c186402610c4eff158d5505ab6fa86eea0692af359`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a51ff7be669db5438ee2d5a09c824f80ad3d9c09bb648c58a7d1fcdefb60223b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:384ad212a4e04dec0e1582702961d3a4b55ddd08593e737c517d8d000edecfbc`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1f8e0ef7d25eff60e11fa3c1a140c4b7ce27cf57558949e98636551681dbdc36`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:6a18063c2eca0e9104385448dd15e7c69be3a28b1342459716761e4c4c6808ba`. Independent certificate: `sha256:a1cedfbfc5f1309184329786c7f8beb97d30fad87fd7c48f7ea1fd7c16180957`. Engine external-validation hash: `sha256:cfb733ced524b6e8b182fcd2d927d1a027f4bc49c4be83605928acf25b292d35`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- equilibrium-1: predicted thermal-equilibrium-equivalence; observed thermal-equilibrium-equivalence; exact match True

- equilibrium-2: predicted thermal-equilibrium-equivalence; observed thermal-equilibrium-equivalence; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:25e7ad1d4802cb8ec058e96d9d018212ba832434e8ac2bd424f290cc6960091e`. Isolation certificate: `sha256:ae5e7367d98474d0d83c43bfecfd9b3a673c7219bed4fd75484574c93d662e5c`. Custody certificate: `sha256:cb1088f4db234ce76c49b76058983fca44445f45939eb61876552df0de22e7c6`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- equilibrium-1 from IAPWS-RELEASES-2026-07-23
- equilibrium-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:767102333f0f294a68e74238a8acb09acffa668a81f929393389a5462cb2d6be`; engine receipt `sha256:6ce39350919988486d96178e51a0bd45264f0c5498519109bdb0e184c29d40e5` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-EQUILIBRIUM-001-6ce3935091998848.json`; empirical-validation hash `sha256:355dafe24741d90270ac9e7b7c41ab85ed0ff2a64c4da56942ea8f9ab0236f6b`; measurement receipt `sha256:25e7ad1d4802cb8ec058e96d9d018212ba832434e8ac2bd424f290cc6960091e`.

55. Heat and work as distinct energy-transfer traces

Claim identity: `SFT-PHYS-THERMO-HEAT-WORK-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Heat is energy transfer whose microscopic carrier label is closed by the receiving macro-observation; work is energy transfer whose organized source-to-response label remains observable.

> *Heat and work are the closed-label and held-label observation classes of exact energy transfer.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`

- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-label-versus-held-label-energy-transfer__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`closed-label-versus-held-label-energy-transfer`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:74033a0c324c51cda11dfdce9d2993c05c1a26b239ee6568ff0d7e209098005a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:522c9c0c66247feb3c51b85bb4689e821a7a799531ab9b68ddadb5de1e8f80d6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:501eb3874ae9c5d8b5ed8a38b0de1fada2ca63887a555bc7e0d0e7f68483133c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b43f038351da4176e1465face97b3eac4b4ed9221038d24fef61fb575fe29283`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1d8e132c7d01dd2d917db2c7bbe6b5d32e5957377f81031e2ea0818c34c348f9`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:692212d6e9d55652d02fb5791a82569e67f831fa33e471c95251c9fea6ac7afb`. Independent certificate: `sha256:d43df8e2076ebc60dc7837d017f9561a29607f73047cab35731c78d92d1ad243`. Engine external-validation hash: `sha256:133e197ab18339a13dd805c1401a6ac561901e9c7553c4c7c4df3b1e873dc542`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- heat-work-1: predicted heat-work-transfer-distinction; observed heat-work-transfer-distinction; exact match True
- heat-work-2: predicted heat-work-transfer-distinction; observed heat-work-transfer-distinction; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:52a3381716ae93ecfd8e95729dd33c5d25f7a54d5d21cf059eee11bd2a04cb25`. Isolation certificate: `sha256:ce57ac9fbbe19e765817a3b77dc06793a33bf4a2087598515d47cc9bf6867049`. Custody certificate: `sha256:9326e5f0715eefcea140b21e12fc722576532ec0c53a217088f898ec8bfa2e3d`.

Registered source descriptions:

- National Institute of Standards and Technology: https://webbook.nist.gov/chemistry/ (sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- heat-work-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- heat-work-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:433edef20fcdce50fae8e8dc2430c354fdca8dbe3c424b7ab6a24a2ac3f83550`; engine receipt `sha256:3db98f953910f921028c19a7464ad19eb2aa44b2888ec82ab1be56d23576cbef` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-HEAT-WORK-001-3db98f953910f921.json`; empirical-validation hash `sha256:7d6de73b7f85e56a2ebb17473aed9ae2420e614e45b7b9ee48f234ec27e98140`; measurement receipt `sha256:52a3381716ae93ecfd8e95729dd33c5d25f7a54d5d21cf059eee11bd2a04cb25`.

56. Closed energy accounting

Claim identity: `SFT-PHYS-THERMO-FIRST-LAW-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For a closed support, every change of retained internal energy is forced to pair exactly with boundary-crossing heat and work transfer records.

> *Internal-energy change is the exact closed composition of all heat and work transfers across the named boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-HEAT-WORK-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__internal-change-boundary-transfer-closure__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`internal-change-boundary-transfer-closure`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0b838b25b123c6367b0fca9efe12dd04beac2bd5294d1bd202f587d1264ca8df`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8d3939a5f370964ef42f4dc5c4475f8ec8afcc953289ae724ab7582161ac8dd8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7f43956afb7cc2b419b15ca6cd4271ae8279b8b1b5f55e8ae05a812653290a5c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ff237dd6c3067a027223bba28bc7fc4200700ad5c54bd10a78a92f9e28fc6a28`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:29c891b7564cbca75ba46f533f7644fc9ece22041d405b8c065259d0ac66c14d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:3f7e45454f55f575cf62b0752260ca31edee7d04bd9f7040f96a1d501cc6c0f4`. Independent certificate: `sha256:4f6cdde720d390368b38f62eaa82b47bcc556c7fa987a4fc2ede5941dff9303b`. Engine external-validation hash: `sha256:3f63f7200bab7ab8d23baed640f89c3bdd69366cb901e77416b24c8729b7d130`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- first-law-1: predicted first-law-energy-closure; observed first-law-energy-closure; exact match True
- first-law-2: predicted first-law-energy-closure; observed first-law-energy-closure; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:5b600737da737e7769e0b3e0cc4528b75a6cc29c33012af23337de9054fa656a`. Isolation certificate: `sha256:4058fbb00286c41adb2382acde521c35dd5555cdd0ea862e2f050985072eb7c4`. Custody certificate: `sha256:fa7b84961f5628d8b997a2cb3e53055a7b9b52bd080fa841a40466a6e78027de`.

Registered source descriptions:

- National Institute of Standards and Technology: https://webbook.nist.gov/chemistry/ (sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- first-law-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- first-law-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:c3e7e60ad32808dc97aaa5544458b40e50de9ca3f17275ec2f17cb2753812fd8`; engine receipt `sha256:18e493c59cbd7979db1bcc2b33a97f582329f92707c16bc6455c2ee55ef2b13a` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-FIRST-LAW-001-18e493c59cbd7979.json`; empirical-validation hash `sha256:fa52340b68dc82d2ea089378f31bc3aadad7fa494d08bb9794add9ad60c54406`; measurement receipt `sha256:5b600737da737e7769e0b3e0cc4528b75a6cc29c33012af23337de9054fa656a`.

57. Thermodynamic entropy from unresolved support

Claim identity: `SFT-PHYS-THERMO-ENTROPY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Thermodynamic entropy is forced as the exact information required to distinguish the generated microscopic predecessors merged by a macro-observation, conditioned on every retained macro-label.

> *Thermodynamic entropy is exact conditional unresolved support of a macro-observation over its finite generated microstates.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-FIRST-LAW-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__conditional-unresolved-microstate-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	conditional-unresolved-microstate-support	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d45decb31bfd567235b15d1f0ff5875757474f9ffcdd0cdb26e008daf9cd4c6b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c5269b08d364fbaa40bf12531c012b795a1463eba0af221214425a401b2c6b66`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ae449ca3e6d245ae018a6d766684ebbf4a52476c94632b8a3bcfc850e0e1bb54`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a0c251a8959265f3da49548a9f5f190dd0e697a490eb0a57bdbd6dd6be29d33f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dd028d42a95245598f80ac0fecf0cc08ac43cfe797abcf2f01a5171440f14a24`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0e840231ff85005cf206bde90bc06180ed9a1f798ea988fe4d7e2c5b15f08e98`. Independent certificate: `sha256:8ed2c912ba7e6fac3828cdbfc755c8870dbb4f24088884e4f4a4a10f6fa1df3a`. Engine external-validation hash: `sha256:7dc109d00a190a5da182ff19c3fd045a03af04958015ea8c534ec9b50ba13f3c`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- entropy-1: predicted finite-fold-thermodynamic-entropy; observed finite-fold-thermodynamic-entropy; exact match True
- entropy-2: predicted finite-fold-thermodynamic-entropy; observed finite-fold-thermodynamic-entropy; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:64b76e6801c73b3c9946a0f76f966c56733d9a2e4a2d3f3abb5ee208f98dc6f7`. Isolation certificate: `sha256:6de0571242ea40dc7676ce7b5fa60f33665d6c3e4105ec0f4c9894d8ce8bca8f`. Custody certificate: `sha256:91264e0a4fba63d40403ce5332242336b735783f3fbe774d3473a88cae9d3e1c`.

Registered source descriptions:

- National Institute of Standards and Technology: https://webbook.nist.gov/chemistry/ (sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- entropy-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- entropy-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:c746cceb4a0536fab9745ff843ca2d6ee21eda56ae9707e3eed6fe98534c0d50`; engine receipt `sha256:18dc9c224330e6542798ebde0073e90d982cd04d30fe0e756869f8148983d4f8` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-ENTROPY-001-18dc9c224330e654.json`; empirical-validation hash `sha256:76bb357cdc75b56d5a332823af4523174d6cd06f6ffe87228b6ed832a083ba3a`; measurement receipt `sha256:64b76e6801c73b3c9946a0f76f966c56733d9a2e4a2d3f3abb5ee208f98dc6f7`.

58. Irreversible observation and entropy orientation

Claim identity: `SFT-PHYS-THERMO-SECOND-LAW-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A closed macroscopic process is thermally oriented toward observation classes with no smaller unresolved predecessor support unless the distinctions required for exact reversal are retained externally.

> *Thermal irreversibility is the orientation induced by closed microscopic distinctions; reversal is lawful only with complete retained records.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`

- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-ENTROPY-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__observation-merging-support-orientation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	observation-merging-support-orientation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4d16b7eafead207076240e8118f52debfa4bc8cbeab2ed4c17e3596aa238013f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bd5d44ca5ac75aae6520413625174d04324f813e3ed129902e5e22ffab16e867`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:89b5457505016a0591d72cd0d779469d88001625cf1c604645660e721a61faca`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:705cb309bd4a45d0bd0d43283d8cc3a6b0fd0ca2ae6d049478e3bc0c95927331`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8a76f82a48d2516bb5d150b93a10eed9104453f1148a75c2790edb45b8119b54`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ceee18ae030e0b6bda80abd30799585a6fce564b94953c5a27fa18b275e7f2c3`. Independent certificate: `sha256:4176b2452cdb078a2fd8544bd43813b1baa59d68052243d8dcd0c60c3badbc8d`. Engine external-validation hash: `sha256:033899701ba3b86b2aaf16432b013dfed77cafbd54d22439c558645423cb15eb`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- second-law-1: predicted second-law-observation-orientation; observed second-law-observation-orientation; exact match True
- second-law-2: predicted second-law-observation-orientation; observed second-law-observation-orientation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:ceab052d9b5040aef06ba2bef0167154438fa87511c4573a57975a3d29a8d591`. Isolation certificate: `sha256:d95004150364294d059993286d2408d576c4f43b28cd79c7bcb4ebf2fa187602`. Custody certificate: `sha256:016ddcec986ead9701ba8676bfd5606876df1a981e6d4619d3f662f1a1e51deb`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- second-law-1 from IAPWS-RELEASES-2026-07-23
- second-law-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:02fcf9468b069998405e2fff909dc5fef14a8c287ad295f5a7c9d454dfc46410`; engine receipt `sha256:b1ec365f5fb8432d4ca9f69ecb6937596ff9e1a144860e58ade0d6a5f6bec17a` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-SECOND-LAW-001-b1ec365f5fb8432d.json`; empirical-validation hash `sha256:3c4186c69899746c010d7918efdf89c44bdf75c4cfadeb977486c08f111d6b05`; measurement receipt `sha256:ceab052d9b5040aef06ba2bef0167154438fa87511c4573a57975a3d29a8d591`.

59. Finite unattainability boundary

Claim identity: `SFT-PHYS-THERMO-THIRD-LAW-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. No positive finite transition sequence can erase the final generated recurrence distinction of a material support; complete recurrence closure is therefore a limiting boundary rather than a reachable destructive state.

> *The minimum thermal recurrence boundary cannot be reached by erasing all distinction through a finite lawful process.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-SECOND-LAW-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__positive-transition-unattainability__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`positive-transition-unattainability`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:19f775c6ec56ce1a2e15c29316400e65b1259c388873f0a478027c79b66253c6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every

generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:14b4815387ff86bbd60b37b5ce2e8305f1297484efb231a9ab76b087ee47487b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:14a67126b7e513be27af787df4da5ee968000098cfcbe746cce45b8ef944f2ec`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ac5e1bef5dbe2e4e07e3e2a331595bb74aa93a46b431b846471762b2ed6bd1fa`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4b5790435184a81b32eaab8dc32a905f9c80c104ab0201d738c3cab6266f62ad`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a4ba57ed42270c8a7ae69418ff067aab619e429c483b71d05d0af94c1c6e73cb`. Independent certificate: `sha256:c7c50729009b9c1eaf672794a191934ca83f87bfd1bffb88d1d982f12e5f3016`. Engine external-validation hash: `sha256:08bfa798c6ae950f3e0ec03b335948a9a18e1bfbf69103a80c7c546b50668437`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- third-law-1: predicted finite-third-law-boundary; observed finite-third-law-boundary; exact match True
- third-law-2: predicted finite-third-law-boundary; observed finite-third-law-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:855ece6421af57b129259ade0b6f65c572b248e809fcb3fcb0fddab1bdc8042d`. Isolation certificate: `sha256:f7d709472482e63022e731f1232abbd3653bde50d8d3769f3100078149a4d3ff`. Custody certificate: `sha256:b97280c0c1752f272b35f17c1ec6f9ba57ce7ef4383f5500e3cfbe4f1b31de66`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- third-law-1 from IAPWS-RELEASES-2026-07-23
- third-law-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:41b7d4b59030442a33336e31cb3ccfc9db10f430919f43cf47bf513ab0612b18`; engine receipt `sha256:a66152169839178e093243202041879920e7d82b50d08fae42cf6c14f464e15c` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-THIRD-LAW-001-a66152169839178e.json`; empirical-validation hash `sha256:411250ec2b850cb5347f9bfb0852a26191fbb8d018f34cf93a43e9bc9645d153`; measurement receipt `sha256:855ece6421af57b129259ade0b6f65c572b248e809fcb3fcb0fddab1bdc8042d`.

60. Exact finite statistical weight

Claim identity: `SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The statistical weight of a macro-observation is the exact positive count of its generated microstate fibre relative to the complete accessible support count.

> *Statistical weight is an exact positive rational census ratio over complete generated support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-THIRD-LAW-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-fibre-count-over-support-count__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-fibre-count-over-support-count`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d4ac1efa48dd0e7bada3923e37cd64c6633edfa64a89544daf2a0adad56efebe`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:20300f111d9dca3814ee908c92f5c0f65a94e71b60770634d02dda75ad148e34`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ae3eb0351361990cc53e46cf3b098f2d373d6267b064dd204dd937ac88f323e5`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ddaf3b086f0fbaa7a731397924b840c450b7228f3903a413bd2d8a5c251bbcd5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b69676863ee2ff8e9d958e57780df3bff7ef82ba16637994944469df9d05ec0a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:81a65c0c1d85723d19ca7c0b2e9103bc8be2e1a3834fd17f20a283a3deb4c33a`. Independent certificate: `sha256:a56d8972ed6beb1ff28103ab146793a9dd7dfa9bf2dde2f83b4cc487035176c6`. Engine external-validation hash: `sha256:4555be56f36f5930731595107d40d12a984c37b19a242640f5797e98052bf823`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- statistical-weight-1: predicted exact-finite-statistical-weight; observed exact-finite-statistical-weight; exact match True
- statistical-weight-2: predicted exact-finite-statistical-weight; observed exact-finite-statistical-weight; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:1d77769023931c2dbdd4012815b6bc4d9b3f7763b52c1c0ef9ede5da8aaec85a`. Isolation certificate: `sha256:833d7b71bb32d1f464d62c543bc9e0b2cd2baf36a33f206913169e194544c6f9`. Custody certificate: `sha256:5f497c555a4ce6696dfc7da414e29eca11ab8a8ca8ac463c5e8897bcfebf6c8b`.

Registered source descriptions:

- National Institute of Standards and Technology: https://webbook.nist.gov/chemistry/ (sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- statistical-weight-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- statistical-weight-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:e5c8a8c575ea73a768b25a408c73950578a9dded6d59d781b870fb94d2ae5691`; engine receipt `sha256:fc2fdc5352d172755221a6fb15a61dc121038e01f7390aae19cdbe4c9334c9d0` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001-fc2fdc5352d17275.json`; empirical-validation hash `sha256:b473c92dbc371b98ccfa9e4b7f5577449e1de07f76492138709dda2596d14a4e`; measurement receipt `sha256:1d77769023931c2dbdd4012815b6bc4d9b3f7763b52c1c0ef9ede5da8aaec85a`.

61. Thermodynamic state relation

Claim identity: `SFT-PHYS-THERMO-STATE-RELATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A thermodynamic state relation is the minimal closed dependency among independently observed support content, boundary response, thermal order and energy carriers after redundant coordinates are removed.

> *A thermodynamic state law is the unique minimal exact relation closing all registered state carriers on its declared finite domain.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`

- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__minimal-closed-thermodynamic-coordinate-dependency__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`minimal-closed-thermodynamic-coordinate-dependency`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d656e11b809197c78f6cd210f98b11a2baeef1bcb0e8d3742d89d2678b5adc40`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2d4ff98f4e05c186b576d6a5c15c2c6d56d46197243299fbaaf7a2b79a4cb926`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:aac21fc9d70da35afbc04c2f4f02367f34bbc59cf10440b79644448a989ccb05`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7eada2f344e23049e34b2caf66a6170bfff87d85a1303e1f3b8f5cb05bd36cdf`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6e2ab2f551a6a20796a7d3d6e7a0cb38cf926f3c1b05915952674190323de8b5`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:46e58c0847d9c306c641b73ccbdfad62f69e06ebb0b62a15605a20decfbcf32d`. Independent certificate:

`sha256:32e7ab8e33d2cdaaa90269d5a5e530a00c55c2cbb74767f7e5f31813ff3e1e07`. Engine external-validation hash: `sha256:7fa13052c0e6557792214b988f1077ff30292dd6c436f65c7fa857aa81ba07b7`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- state-relation-1: predicted minimal-thermodynamic-state-closure; observed minimal-thermodynamic-state-closure; exact match True
- state-relation-2: predicted minimal-thermodynamic-state-closure; observed minimal-thermodynamic-state-closure; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:ab0aa384e9b13ed17f21031e5eba406bc1c9b93bfd5c5f5de5fd1c6a868c454e`. Isolation certificate: `sha256:65dc12ea74c2f7cc9e26067c0fecb61984579c9f0afae668f2a0b4f9ee7c2589`. Custody certificate: `sha256:f10be60f8dba1c824a1b30e71edd30cd47b66a9bcbd77abdaca784c51cccb958`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- state-relation-1 from IAPWS-RELEASES-2026-07-23
- state-relation-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:83596a70b3b622fc58c4f2694fb31f764aea5895333f19dddec9148a9d267987`; engine receipt `sha256:e74262c55559b31cb22638ee24e9203e34a582937ebb8053d54e55268cb707a8` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-STATE-RELATION-001-e74262c55559b31c.json`; empirical-validation hash `sha256:2a5bd1c195abe5b66e19ebb70d273a115ed3ee1aaf309aa3dbd37240053b73a2`; measurement receipt `sha256:ab0aa384e9b13ed17f21031e5eba406bc1c9b93bfd5c5f5de5fd1c6a868c454e`.

62. Phase coexistence and transition

Claim identity: `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Distinct macroscopic phase classes coexist when their exchange boundary preserves a common thermal order and reciprocal transfer balance; a phase transition changes the selected observation fibre while retaining total provenance.

> *Phase coexistence is balanced exchange between distinct structural observation fibres; transition is provenance-preserving fibre change.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-STATE-RELATION-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__coexisting-observation-fibres-under-balanced-exchange__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`coexisting-observation-fibres-under-balanced-exchange`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2ea8645395888cb98611995be975d320e16cfbe1355bdc5cdd20b9f8e342c64f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every

generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f42cfff2d91851d91717f421aeae259d02959ed2a9a736586d1d333e968d9723`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e8b2e3d4a7c16422e8b1867ca2d6dcbc36f3f90ecbf6962b6ccec29e53d2d3ed`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:312eb8573d58e786edfd4a14765ad2fcb8bad4ec82de9488659ff013dfa0e92b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b7592e2613534291ef6f2b6dd187cc4a2a88ce205c213f5c8265ce95707b5998`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c5ad6f16c8ccb1c0195b7dcd450794abf0a6582d0801d755e1d1cee852687b5b`. Independent certificate: `sha256:46defb37258a5ca944a8d3954d8d25998ccb82efeaa199a0c98458f785896355`. Engine external-validation hash: `sha256:5cac7be29a9861ed564fa20e12133e3b1e74e3c52fa323c9bd088c55c20ec720`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- phase-equilibrium-1: predicted phase-coexistence-transition; observed phase-coexistence-transition; exact match True
- phase-equilibrium-2: predicted phase-coexistence-transition; observed phase-coexistence-transition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:8d7e1705c61cb89e775e56f2bd970529f4c51505e8cdda5a4966c3972736e18c`. Isolation certificate: `sha256:e65dc19612a5d5e476aa80c965b6024222ee83334fd6f5e73163134f47edff48`. Custody certificate: `sha256:63365f3340988042c817187843078O2ae1e5fd1d5c10b7d3a792710962be5582`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- phase-equilibrium-1 from IAPWS-RELEASES-2026-07-23
- phase-equilibrium-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:500813f6ac91be300d730b71689ca9330667887f4ebd6d97ac3b77914245ad2f`; engine receipt `sha256:804bbbce151a9807129d17226d039f46d4b5ad9fb4cd7d90135bbb01fc50da79` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001-804bbbce151a9807.json`; empirical-validation hash `sha256:3507c316bca528f72c7e066c7e9b8da9bcf31bccfdd292be2258f80106b712e0`; measurement receipt `sha256:8d7e1705c61cb89e775e56f2bd970529f4c51505e8cdda5a4966c3972736e18c`.

63. Kinetic transport and collision mixing

Claim identity: `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Kinetic transport is the exact propagation of held motion and energy labels between generated cells; collision mixing is their closed local permutation subject to conserved transfer records.

> *Thermal transport is adjacent carrier propagation with locally conserved collision permutations over complete finite support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__cellwise-carrier-propagation-and-local-permutation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	cellwise-carrier-propagation-and-local-permutation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:970188e713877d4ca242224e6ee4bdbd89d64d5cdcfd9a024fa03e9b76183a55`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d30fbb38fb12fda20dddfb163e4dfd0434415120f7118b225279220c17463cf7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cd84dd2bc9be81e40abe6345768b97a1c0f4d9fdcd7a310121d3935a08eb4d9c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b510d9c52a16063ce072ec1ae96bbc6a002abe947f74d692ed979787fecfb4f6`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d94f1208accd846d0e7148bf3577939ac944619e4279e87006453d03f5b2f579`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:83e6a6d6a4a84aa01fc03b09986a11fdce29c319f2018c0f199a7f2fa0cade8c`. Independent certificate: `sha256:3d10eba440c2b0fc41c95cdf04539a180db711dde6c55e71a42c60bc9cbb9d6a`. Engine external-validation hash: `sha256:8819d0ab3dc2bd3bec32854c33559c8a448d22a05cbaa76d386092e589127985`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- kinetic-transport-1: predicted finite-kinetic-transport; observed finite-kinetic-transport; exact match True
- kinetic-transport-2: predicted finite-kinetic-transport; observed finite-kinetic-transport; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:8e81fb917a5769ce9d1477e1fb113e43addd31d4cdd6a267e5f8a7e8b0114e98`.
Isolation certificate: `sha256:bd4b00e6c3dc34dee8920efb54c94fe0707ca3f051c04097f64f4eaad9dde8bc`.
Custody certificate: `sha256:f9ebde4d1266eb476bf8be09200374a0ab79672ae4e591700006df918d88e4fc`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- kinetic-transport-1 from IAPWS-RELEASES-2026-07-23
- kinetic-transport-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:27dc7b5558723466a12162ff23a4c83eb32bb10ce9053af043919dae22f6d8d4`; engine receipt `sha256:0001a2d7176cae05e6b1ad4ef9942e474d14916872262d377350c7a465823c18` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-KINETIC-TRANSPORT-001-0001a2d7176cae05.json`; empirical-validation hash `sha256:aa0023c1ca2b4b546c7f8680153d6c14d572396674e067622c2a5ee39399302e`; measurement receipt `sha256:8e81fb917a5769ce9d1477e1fb113e43addd31d4cdd6a267e5f8a7e8b0114e98`.

64. Finite fluctuation and recurrence

Claim identity: `SFT-PHYS-THERMO-FLUCTUATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A thermal fluctuation is a finite departure of an observed fibre count or transfer count from its recurrence class; complete deterministic support fixes every departure and return path.

> *Thermal fluctuations are exact finite count changes across deterministic Fold paths, weighted only by complete support census.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`

- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-count-departure-and-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`finite-count-departure-and-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1018628409f455d8d475562031fbe08345c74b74652c2d2532dfa4f754b2164f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:00138e10f0a178a63731e18278eccfda8318dbc158bd82da5420e550a30a5c76`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b0e40ef20b9a4436701c2356257b61ba295912e51754d039135fdf2fad587680`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c1f74423658eddae370554c1aeb672cd2b0738d8683ca9eba6e8d6e78ef88e87`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:087c5bd21117695a5b7de997a677d07134aaa675ae69aed071669d6feba944eb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:03fa0b411cd53d9dae61566fb1f06b5e587d565c9384c0f8799eb196854cb1fe`. Independent certificate: `sha256:9022eaebaff4f04b2b66185a6fa3642d43f39cb420d9e29868b084491ee72d67`. Engine external-validation hash: `sha256:21cd8ec2459dfd3b1b895655ea8ef38fef71ae97b210d79505e9e3e3bb2bec34`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- fluctuation-1: predicted deterministic-finite-thermal-fluctuation; observed deterministic-finite-thermal-fluctuation; exact match True
- fluctuation-2: predicted deterministic-finite-thermal-fluctuation; observed deterministic-finite-thermal-fluctuation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:c24fce3c578e4d987bd8cd1890a1888daa511269141b9f1dfac7e3d153619c23`. Isolation certificate: `sha256:15a12ba1249c8fdb3325c1b9cff562958f45b7c8eb979ca716e0569508249c02`. Custody certificate: `sha256:2a54b66f71698bd7cb2234ceee3b35417c2eb375d1a8722952bc82f17346b888`.

Registered source descriptions:

- National Institute of Standards and Technology: https://webbook.nist.gov/chemistry/ (sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- fluctuation-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- fluctuation-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:9e7385a5320d843028b9c3cffb918c0cbcbc4b5ef29a98d10c68f48b2183297a`; engine receipt `sha256:c39f22eaadf319ad2c29d582ffc035e542f8df1e7665956e9e3d50506ce26d07` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-FLUCTUATION-001-c39f22eaadf319ad.json`; empirical-validation hash sha256:0193016a0f6bdb0b5d0349f7a7e4562b81d57c23bc26cc8a9d05f83923ab49ef; measurement receipt sha256:c24fce3c578e4d987bd8cd1890a1888daa511269141b9f1dfac7e3d153619c23.

65. Macroscopic irreversibility with retained microtrace

Claim identity: SFT-PHYS-THERMO-IRREVERSIBILITY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Macroscopic irreversibility occurs when observation merges distinct microscopic histories; the complete Fold evolution remains reversible exactly when their fibre labels are retained.

> *Macroscopic irreversibility and microscopic reversibility are corresponding projections distinguished by retained predecessor information.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-FLUCTUATION-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__macro-history-merging-with-microtrace-retention__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	macro-history-merging-with-microtrace-retention	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:06cf6ba1530f9c8dfac16b38c8aec8731e075a8f595e8680295fc0906b8c57ad`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a951e838910cc42db069eb2b5f345c610e61552ae58bee73b9c3555a03f303b3`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e4a5fa3374d35a5a5b6c7b05e6ec9a06ad4ce856e8f25bbf68fe2c43704dff22`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c58c5081d1760e12d16d32be04067b042d8b4489f64006681cc3ae974d5fc0dd`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4c4cbcf15512184cf5cc9dc5122d793badd4085ca55356feda189d5f580cfa8e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:208cabe0f6efc2a2901bb5510b1559bd30ab859ffdc6a6f1d9e5bada19fac7c0`. Independent certificate: `sha256:2b6ba3c9dbbc8057e7d929f2e3bc6c310019c4451f1f1e0000d3de1e6209457d`. Engine external-validation hash: `sha256:848fef281db085300df71fe607ec80a438182f91d4d6037e05c84183bb7fd321`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- irreversibility-1: predicted macro-irreversibility-micro-reversibility; observed macro-irreversibility-micro-reversibility; exact match True
- irreversibility-2: predicted macro-irreversibility-micro-reversibility; observed macro-irreversibility-micro-reversibility; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:ac90cb1396e2257b13162cdd1249089686047e24f83b2226b1357f983a0710ba`. Isolation certificate: `sha256:efdba3d89f4b065bdeedeafb72ce9fccf4b3d2f662fb7d108cb6e25012a09ea6`. Custody certificate: `sha256:ece97206b53986eb944a1e4d39735d49efaee16fbf27920fb982a958d5a837e5`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- irreversibility-1 from IAPWS-RELEASES-2026-07-23
- irreversibility-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:47d423e50a3b67c092517b42de9ef930ab20226a2be315c511c8d4e1ce7f4078`; engine receipt `sha256:10ae5d449758f74df3bba03643a4b2f3f48f58096812e5052dffe56e1b0ade69` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-IRREVERSIBILITY-001-10ae5d449758f74d.json`; empirical-validation hash `sha256:62adf31b0648d8e60afcc5d5009c749c3f5e9b5f409f3c878323efd055f95f03`; measurement receipt `sha256:ac90cb1396e2257b13162cdd1249089686047e24f83b2226b1357f983a0710ba`.

66. Susceptibility, response and transport correspondence

Claim identity: `SFT-PHYS-THERMO-RESPONSE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Thermal response is the exact change of one registered observation carrier per exact change of its source carrier, conditioned on the same held support and recurrence class.

> *Susceptibility and transport are exact source-conditioned response ratios on a declared finite observation support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-source-response-change-ratio__source-bound-proof-trace__capability-closed-prediction__separat` `e-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-source-response-change-ratio`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d65cb90346dfc94d9cd5a3654979e1098dd16aaba415bdb46f3024797e32d35c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:eccd392239638a2f850d68b176fe90134f718e514c3d996a6575f82bfd601890`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:415c4fc3248360a54fff8b3877d91378f6901ddb21ccb24a44977cd058c71cd3`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1f5e7ec3a67a74630e6a6955565c03bb4cda052fb88b4976c9c91db0458b06af`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:554006a67b3012c6996aefa6658c90e5eb362ee1467800525bbdac3741313e81`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:aba9c526a54975949c1a2800c2ea4f06406cea4ef6a10c7e60679e922e20e072`. Independent certificate: `sha256:554a7ad79538b78eb9bc6ff390b0bc27dc162c20a834a55108f16d138ef8eec9`. Engine external-validation hash: `sha256:eec549428de2dcdc92dac63d5312e065c53c037e7ac0f9d8dccdfd83926826b9`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- response-1: predicted exact-thermal-response-correspondence; observed exact-thermal-response-correspondence; exact match True
- response-2: predicted exact-thermal-response-correspondence; observed exact-thermal-response-correspondence; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: `sha256:e5ca7d358c3abb7496c905cf1f57e75196354b55b78e843ac022e7f01592c1ae`. Isolation certificate: `sha256:b1134e2384835c5a56f4b0c2eed231e084de016745ff87843ef63a3483c2122d`. Custody certificate: `sha256:eb1eb989aafa52313661a345fe973c6a70b2893a9374fca124d5a221fadeec90`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- response-1 from NIST-SRD-INDEX-2026-07-23
- response-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Thermodynamics is reconstructed over complete finite microstate support. Temperature, entropy, equilibrium and irreversibility are exact observation and transfer relations; ontic randomness is not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:6e79304fa667cd6f93ef31a4f8ff35ec930d6cafa75c59f2bbbf91dfe3ab074d`; engine receipt `sha256:b5a02f181350f74161c9c1041572b66e4a180034f99401bdad07bb9ae177c262` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-RESPONSE-001-b5a02f181350f741.json`; empirical-validation hash `sha256:742f32e96d6a7c55270447d00d5410b54ddb51f4daf50b60f1522d80e87efd1c`; measurement receipt `sha256:e5ca7d358c3abb7496c905cf1f57e75196354b55b78e843ac022e7f01592c1ae`.

14. Quantum Physics

Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level.

67. Physical quantum state correspondence

Claim identity: `SFT-PHYS-QUANTUM-PHYSICAL-STATE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical quantum state is the complete generated support of distinguishable Fold words together with held phase/action and preparation provenance; it is not an answer-only amplitude list.

> *The physical quantum state is complete finite Fold support with held phase/action and source-bound preparation trace.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__complete-word-support-with-held-phase__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`complete-word-support-with-held-phase`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a66a1c035a02aac6b4092403632107eeacca1d696e5f6dda52c4848a19601c81`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f11e96e98ea962641d29ecde80b00416bd1674538139bbe194dd49859d02988f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:672a07dff9f246bfddbf32a064c4aa643ffdcd3d7a7f1b0603c5631ea289fcf7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:815054d543d23cb8762114e29be753c6caafd2edb3526a773cb4c792281373f8`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8cd47b523aba7d4c02f62e90f0dbf213234e179910d3affff1ab8752f8ddcf89`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:fa75461c4e288e36ede8428abbda45cf621831355500de7054dc6b6f4e0868ac`. Independent certificate: `sha256:5efc8bd57bf4ead54b60a7a9f2b23c13a85041c03edc946693ded1dc62e03b6e`. Engine external-validation hash: `sha256:05266c971285714a284201a1b440e9c615418c345e0c843036039631cffa8c5b`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- physical-state-1: predicted physical-quantum-support; observed physical-quantum-support; exact match True
- physical-state-2: predicted physical-quantum-support; observed physical-quantum-support; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:557cb140756b84937cff842ca98bc46a5ba17f33b66ecd390541eea96ab3e8c8`. Isolation certificate: `sha256:bc978552bd11060de1127ea1f2c8b39fe01b33a7496f7fadda7066d388989b25`. Custody certificate: `sha256:6aa6b08641be66190a4907283b531f844e8d10f1bbd5387be9dbc70a336ba149`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- physical-state-1 from NIST-ASD-5.12
- physical-state-2 from NIST-ASD-5.12

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude

- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:646d05ecdd35f77c6440a8725ed08c85e6a8b59ed17c7a999e30c649417ad9e0`; engine receipt `sha256:66bb63008bf5c34e9b3d260cf59890ac2d05f4474830c998296449725445145b` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-PHYSICAL-STATE-001-66bb63008bf5c34e.json`; empirical-validation hash `sha256:0b3e2b8b03aa236e15c72bb453856c5055d6b5996c4f8c5cf9b987ec2ca2f54f`; measurement receipt `sha256:557cb140756b84937cff842ca98bc46a5ba17f33b66ecd390541eea96ab3e8c8`.

68. Reversible physical quantum evolution

Claim identity: `SFT-PHYS-QUANTUM-EVOLUTION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Closed physical quantum evolution is a one-to-one Fold transition on complete support; its held predecessor labels reconstruct every prior state exactly.

> *Closed quantum evolution is exact reversible transition of the complete physical Fold support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-PHYSICAL-STATE-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__bijective-complete-support-transition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`bijective-complete-support-transition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:73d98d6a2c61130e167ae21e7c0dc1228a43f3734d500dc67b73b587ee99e228`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5aa3839cb9d2fa990c5eef22d1b1bb2a10642f726af0ffdbf0a071f8fc7ba3ed`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7577729d7390f62b2fdb9ae943374aba6baaead28cb6bb4a7caec07bd7d79837`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:646833aea82f4db8f7b9a7b40a8b4d8d66ccd6a60744c865a76651b68d27500d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:03664f9fbe104e20bee6d232f787bdffdaecf7eb2fcbb41a903dd035e1bc078f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b88b6661761e1575417e3ced3cbbe436daefa0efb420a72ffd6e9d29cf0d87d6`. Independent certificate: `sha256:c5f7cee6cb315e49e0e49a0d5bd83897c9fe20a141537f38d1d96ada4f489557`. Engine external-validation hash: `sha256:3b7ede270308eaa7871b044a15f9f4cc8dc31d8439b88625aeb358ff71b8af67`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of time: exact predicted interval [81120909/3353649786316000000000000, 527285909/21798723611006000000000000]; external interval [12094421632919/500000000000000000000000000000, 2418884326589/100000000000000000000000000000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact action/energy interval does not overlap the released CODATA atomic-time interval or the displaced control is accepted.

Measurement receipt: `sha256:41780708e62d7216e0211cf8c8c7e026ec72ec34530c9d843f5e0051cc445edc`.
Isolation certificate: `sha256:74a7416c4ed12235fb4c2277ebc1a12fd5d38ffb17665047605bc0a9c643486b`.
Custody certificate: `sha256:f5660469cc66b1b704b11ab57d4d9f7a704e73ce923e5b194332bac8c74f0530`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- evolution-1 from NIST-CODATA-2022-ALL-CONSTANTS
- evolution-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:c5f29458763791f86a617642bcdf6b2e8e97f01f8baf5d689e702d136424a983`; engine receipt `sha256:0744b73d5436d955436b9dc7c39a4a63f229d0ab019c41409dccc37663148fdc` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-EVOLUTION-001-0744b73d5436d955.json`; empirical-validation hash `sha256:7665bd0ab9d73de01f2b4ef090839dc749773336cf8a7d28585bb22bcfe5b1f4`; measurement receipt `sha256:41780708e62d7216e0211cf8c8c7e026ec72ec34530c9d843f5e0051cc445edc`.

69. Observable and measurement record

Claim identity: `SFT-PHYS-QUANTUM-OBSERVABLE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A quantum observable is a generated partition of physical support into distinguishable record classes; measurement creates one retained source-bound record for the observed class.

> *A physical observable is an exact support partition and measurement is its provenance-retaining record transition.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`

- SFT-PHYS-QUANTUM-EVOLUTION-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__support-partition-to-retained-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	support-partition-to-retained-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:12d0fb73d6f5134644ce0e8a9714cb56e7d21e113e4502e9e1d3a8a92238249f. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:31e681c5a7afc9a3487c9cf195df85aa683e030ef0fb4c5d807f5b18271593f4.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:958782c82e68ed494612530e55434209d317595a5ddbf1f8ddfb1058552a7d5e.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:b9d57ba62cdefebfb656007ac936b9d0af7c745999bfa855115292ca30a28787.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:f82689babc0a8c4eb4d3bf59e677a2f10f20e0007a87075032177ebb2324c442.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:5b205bb4f52f6c844d0dd0b60e252d9c5fe684eba16878c5d62c82453b782472. Independent certificate: sha256:d856dda98523631331ca9e6d52d2825b95a3d832172d3fff08ed4158e6978057. Engine

external-validation hash:
`sha256:eafb213ceb10b58f0c1608cb8302ad1e1f376def86e2786fbbf9bb0460904d58`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- observable-1: predicted quantum-observable-record; observed quantum-observable-record; exact match True
- observable-2: predicted quantum-observable-record; observed quantum-observable-record; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:2871381c6705550221b73749a4698fea5746c13528f010e96af8cc00cdb27470`. Isolation certificate: `sha256:40eba5543b5dfbb2f077246045519bfea6f4025844baf9529c4430cf00919662`. Custody certificate: `sha256:5059ae0c8ccc527ccee63e88fc8f8bf516b812e8ed5c90d7ca724f77f02081dd`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- observable-1 from NIST-ASD-5.12
- observable-2 from NIST-ASD-5.12

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:a15964f14ab058b547e8eb3b2e3238bfa7a09a9ed447f178e78f39196dee4579`; engine receipt `sha256:a7fffafe809c0f14f4eeaf1b32dbb4f54057f22c2ec06450c9f6d42450df1be1` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-OBSERVABLE-001-a7fffafe809c0f14.json`; empirical-validation hash `sha256:93cbfc103be9c659299bdfbe057e2d895f8d7026be8e8dee44eb7b363d2f1087`; measurement receipt `sha256:2871381c6705550221b73749a4698fea5746c13528f010e96af8cc00cdb27470`.

70. Measurement weights from exact branch support

Claim identity: `SFT-PHYS-QUANTUM-WEIGHT-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The weight of a measurement record is the exact positive count of its phase-compatible predecessor fibre relative to complete generated predecessor support.

> *Quantum measurement weights are exact rational branch-support ratios over a complete deterministic census.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-OBSERVABLE-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__compatible-predecessor-count-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`compatible-predecessor-count-ratio`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9f0e7a4dd0856f15844b0912c6abfcc6289787e87abbb1de36cd1c994a45c7b3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bf51dde0d20e4b318bb6a0714ec5fc0cdce3ab2b91178e3c43fdeb812f2d19cc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d9fc7d487a086cbfa2b9d9e69dddafb926af1837fc1d891ddae7054521e63f3b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7e5b72118aedf409fbf527aa4d17e4b4fceab7d8803cf0bcfd802da9548754db`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9869064001c5cfb9d3cd21d88947abeeacaebb42c5b91b62494c58a07c5f5b69`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f9af1074be87591526a6071e6a4804494c8b408d6d5e87a104a67f0d57271c1f`. Independent certificate: `sha256:daf893f865417ee3cdfc25605e989b9074346c5345fe92ab2cfc268bb782aa78`. Engine external-validation hash: `sha256:5f5a4400f4a09026d6210fb06e32594bedfb54bed54a299568b95d1d54bba89c`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- weight-1: predicted exact-quantum-branch-weight; observed exact-quantum-branch-weight; exact match True
- weight-2: predicted exact-quantum-branch-weight; observed exact-quantum-branch-weight; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:535fc025ce1999bed3c30aef0f2ae77cfba6395ee37f9ffe816aa59b2c736dae`. Isolation certificate: `sha256:6cc773d44b9bbde7ca65d80e0e7ff2f9a18a859e52f9d8cc5a1d184c566646b3`. Custody certificate: `sha256:d0a75a812468b077f05fbf9695e654ee8be64cee230eaec818d6b21f110591c6`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- weight-1 from NIST-ASD-5.12
- weight-2 from NIST-ASD-5.12

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude

- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:3bde7c0d35d4ce9a0fd5c01b7fb760d6fbf328b60b0152b2e97f962527dc810a`; engine receipt `sha256:ac65de8a653d0efe53bca999f07d44dcac07e42d121a74974e8ea0ecf49ad101` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-WEIGHT-001-ac65de8a653d0efe.json`; empirical-validation hash `sha256:e92f51b52deb77493b1cdc52cd55bc05c358afa0c818375b3a9db0d54132d265`; measurement receipt `sha256:535fc025ce1999bed3c30aef0f2ae77cfba6395ee37f9ffe816aa59b2c736dae`.

71. Incompatible observations and uncertainty

Claim identity: `SFT-PHYS-QUANTUM-INCOMPATIBILITY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Two observations are incompatible when their support partitions do not share a common refinement preserving both held phase/action records; resolving one necessarily closes distinctions needed by the other.

> *Quantum incompatibility is absence of a joint record-preserving refinement of two generated observation partitions.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-WEIGHT-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__noncommuting-observation-partition-refinement__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`noncommuting-observation-partition-refinement`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:35d09d2429fe2535d01ae8f4c3af7575b3a3a80ae8da112567c935fc4752dc2f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:158811357c5a538c1fca4b7fdaf9bc1fb2c2753248c18b4a1b9f19a16d714de2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:30262f754cbdae3c420a4809a86515b4fc9ec2e6b0f393d644af5f266f1b82b2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0de745408f2d5a993ebeb485c54819897e0b5ff04972894f1a2a0d6b42961c55`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a2c93fe9bb2f155be22a81fa63369500fcd12366413eeba1836870243c1f7595`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c35b641f13953583e4e48f397470b0a9d440416f2c0bd73e993ac6d7033b8464`. Independent certificate: `sha256:bd6f7030405d0751463d75d2a47c86bb88207d4f8e6cff53d7048eb753b2927d`. Engine external-validation hash: `sha256:fdbd79494dee7ffb5e8061edd0a2c011155d467053a97cc2f97ba7e3d86d4fd1`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- incompatibility-1: predicted quantum-observation-incompatibility; observed quantum-observation-incompatibility; exact match True
- incompatibility-2: predicted quantum-observation-incompatibility; observed quantum-observation-incompatibility; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:3c0bc44b43868fe92dff7a8b258cdf6895f9adfd65f43532e528e7e1c3fbd205`.
Isolation certificate: `sha256:8249451c3e3afa9a18b70a43af14ead80b8c1da21fc7b8bf6975d4722f8d8501`.
Custody certificate: `sha256:f7dea63a65493f7ca9a1e266646684725b37edd5536f07d3e931c8d2616ca79e`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- incompatibility-1 from NIST-ASD-5.12
- incompatibility-2 from NIST-ASD-5.12

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:a35454ea1d2cc3324b041bfa1e5a343337b71ce6fc3ef953d24d47cdd192b269`; engine receipt `sha256:fd15b51c35d435c048c1f8d3d05bb80e67fe5ab8f706df40e6a2af13e04fd430` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-INCOMPATIBILITY-001-fd15b51c35d435c0.json`; empirical-validation hash `sha256:f925f8612be1d3bb5ebac1b74f17e66b9c7a3c4d00859d8425e64e5c8cc8d04d`; measurement receipt `sha256:3c0bc44b43868fe92dff7a8b258cdf6895f9adfd65f43532e528e7e1c3fbd205`.

72. Spin and finite cyclic label action

Claim identity: `SFT-PHYS-QUANTUM-SPIN-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Physical spin is the finite cyclic held-label action of a constituent under generated orientation recurrence; opposite readings are fibre labels, never signed proof magnitudes.

> *Spin is finite cyclic orientation action on a physical Fold constituent with exact held measurement labels.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`

- SFT-PHYS-QUANTUM-INCOMPATIBILITY-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-cyclic-orientation-action__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	finite-cyclic-orientation-action	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:86b0dd83b47ba632c10f4e9c87ce9c6bf978a392949102cb2243f619da59f0cc`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9abc35415e4749e0e224f44a961bf7e30269d12a9fb7256953f3c14186f3364b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:db916532b2cb153c2f330fa206cf12fb490d61af66dc238e59275ecb4bc77611`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d17d43683fff65b3f9dfdcd5236679043af5f399108fa0cc78c365dc5861b49c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:963828635e8ecbdebf873622872f9b3cc1a16fff70c650a55b7339c0e0c695a3`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ecd65100aa6616719ee338cf1e710ec95cded237bb9586bdb45c8f532803a5ce`. Independent certificate: `sha256:371f7e7b225aa68088362c332d7f831a371291b1bf96280fd57333b2cb9ea08b`. Engine

external-validation hash:
`sha256:f24c73aefbbebf32a9b57b99522822d4dc424d4bdb1d6078ae1b96e8b518d240`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- spin-1: predicted finite-cyclic-spin; observed finite-cyclic-spin; exact match True
- spin-2: predicted finite-cyclic-spin; observed finite-cyclic-spin; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:7d20ee9ac2ed5843b1a09f6ffa54fce08fb7bb59c53e19903c735c772dbd3ffa`. Isolation certificate: `sha256:b820f1e1246c234f6aea5595ded40d7f1e2fb323ad4a542047f8da18f584269d`. Custody certificate: `sha256:d8d97712140f7e6c0713dc45d85143e04903db389200f5cf3f127d18daea222e`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- spin-1 from PDG-2025-SUMMARY-TABLES
- spin-2 from PDG-2025-SUMMARY-TABLES

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:b9ec52945330138564cb3a577cb2440ccc4d25c872102ed9e1362f569832a8e7`; engine receipt `sha256:f233aa5e16a279c3803209c902993252d6631531d91fc5959f1e3123c13f5351` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-SPIN-001-f233aa5e16a279c3.json`; empirical-validation hash `sha256:0f6db2e5307e2cfc3bfb7007770f189e38455b9b82594f11e4ce3c2d2f95e5da`; measurement receipt `sha256:7d20ee9ac2ed5843b1a09f6ffa54fce08fb7bb59c53e19903c735c772dbd3ffa`.

73. Identical physical constituents

Claim identity: `SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Constituents are physically identical exactly when exchanging their unobserved identity labels preserves every complete observable transition and composition trace.

> *Identical quantum constituents are the exchange-equivalence class of complete physical Fold traces.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-SPIN-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__constituent-exchange-observable-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`constituent-exchange-observable-equivalence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:94f7c002b3ea517a33a036afd2a422103c08e0dbbd2c5874b4c564fc97ebe8ce`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:aab2c92e7b762d31cb25bc68bcb80ed971da4eb2051a50fa55e461fafd318bde`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0efa8b7889317c183594f5168fd1f282182dab407ced9e46aee80da270edb6d0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d87f052ae46189313214dd64b3e37c5561f0f0ed21d4e72b9fb960bd430e254c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c1e08af79027067b143e65b5f88a280c20655a915fba624dabe47979298896f4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:85d4bb3454e1a15386cb5030a57a6e6279dc9115b659e2f00377e143c77df2c4`. Independent certificate: `sha256:7384ac457d91ab3248d9d5bdf1921a272b92984ed1cb3ea17355a246af5627b9`. Engine external-validation hash: `sha256:391793959adb5df11e16b2f1581e0e5ff20a62094a3163eb54ea6175044ba4ae`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- indistinguishability-1: predicted quantum-constituent-indistinguishability; observed quantum-constituent-indistinguishability; exact match True
- indistinguishability-2: predicted quantum-constituent-indistinguishability; observed quantum-constituent-indistinguishability; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:dad17c56682caf0c7113d28d618515d4903b9790953166b4479fdcb8208cc46b`. Isolation certificate: `sha256:bf9f554f65c9b889862ea7a292c05ef0e64d626223009e593fd887dc644bd453`. Custody certificate: `sha256:48c4eef98bbd600d413d9dd9786e4b9f37d859086c1cc930c77a1848c03013b2`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- indistinguishability-1 from PDG-2025-SUMMARY-TABLES
- indistinguishability-2 from PDG-2025-SUMMARY-TABLES

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:4f56f583db68635a253fb89c468c68303a45f12623b0f532f3b24b756bc27dab`; engine receipt `sha256:7e67d989399d97c898d0a1aad60aa27d5f62224f323adaec9159f8f38dc378d8` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001-7e67d989399d97c8.json`; empirical-validation hash `sha256:89f7c99135492c64d116908f9bab5ebf019a4b4d4e8a89c7b034ca1613bdb830`; measurement receipt `sha256:dad17c56682caf0c7113d28d618515d4903b9790953166b4479fdcb8208cc46b`.

74. Exclusion from antisymmetric composition

Claim identity: `SFT-PHYS-QUANTUM-EXCLUSION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For the alternating held exchange class, two identical constituent labels cannot occupy the same one-constituent observation cell because exchange would erase the only distinguishing orientation trace.

> *Fermionic exclusion is absence of a distinct same-cell joint word under alternating held exchange composition.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__alternating-exchange-composition-exclusion__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
`relation`	`alternating-exchange-composition-exclusion`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0cd2200a9acc1547dbc642a6ea043db99f30ec0df5cdf1fa1972a8138af8947a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8bfe4d1671a6378a9659881a08b2ec43b7e8ea2de48f7d6256b5b2e2f13fa1e7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5a37d4decf9f8cfb19f4158b0ec17548bb010b57567a48cb269d18f322ccf018`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fda1464760f59392dbe0dd19fc04d4affad8b926f222732fff7b158093adc211`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0c3883c36033a3fc149220284e3b199a88357eee0385de508566454e6fe4b5da`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d3c390a960261b1252d41e7ce98d06d23696d6ca0d172141b31fd3a59f3c354f`. Independent certificate: `sha256:26695bc4bcec93fcb1cb6ff194422727859a22fa8a4589dca768f75e6abc9536`. Engine external-validation hash: `sha256:1746eec6ce34fc98f992695265b38dd23d513a4ada8ed3ca065b48b256cfe8e7`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- exclusion-1: predicted fold-exclusion-law; observed fold-exclusion-law; exact match True
- exclusion-2: predicted fold-exclusion-law; observed fold-exclusion-law; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:9d4c868e9c114a7b9fcd325b6ca8cea8a1497590cdad5e6ef70e03aa66c382a1`. Isolation certificate: `sha256:b624470e88af677e4189745c8206e482595e421d8bbd783c47349f9cb05ad724`. Custody certificate: `sha256:bfac40a5753c85004ab5cae4e30e48e561ab4fc458180e1dc5e232910baf6251`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- exclusion-1 from PDG-2025-SUMMARY-TABLES
- exclusion-2 from PDG-2025-SUMMARY-TABLES

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:4ab498dbd8b0d9aa4e62db6bca8e8e87eb80c4ed58fea206c06eb538c21f1973`; engine receipt `sha256:27239f90141d1287616ee28f368bcd94cf1f94b549e9b65a0a0982afab765eaf` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-EXCLUSION-001-27239f90141d1287.json`; empirical-validation hash `sha256:3802d875f3aef69b91e7bbac3eb7f17bee7c6b599c0fabaa61b92e5bdca5f129`; measurement receipt `sha256:9d4c868e9c114a7b9fcd325b6ca8cea8a1497590cdad5e6ef70e03aa66c382a1`.

75. Barrier traversal on complete support

Claim identity: `SFT-PHYS-QUANTUM-TUNNELLING-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Barrier traversal occurs when complete reversible support contains phase-compatible paths through generated barrier cells even though the corresponding coarse classical observation closes those paths.

> *Quantum tunnelling is observation of phase-compatible generated barrier paths present in complete support but absent from the coarse classical projection.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`

- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__phase-compatible-barrier-path-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`phase-compatible-barrier-path-support`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:8354dd1efda50ff11c80bace128dec0d52c753e8e592b3ceff7237c5a4d3c975`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:054d47cba32d67859017762ff6bf14c2f0b544c41418c091bd0983deca5446c7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:10a111e522af07c23409f5512cbfae2bf9b26e5c9b79a06caa1a9cfc1889f632`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a4364fd900c8a3a7d2940706b9158a806370b044965df03083717e810be4dd1e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9a065d74a8c2e4777619141c741c06b2d5fc12f379b8ec06fdc2aa9170236900`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f68922fe77299126dd1219fda584ddd255588262561a49164d3d2d6baa4febb6`. Independent certificate: `sha256:6c13114affef6115891454d71a84bb6273b1ddd16c4d33f8929c0847e9f3de7e`. Engine external-validation hash: `sha256:3912d0e1650e14bb5a40a2160052126d40f35d9b5b89e9a121c52a01a24883ee`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- tunnelling-1: predicted quantum-barrier-traversal; observed quantum-barrier-traversal; exact match True
- tunnelling-2: predicted quantum-barrier-traversal; observed quantum-barrier-traversal; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:74ad6683ed699277b5bec118e07af8ca5d095679bce121b28a5991cf24d5b1b2`. Isolation certificate: `sha256:1e114351c80ee50e3895aca59986d5ba43b4c10ad0cf6572780c8c419dfb08ad`. Custody certificate: `sha256:c7debcaa62135bfbffff29279e828139b24ead3be533bd7bbf1d0044e5892784`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- tunnelling-1 from NIST-ASD-5.12
- tunnelling-2 from NIST-ASD-5.12

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:d4dcc2f2b5cbbe69ab4db94e0813a7745106099cda2bad310876b7d29526d5b1`; engine receipt `sha256:a07368612ae686daeb2371484fa843364d9da57a9c8a8ade54dc8957c5841fe3` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-TUNNELLING-001-a07368612ae686da.json`; empirical-validation hash `sha256:86d898a1e550e1d68a3c624d7d6e1994a7960621a6129f49e0112a1b7f2f6701`; measurement receipt `sha256:74ad6683ed699277b5bec118e07af8ca5d095679bce121b28a5991cf24d5b1b2`.

76. Discrete bound-state spectra

Claim identity: `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A bound physical support admits only recurrence classes whose phase/action closes exactly on the generated boundary; transition records are exact distinctions between such closed classes.

> *Bound-state spectra are the generated exact phase-recurrence classes closed on finite physical support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-TUNNELLING-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__boundary-closed-phase-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`boundary-closed-phase-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:895e71892702f582fb925e974066e9a49464f785fe2b0ebd94c167b255deead2`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every

generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:422b87a303403ab306d1efc983d71bf64e249005e7626ec235958c406b1d08c3`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6919d017b7fdd311bd79696d8ebe35689cd2ae098a3c871b3d9d95876d9084ab`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e3d63ad84f86ab21de692f6bb037b798dfe1b99d2fc9e29964caa3a00c5b7680`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a4286e85fa7e5a0c26c0671a240ca4971459ae6c3f05323ecdcbf5fd916a0fd7`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c92ea98d77b307cc0c0a2b4fad119b5a338d78ab94afcb130153cc1ae7ff6330`. Independent certificate: `sha256:a1ff6a07fb1629118374bb4fd5751dedff5fd2b41f04b407bade1fdaff80f379`. Engine external-validation hash: `sha256:2f31eb94c1b9bd95852527400d37c08126ecf50866aaaeb219ab5bb8b02f6a43`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- discrete-spectra-1: predicted discrete-quantum-spectra; observed discrete-quantum-spectra; exact match True
- discrete-spectra-2: predicted discrete-quantum-spectra; observed discrete-quantum-spectra; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:b674c815c986f459beac866858a0cc50e67908b3044aa2d8a5e68e72e13cd720`. Isolation certificate: `sha256:4923a9f29b96d66b7f725d7d78b01ef5979d4e2816a594db85234d4c8cb3bb5b`. Custody certificate: `sha256:6c8738676619e3e7adf23fb5a3f544cf6630b181f48edf89bf2afe219c936119`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- discrete-spectra-1 from NIST-ASD-5.12
- discrete-spectra-2 from NIST-ASD-5.12

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:a8581a5d686b58f9a0bfee85d11be0a91a13e3b6d1ebc5fbfb40da70d49c68f8`; engine receipt `sha256:40bcca8d05aea67f6b2812e426772bdf7b973d0410a1a24129afb79ab1650a57` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001-40bcca8d05aea67f.json`; empirical-validation hash `sha256:bbc82779ca094ae3dec89dd18b17ec8c653228a0e5072e95cf2dcd1169e8ff1c`; measurement receipt `sha256:b674c815c986f459beac866858a0cc50e67908b3044aa2d8a5e68e72e13cd720`.

77. Physical entanglement correlations

Claim identity: `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A joint physical state is entangled when its complete word support cannot factor into independently prepared constituent supports while its joint preparation trace remains held.

> *Physical entanglement is nonfactorable complete joint Fold support with common preparation provenance.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__nonfactorable-joint-word-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`nonfactorable-joint-word-support`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:89c06b1637317cb2d824dd49c5780e50123ba82a8291b611eeb302a094b10e01`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a885931cb76ddb45fb63b53ba90f23207ca655e03daa2d7193e42a798446a8c9`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b3951cec8c44b8bd6c767ca92ee5c91df5cec9d2f5b115a19ebc6c778779c0cf`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fdd02410e8f55f94f279bb3939bf2828f51933a4b9cd9ff275b0e6d36c1b8f96`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2221140dc52e9b9d8091d2b19d0e34c3f6027d8d989fc7d1c13426540cf8ba54`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:69341d35615e16f99eb1c1728d5507acfe896271668491a4a0e883f961d85069`. Independent certificate: `sha256:cb2b51584c68f7af8a20930aa1af9382548a8b30fa84bc2f501af83b1f3f3054`. Engine external-validation hash: `sha256:49486b7ca96bdf4e55459e39a8480e81e8fb55cc6dad29964eb0a29e6b40cec9`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- CERN-OPEN-DATA-DATASET-API-2026-07-23

Observed comparison records:

- entanglement-1: predicted physical-quantum-entanglement; observed physical-quantum-entanglement; exact match True
- entanglement-2: predicted physical-quantum-entanglement; observed physical-quantum-entanglement; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:738d451159d76be70e460472ad139ae20cad76767c7de8959369e12ffbd2f299`.
Isolation certificate: `sha256:5cca2cfbf4c5742d79331a3151501296e75890450fab93eed2fc8b52f98c2c2e`.
Custody certificate: `sha256:484506d3fad808fbf7ab825c09c56342ef01fc3b192a50bec882d71909afcac1`.

Registered source descriptions:

- European Organization for Nuclear Research: https://opendata.cern.ch/api/records/?type=Dataset&size=1 (sha256:4e232ef39985c02005828f97c43d903381411e7c0a762af768060343d17bd57d)

Registered target identities:

- entanglement-1 from CERN-OPEN-DATA-DATASET-API-2026-07-23
- entanglement-2 from CERN-OPEN-DATA-DATASET-API-2026-07-23

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:9c33166d4650e2b66a1daf713804f5c025a21c157730a80bee1645131ebca6f8`; engine receipt `sha256:4fe300f9b7c68fa7877536f5a6b517ad868f51f653621adf0ca0a16db177cebb` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-ENTANGLEMENT-001-4fe300f9b7c68fa7.json`; empirical-validation hash `sha256:2a22e56f177817dff8555448a08e6bdb6f9c740557a06fff6e4873f380843f6b`; measurement receipt `sha256:738d451159d76be70e460472ad139ae20cad76767c7de8959369e12ffbd2f299`.

78. Bell correlation and local-hidden-record boundary

Claim identity: `SFT-PHYS-QUANTUM-BELL-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Bell correlations arise from joint generated support in which preparation and measurement-setting records belong to the complete deterministic state; a factorized local record omitting that dependence cannot reproduce the full joint census.

> *Bell correlation is exact setting-inclusive joint support; the excluded local-hidden-record model is the incomplete factorization that omits a generated dependence.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`

- SFT-PHYS-QUANTUM-ENTANGLEMENT-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__setting-inclusive-joint-support-correlation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`setting-inclusive-joint-support-correlation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:758ea149915b6ed73f409d98a8fb8d1f4f618e42e9c5ecfecda05a31e027cf98`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b09299bc18140e5c855348082952106725ba8320a83c8cafd1ab997351ddf377`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:538825205848b5bdac60332d5d6e1737736e12e6a7bd2068badb9f674bbd2c93`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:939c55bbbbf66a6d3fd049812da4f9361ad4b43066d21432229ec2ae79449d78`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9fde24c371247082d6f4f220726cbb0fb1a12324e0645156b6a6bf934926108b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:7367a60e4c819fe426e6407cfb2dd2c0a0ac39d3ce82aec4540244a4f6914860`. Independent certificate:

`sha256:03bf661944d56afc47ced421769684b15f988ec4f3b90cd89073e1ceca635ede`. Engine external-validation hash: `sha256:9fadca435dfdb4e3059108d93f1fc5358c80359ec812367e7c2bf90c0945e344`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- CERN-OPEN-DATA-DATASET-API-2026-07-23

Observed comparison records:

- bell-1: predicted bell-joint-support-boundary; observed bell-joint-support-boundary; exact match True
- bell-2: predicted bell-joint-support-boundary; observed bell-joint-support-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:36ee32c42f1452ca3d8556227dc36fe48b04c0e2f7875de74807377bbfe4636d`. Isolation certificate: `sha256:fc7b631cebb31e370d7d74e5ee1046cfc152743fb555cbb7df6952ce1b22fad5`. Custody certificate: `sha256:294fc3d6fa9b521818de1299efb46f279048a6964320b9e90aa7a24aa81cdf98`.

Registered source descriptions:

- European Organization for Nuclear Research: https://opendata.cern.ch/api/records/?type=Dataset&size=1 (sha256:4e232ef39985c02005828f97c43d903381411e7c0a762af768060343d17bd57d)

Registered target identities:

- bell-1 from CERN-OPEN-DATA-DATASET-API-2026-07-23
- bell-2 from CERN-OPEN-DATA-DATASET-API-2026-07-23

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:8480c03a3eb947112387ce6883c18aa824b617284a2bb64c67170422b924097f`; engine receipt `sha256:2c368d4a200d2566c3c78014d6927a9a64017b2fde91972501b7cf865fb47aa7` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-BELL-001-2c368d4a200d2566.json`; empirical-validation hash `sha256:130c9d0eef669e97bdb02886a33ad59beeafb42924b6b27463c70fc42c9ea016`; measurement receipt `sha256:36ee32c42f1452ca3d8556227dc36fe48b04c0e2f7875de74807377bbfe4636d`.

79. Context-dependent joint observation

Claim identity: `SFT-PHYS-QUANTUM-CONTEXTUALITY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. An observed value is contextual when its record class depends on the complete compatible partition measured alongside it; no context-erasing assignment preserves every joint trace.

> *Quantum contextuality is dependence of an observation record on its complete compatible partition, with no universal context-erasing assignment.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-BELL-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__context-bound-observation-partition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`context-bound-observation-partition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:5f7527ff4f10d355ab298a5f09703479616714faa258ee1703d3a5948921ae41`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f66538e9706174e08adb910288533ae32057f5077a78e705b803aad5888f1a1b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0feebd50f773f44a1e5e8c31a4ad6fee0aaae9cb2b1af74b004e7899bcb4495c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:24a148617bb3fb6972e20eeaeeaa4f7130699c648ebefb76023bdb68b76a0823`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:098b5f19652af1197d4c456245a7ce41033ce299cdec9b34a3a6a35fe53d4d28`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e5437e0655b540bc5cbbb7e119227c0519503d0644c3a3e649ca6b3cca31c80`. Independent certificate: `sha256:b00a502d9248194ee83d3dd6f506e0298891aef1c17ecfbff08c39e13e7f113e`. Engine external-validation hash: `sha256:b6251d620ac893b10e3651fa00d8758ba9716ac0347d768d4fbca62e6ace2e56`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- contextuality-1: predicted quantum-contextual-observation; observed quantum-contextual-observation; exact match True
- contextuality-2: predicted quantum-contextual-observation; observed quantum-contextual-observation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:009ae275af7ed879270ba6b817d4e6db440ee62d4b6123a81bb8a026c8101bb3`. Isolation certificate: `sha256:f12380f13a0918d84a2862bdeefaf7418abe87793b793c6bdc601c6c48a83303`. Custody certificate: `sha256:ce623c10e91a407067a43df273267540c6917b73d734454445a41e42ff10d8b3`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- contextuality-1 from NIST-ASD-5.12
- contextuality-2 from NIST-ASD-5.12

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction

- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:dba339b5c3b9f5f131240bc554ec7d013eec62b16b1163a31a7c94c5291437e3`; engine receipt `sha256:8b6083f43638b7416c9639697971b3e176c77f6db2858f00b927bcf0d8942946` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-CONTEXTUALITY-001-8b6083f43638b741.json`; empirical-validation hash `sha256:51a645d0762908514a2c86f29988424b8858318ccc8ff71aa2ccfc6907325d10`; measurement receipt `sha256:009ae275af7ed879270ba6b817d4e6db440ee62d4b6123a81bb8a026c8101bb3`.

80. Decoherence through environmental record distribution

Claim identity: `SFT-PHYS-QUANTUM-DECOHERENCE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Decoherence occurs when phase-distinguishing labels are reversibly distributed into inaccessible environmental words, closing interference in the reduced observation while retaining it globally.

> *Decoherence is reversible distribution of phase records into environmental support and their closure under reduced observation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-CONTEXTUALITY-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__environmental-phase-record-distribution__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`environmental-phase-record-distribution`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:16e18e9d72bb357c0e45f9521e474ff93dd8648a2356404332dacaa261aaf80b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a3cf5ecafd15bc2a406db28ad3834d285d09fc7a8c70a3b1bdde10fab6ea314b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b0d671dbe216be2ae2b9dbfb167004711fd0b12487cf958d5d4f49741f6ac73b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b7aac10ce1b3e9e4e646b126c82c729b0878336745423529f2acdcf6ea8621c3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8ea6ddfbc48eb65b1fdd13fe12f252413f61491c7067bced5cadc06b43e64ad4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:92fe5a2bbcc2e642707032bfaed0f2db0f7f0705c2f481da74f3282d0b1ec3e9`. Independent certificate: `sha256:c00a9d191b43079e1bcc5c4a74d8d902662444b646199961a5ba04ea60a50252`. Engine external-validation hash: `sha256:e3587715eab421a9b3582588febd463ff2cf8ee725e0ceac3032e1fbb5ccc550`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- decoherence-1: predicted fold-physical-decoherence; observed fold-physical-decoherence; exact match True
- decoherence-2: predicted fold-physical-decoherence; observed fold-physical-decoherence; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:0a29356b051ea36807814c65d4ced93db31a45d9c91278e0c43a2ad417c67a10`.
Isolation certificate: `sha256:4ea72fb7a3ae7fb8881d4e9b23307e635f0a74181c0b113729ea53511f24d3ba`.
Custody certificate: `sha256:01ab93b3d8fd6ba6807073233ea08cd3f7fb72800659856c586708ab9d732aa5`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- decoherence-1 from NIST-ASD-5.12
- decoherence-2 from NIST-ASD-5.12

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:40b2ad33193d36d486b8c6fb4edae22fbc2632b36e355b165a2296ea25070939`; engine receipt `sha256:d55a9454999245b5fa0177305095d9fcd1e8c0bc6f0d75c9541ea1cfa58d8325` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-DECOHERENCE-001-d55a9454999245b5.json`; empirical-validation hash `sha256:8b313ff949f29e4e2e1c59211c96794d2a79f01ed2dfb32897411483ee92830e`; measurement receipt `sha256:0a29356b051ea36807814c65d4ced93db31a45d9c91278e0c43a2ad417c67a10`.

81. Operational quantum-to-classical physical limit

Claim identity: `SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The classical physical limit is the observation quotient in which environmental records distinguish all interfering predecessor classes relevant to the retained macroscopic variables.

> *Classical physics is the record-stabilized observation quotient of complete quantum Fold support on the declared scale.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`

- SFT-PHYS-QUANTUM-DECOHERENCE-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__environment-distinguished-classical-quotient__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`environment-distinguished-classical-quotient`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:14f71f5fff82287d9285688725670bdfd11b58231e86c98d6eaea79271206cb8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:548a2a84e76b692c08a361e060e76119136c52b19bc29e977c10e85ba345eca1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e34d504d275c746a0589c9eae6f8c894578a1f6691b08044db822366b58b22a2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bb4d0b3743489900ecc2943b7e7a02be2a24b9c4135e523ce740f2e15ed6a769`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b194e483b537990a34a2cfec1a0b01fadb9bb869aff1f526858c510ad6b1a1d7`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8c3401dc997205e7cd4e8a8ea930ddc09f7f56f8800443b67cfb2483a1333a3c`. Independent certificate:

`sha256:4585eff2f4b1d7856d596067f9882bea67f448f44cfb2848221722c3c214803f`. Engine external-validation hash: `sha256:2fa0cee18eae4479641c0766b9a98f829c23e45475af16c20e12415ec1d6fade`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- classical-limit-1: predicted operational-quantum-classical-limit; observed operational-quantum-classical-limit; exact match True
- classical-limit-2: predicted operational-quantum-classical-limit; observed operational-quantum-classical-limit; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:51666f38ecfba3745a6e04f824e24e0e2cf0c95104722ab8b5f1dccd6d2e59e5`. Isolation certificate: `sha256:cc6dcabadc29290b38300f49dc4b819220dfca95cd7ba16be977e05b592f2c28`. Custody certificate: `sha256:742d020801434c924e3d391c6f88bf836adee9e7e59bb48612aeef2e3732de4b`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- classical-limit-1 from NIST-ASD-5.12
- classical-limit-2 from NIST-ASD-5.12

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:316655c872665410113abbe9d5e347660917473f2df1466f53c8ad0fd014f64e`; engine receipt `sha256:b87f06d0e0edbee677f693e0480d07cbad931e2a07a64841f47a204726cc9020` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001-b87f06d0e0edbee6.json`; empirical-validation hash `sha256:4813ad6d27220db24bec1ea9c440157ef9a5ded7fec9cce1f48408ea7d69d5b1`; measurement receipt `sha256:51666f38ecfba3745a6e04f824e24e0e2cf0c95104722ab8b5f1dccd6d2e59e5`.

82. Entanglement and no-signalling boundary

Claim identity: `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A local observation of joint entangled support sums the complete remote fibre; changing a remote partition only relabels that fibre and cannot change the local exact support count without a causal transfer trace.

> *Entanglement correlations preserve exact local observation support under remote setting changes and therefore cannot signal without a generated causal path.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__remote-fibre-relabeling-invariant-local-count__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`remote-fibre-relabeling-invariant-local-count`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:74316a1ebb4a384bf4ea3fd4122b4f47673c690dc609d9de48750c88c8c25bbc`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated

by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:50ef85a22d59552c7f977b851d3dfe881ee9bf2e8353f75918709b57389cbe51`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3aaedc89dcc240da20f0214d2aaf19790e070bdde550e9c704f7d7f159c0c460`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7fafa2cec4d1912ec287601653ce89a461e18341a346ad5c38fe17d0bda33c0b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9070a95e7db91df8731327226a744baa43d0621b5df358e3b261a617508160c4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:80893f62bbf80046828c9dcbc41c2c93c22d00d2c6270a5dbfa60785047a03c4`. Independent certificate: `sha256:d1f318fb1e4a7bb9503e6359296cfd7b940932aa9767cb17a5eb8642efbbe15a`. Engine external-validation hash: `sha256:e7f1bb01eb155da13830bf10b3d0d0494a9aac0e343141440f3ca19d5bd4fc79`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- CERN-OPEN-DATA-DATASET-API-2026-07-23

Observed comparison records:

- no-signalling-1: predicted quantum-no-signalling; observed quantum-no-signalling; exact match True
- no-signalling-2: predicted quantum-no-signalling; observed quantum-no-signalling; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:450eab3defda05b7ad82cccc90bbe07e93564b75c6117d69cf907932c4348215`. Isolation certificate: `sha256:b206d1d5ed8ea02f5e9078fa88e2f325493da9c8572c6c08030d9d8c3c1856ef`. Custody certificate: `sha256:90256980f5bb8d51a621c5d719aa16cce1f907225e2012a6791de649fef8a3f0`.

Registered source descriptions:

- European Organization for Nuclear Research: https://opendata.cern.ch/api/records/?type=Dataset&size=1 (sha256:4e232ef39985c02005828f97c43d903381411e7c0a762af768060343d17bd57d)

Registered target identities:

- no-signalling-1 from CERN-OPEN-DATA-DATASET-API-2026-07-23
- no-signalling-2 from CERN-OPEN-DATA-DATASET-API-2026-07-23

Meaning. Physical quantum theory is the empirical correspondence of the already closed Fold quantum computation model. Support, phase, composition, observation and environmental records remain exact and deterministic at complete-state level. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:38c8e09824b751f13b2c7841ce8b49995fc78f0cb862bf570a4909621599fbe1`; engine receipt `sha256:984ce2195db3fdf46ca0370f06565f79c7f1469bdf581f7c2ba99020827cd2fc` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-NO-SIGNALLING-001-984ce2195db3fdf4.json`; empirical-validation hash `sha256:173268462d4ff03fc8b60527fcf3a464e24cda0ac2a7dc90e9c654901931c602`; measurement receipt `sha256:450eab3defda05b7ad82cccc90bbe07e93564b75c6117d69cf907932c4348215`.

15. Matter Particles Nuclei

Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed.

83. Mass-energy correspondence

Claim identity: `SFT-PHYS-MATTER-MASS-ENERGY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A retained inertial recurrence composed with two limiting-speed propagation transfers is the equivalent transferable energy carrier; conversion preserves the complete source trace.

> *Rest mass-energy is exact composition of the inertial carrier with two limiting propagation-speed carriers.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__inertial-carrier-times-limit-speed-square__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
`relation`	`inertial-carrier-times-limit-speed-square`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f1dcc64ee584cc011237e0a5238afcdb7d5581b36b3f37cc5360b07f38daaeca`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ed0bec154485f63255b26601114771f2175e5c9c127aa857b0e159ebb53d3819`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0d3b8bf23c1d3242a5b5801dad3467f0698bae611dfca54dd64fc741e69eb04e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:aa11d50ff9abd65f109a711ae145bcaf5ce16355a23aabea67ce51f63cc71a1b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:5fe31bb7a21737a2dbf16c212f31f9e68ee0f67bb7ecd1330de2ae306d6c9945`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:40cd4a67fb937a98d196b2c5a836586e9cfb46c3a74c27fe4d1692c39c27cb7e`. Independent certificate: `sha256:e977626f21f6e22faac2b47e0d2f32e48bd60458129716fff62d1771b3cd8e2e`. Engine external-validation hash: `sha256:cee15ea8bac655817aa738d2fec855214375d63b62026c25f19645a42ba8cd20`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- electron mass energy equivalent: exact predicted interval [20467764463629839209167859 51/25000000000000000000000000000000000000, 20467764476212411711483306 47/25000000000000000000000000000000000000]; external interval [40935528927/500000000000000000000000, 40935528953/500000000000000000000000]; overlap True

- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact mass/speed prediction does not overlap the released CODATA mass-energy interval or the displaced control is accepted.

Measurement receipt: `sha256:8bdb0451dcbb1ef48b8ff41f8847fd33e3892e527e0b6a7eaa3c2b62513f8a6d`. Isolation certificate: `sha256:342a9584b5be2165873a97779bbe074cb9dd04bdb169d8d9ef0335d39b9660ba`. Custody certificate: `sha256:6f8b83b11a77161dab0cbc0eed7e4edaffafd241915c40423b159f015a1d602c`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- matter-mass-energy-1 from NIST-CODATA-2022-ALL-CONSTANTS
- matter-mass-energy-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:205e2ecd215cf16dae0cb8afaecb503ddde7c5ec132379c728020e8f87c80147`; engine receipt `sha256:f812b0091bd3e7f6fac435253a1e3dcf81af7a36de5b57f25c4b641cd3fa13cd` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-MASS-ENERGY-001-f812b0091bd3e7f6.json`; empirical-validation hash `sha256:c92bb3e6d747fab7f9feccef2a19ea88774b68e10431891b9b22ca0af330677b`; measurement receipt `sha256:8bdb0451dcbb1ef48b8ff41f8847fd33e3892e527e0b6a7eaa3c2b62513f8a6d`.

84. Particle-antiparticle held orientation

Claim identity: `SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Particle and antiparticle are opposite held orientations of the same constituent word under conserved-label reversal, with identical positive inertial support.

> *Antiparticles are conserved-label orientation reversals of particle Fold words.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`

- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-MASS-ENERGY-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__conserved-label-orientation-reversal__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	conserved-label-orientation-reversal	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d0c020e5bbbb2a863a14652173767dc88149cef47ddfbf32699d56cc043b5601`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:991779226ade641f899769964c870af0f0c4de28a230876d1cd633ba35b19e5f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4eaff1ede6a4c124348e1a40c781d70d96d04dd6ba35566819589270cf3ca56d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bf21d14f456d343ce6a02b72bcccb0b561869c44afa8d3dfd6b77c4f0e0e7956`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1cd2a34c6959a22862cd410adb85dd16cf0885b2d555bcfa80ed0b66811afb0f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:66fbd5e270b64c302f3c7c195598eb54e596e438cf610d36a1a5fcde0b818096`. Independent certificate: `sha256:746cfe41e7346ca3f8fe778d89bd096037143f9bef8199d0fd4871711ef88768`. Engine external-validation hash: `sha256:861e1634065625f513cfc6fb505d7e7d77493a092f049c219057f75b4be87da7`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-particle-antiparticle-1: predicted particle-antiparticle-orientation; observed particle-antiparticle-orientation; exact match True
- matter-particle-antiparticle-2: predicted particle-antiparticle-orientation; observed particle-antiparticle-orientation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:114ac7cfe3c8afe08e4ded55d36915cd0f9c6b736def7d0c28cc69522b3df520`. Isolation certificate: `sha256:fb1def96c1f0e769619cfa260e5cd5438abc0821762523f235e569d98494c9a1`. Custody certificate: `sha256:2a2a0cc1a5d8001f326b7c04495dacd65be095c29a4b7dfee0320cceb8828c9c`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-particle-antiparticle-1 from PDG-2025-SUMMARY-TABLES
- matter-particle-antiparticle-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:893742fb4e72ad76d3698e0dc95990485d8a2282321c2fc79ee6fc6528535dac`; engine receipt `sha256:4840b7076ad83c2c76882af2e81a8309d6ee80028e61796c2f91a896fea2ebfd` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001-4840b7076ad83c2c.json`; empirical-validation hash sha256:21fadf4dd9de421fc4ef535a9d2f551fb9dfd5d59a0365c85c79cff5d3f8dd1b; measurement receipt sha256:114ac7cfe3c8afe08e4ded55d36915cd0f9c6b736def7d0c28cc69522b3df520.

85. Fermion and boson composition classes

Claim identity: SFT-PHYS-MATTER-FERMION-BOSON-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Constituents divide into alternating and preserving exchange-composition classes according to their finite cyclic spin action; the former exclude duplicate one-cell words and the latter admit them.

> *Fermion and boson are the alternating and preserving exchange classes of physical Fold composition.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__spin-exchange-composition-classes__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	spin-exchange-composition-classes	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:ebb1423777328bdf7e4eace13e90f246ffbcd726fff5156fd72f7b219994860c. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:156fcb775b2aff5eca6d29287f58eeab4ab91b456fb3d64093f8169f35b9f222`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:39c0be052afd07273ef0ad9a013e145adfab5e66c08807f1c72adc015286643f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6213645d9d40a9519d4cc65ec8a058d98c3f73b8e70cabf0db75f458b8c8b1b1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9f77efc02091736508973db62a1945e1adc476f5486ec713a061d46cb177c231`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:347ed8537673fef317ddf0f5ab470dfb67c15efb12d37358fc4db2d8fc5e4773`. Independent certificate: `sha256:9de3b2995bd56763940473afbf734a3bf257024a3d7614121e2bbdfc890fa227`. Engine external-validation hash: `sha256:1206b2c448c117ea0104acef16c31c9400275abc30c3beddc7a1334de5825a91`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-fermion-boson-1: predicted fermion-boson-exchange-classes; observed fermion-boson-exchange-classes; exact match True
- matter-fermion-boson-2: predicted fermion-boson-exchange-classes; observed fermion-boson-exchange-classes; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:b62d84281d62a3bd43e014f64e0fc119c7173a712656a2e1c54a5305b7a089e4`. Isolation certificate: `sha256:bdcf6d556e0634792d59829d64ea70f32749a0eebe036193b94775b8e9cb26ee`. Custody certificate: `sha256:47ff6011a01dfff2f7d7726254ec7a0d7dd02c9013e874699e4fe314b292c58c`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-fermion-boson-1 from PDG-2025-SUMMARY-TABLES
- matter-fermion-boson-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:84e502c03f5b1e7f8ffdd767e97ebba9c71a066725b2500625ed3fa8817213c9`; engine receipt `sha256:8e8738ba7238b2b8216d04aec21ebeb1d15ab6ce2e919b3b265f5a65d50c3951` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-FERMION-BOSON-001-8e8738ba7238b2b8.json`; empirical-validation hash `sha256:a27494b99485e534973865ca0648ff11ae6cce9e9cc3f0ec7bedabf42782cba3`; measurement receipt `sha256:b62d84281d62a3bd43e014f64e0fc119c7173a712656a2e1c54a5305b7a089e4`.

86. Conserved physical labels

Claim identity: `SFT-PHYS-MATTER-CONSERVED-LABELS-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A material label is conserved when every complete generated transition pairs its disappearance on an input word with its appearance on an output word or named boundary carrier.

> *Conserved particle labels are exact held-label flow invariants of the complete transition grammar.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-FERMION-BOSON-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__transitionwise-label-flow-closure__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	transitionwise-label-flow-closure	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3250981cf6d63ab0da7638f97c224fc22a0c3276f9655b03f6ffdecae9314435`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:baf66a25d81ccf81aaa130e0732edf6d70758e68cac88ca87f1b06e3beaa1602`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6f149ce11d6661b1bd2c56ef88117bd66516923c70eba8e0377be09be835918d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b3c1dddcc5408913462dfdbe70490ee0d8e0144e9f670c69d992958154620a30`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6f479d740b225e7286d71a8a4d28f81e285bb80433f72f011ad1cb78006a0e72`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:7ebe6010d8a85c0fb10f770a73d7ca4fbd8db55ac00f6cb454d8fca33966cce2`. Independent certificate: `sha256:09fbb8d4b6458db759a9b634a65db73ce94ab658a72b96bdddbd9923c5621465`. Engine external-validation hash: `sha256:fd5e7a453ae51737b1b9750c5a3062a9700c0e108971f0cbe639a64c78c17863`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-conserved-labels-1: predicted particle-label-conservation; observed particle-label-conservation; exact match True
- matter-conserved-labels-2: predicted particle-label-conservation; observed particle-label-conservation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:a37fbf21c2c1532ceb7477f56a5acd1e4ce9cb6c5588899a05175aec3ac4e588`. Isolation certificate: `sha256:fd1debefc10a612117b4a30ddb4ba04b1a4ad115b85018c2a0f4f84bc46dd08c`. Custody certificate: `sha256:ebdba44b843f9833a0db8b9ee44a89464fc6aa1cbcba27a787ebab8368fb0ec1`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-conserved-labels-1 from PDG-2025-SUMMARY-TABLES
- matter-conserved-labels-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:73ef40384dac7f3c76495576bf9f808e57c5fdc63c81a989c914d82c090bc2d1`; engine receipt `sha256:83772ecab34a176c0e87265993183d8c445be8f49ec0d86e615c9d0c1740e945` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-CONSERVED-LABELS-001-83772ecab34a176c.json`; empirical-validation hash `sha256:b44d6261da8de14792040b7131fda6cf2219d91d5632dfc2b4b3ca60fd0ba8c0`; measurement receipt `sha256:a37fbf21c2c1532ceb7477f56a5acd1e4ce9cb6c5588899a05175aec3ac4e588`.

87. Observed elementary-particle spectrum

Claim identity: `SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Elementary particle kinds are the non-equivalent minimal recurrent constituent words under complete observable transition traces at the registered measurement boundary.

> *The observed elementary-particle spectrum is the finite census of minimal non-equivalent recurrent Fold trace classes at the PDG boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`

- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__minimal-recurrent-trace-equivalence-classes__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`minimal-recurrent-trace-equivalence-classes`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:30471ffa0c2b9f94dc70357fa22a67e4d1ff45623cb94929b781ac28f146f923`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1111b793deef38ccd05162c2a85883eef2417a3e7762c5c86e6d8f3d38c03cf4`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ddcc358eb534bdd1f240bc3694532fb8db0b67e6281b95bc4c3c6eaeb0c6e5db`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:512ad96f068b575ad0544dd3c8797dd7f8893144162266f9d2f3d8abb440f5c1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c110db659ea14ee96663e2ed3e8c5dbcdcf57d20d7a4d67e624d15c7cd6be233`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5b513a50d02c9708abf487a28feebbe002b36f0f797c14df1948fb0be98f34af`.

Independent certificate:
`sha256:f5851afd8e7790d039c120b8a89d75a1b5f34eb15ec0422ad89d16aa202a775d`. Engine external-validation hash:
`sha256:d10ff651bcccdbc2894bfb2cbcf8e09a7eebaba0885a3ccb0d008c36cac0d491`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-particle-spectrum-1: predicted observed-particle-trace-spectrum; observed observed-particle-trace-spectrum; exact match True
- matter-particle-spectrum-2: predicted observed-particle-trace-spectrum; observed observed-particle-trace-spectrum; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:c6c667a66b77d96b117f8e97fafdf3cb9c40c0423908f5f9c6b8ea5f27fa763f`. Isolation certificate: `sha256:8487cf41ea71e8a870132223c040819a2b4e46c48e13136b029d6f2272a0728f`. Custody certificate: `sha256:3a1e08e0415632f358fe4ba66ee3fce7d967e81d128d2a5c7029a51f299707c1`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-particle-spectrum-1 from PDG-2025-SUMMARY-TABLES
- matter-particle-spectrum-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
`sha256:17408d0e7ea308694f773c1adfbd532ab0a6f5304c8e9088f7eff63cdddecfca`; engine receipt
`sha256:a5d2f4b06e01f8d5acabb36577883d2ad220e3dd5aa6aa2e170b52ff5273fff0` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001-a5d2f4b06e01f8d5.json`; empirical-validation hash
`sha256:72e141aef98afe3d189fe1fd563e8f46123cbe94f14447d62931d82925f53772`; measurement receipt
`sha256:c6c667a66b77d96b117f8e97fafdf3cb9c40c0423908f5f9c6b8ea5f27fa763f`.

88. Composite hadron organization

Claim identity: `SFT-PHYS-MATTER-COMPOSITE-HADRONS-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A hadron is a closed joint constituent word whose internal interaction labels are not separately observable outside its boundary while total conserved labels and inertial support remain observable.

> *Hadrons are boundary-closed composite Fold words with confined internal labels and observable total carriers.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__confined-joint-word-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`confined-joint-word-composition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:7d15922136da75a509275a76b6cef216c7ac8a155b8833094706a0558283d24b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated

by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:794134875ca65cc8d22045a544dc4a94357c2b1cc7965734eaedfae649186334`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8fc2f5f15a7a26710f8f241bc15eaba1833240845edaf6b7a334130a1450ad09`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d5ce26b1e74bebc5ff35a429fd4f17b9b392d39ed94a7a0bfc31b39b17b64f85`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fc3f7576bde48ef11fbe9449ae4006d0b914605fa8f399aa5036a7ab3829ed40`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0bfc340f875b4d4ee94d7602b2670fe3e8c1a1b8dfd0814b98bdb78bbc2b9876`. Independent certificate: `sha256:22d961672ae7e26b6566dada76865b9771a018c67e77c2c2299f96c1a96be5fb`. Engine external-validation hash: `sha256:bfe345229b0d776a8998d0458b0f1c52795ecdb8d76ec08edd612f2ee907c1f7`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-composite-hadrons-1: predicted composite-hadron-closure; observed composite-hadron-closure; exact match True
- matter-composite-hadrons-2: predicted composite-hadron-closure; observed composite-hadron-closure; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:84b6999676d9e3786fb32a7bdca5efdc93ca0d9596b75fbfeae52b80a0bb3bb6`. Isolation certificate: `sha256:e487257b9dfa92658e4d73467970da58fc7f9d08e27ce42049a23cb593ef567a`. Custody certificate: `sha256:60521bbad490f550ac641aba82b204003ff0f812c605a61197607ca7070d9581`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-composite-hadrons-1 from PDG-2025-SUMMARY-TABLES
- matter-composite-hadrons-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:98f3c1986b69b272fabf8a95f88c72cb8ecfb48edd659776bd5c16fa73fe7d74`; engine receipt `sha256:f99b0600faea8923a9b335972b511b488ad0a252566da40911ad5fd2d9136697` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-COMPOSITE-HADRONS-001-f99b0600faea8923.json`; empirical-validation hash `sha256:02b815af214cb37ac677f126abf0c880ed63cebda6397dc7704230c5fb42b682`; measurement receipt `sha256:84b6999676d9e3786fb32a7bdca5efdc93ca0d9596b75fbfeae52b80a0bb3bb6`.

89. Scattering and cross-section measure

Claim identity: `SFT-PHYS-MATTER-SCATTERING-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Scattering is the complete map from prepared incoming joint words to distinguishable outgoing channel words; cross section is the exact positive successful-channel count relative to registered incident support and geometry.

> *Scattering is a complete transition-channel map and cross section is its exact source-normalized observed support ratio.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-COMPOSITE-HADRONS-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__incoming-to-outgoing-channel-census__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`incoming-to-outgoing-channel-census`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:bde4ec9cbc784e2c29a227b2901c09e7461898de7cecaf6aeba557b173d80f10`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5c420945a230b02e98d4dfa2a708774bb8984f0abebc9d5c98f193adaf32def3`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:999d57cb70e017713c5af6e41f6f7e0459044c050f9ddd56eddaba7cdb1220b6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:096cbfc59e330a05b88b0554bfa19788f10aa302e7d08849e159857d7ac7e063`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:61ca7df4431fdfa7b0090df0baab5bf4c030853e341a00f3fcb1d36be9f00886`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:80d091c6fc94b6f1aced9db1f5b7b7d92be9b0ffb9cea55c5347a8d0dd789c84`. Independent certificate: `sha256:3f803d9f31e1300c8f38daf92f03201cfbb1a1ef21d0f0565a17db3cc385108d`. Engine external-validation hash: `sha256:1aa9fb8df8af6ae11e239bfd8544ce8b8cb82d5a6e3539c4c6cb51acae494b2a`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- CERN-OPEN-DATA-DATASET-API-2026-07-23

Observed comparison records:

- matter-scattering-1: predicted scattering-channel-measure; observed scattering-channel-measure; exact match True
- matter-scattering-2: predicted scattering-channel-measure; observed scattering-channel-measure; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:9fce4306e8bd1dc729a61f34e3166eb388153ff8bd3fda93842ffd1258dd5f6e`. Isolation certificate: `sha256:9fb6b4e31148bb7309035fce9e6ad14a1c21ddd52d7e086bcce6b31d4d220ef1`. Custody certificate: `sha256:25e982819e4e752fe6991214a1c164a26c8d34058ad2adc25dba38093cabae8b`.

Registered source descriptions:

- European Organization for Nuclear Research: https://opendata.cern.ch/api/records/?type=Dataset&size=1 (sha256:4e232ef39985c02005828f97c43d903381411e7c0a762af768060343d17bd57d)

Registered target identities:

- matter-scattering-1 from CERN-OPEN-DATA-DATASET-API-2026-07-23
- matter-scattering-2 from CERN-OPEN-DATA-DATASET-API-2026-07-23

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:cfd674c28ced7dfb9302c3ef44c582c57bf8fc35d77a6525e9f426c250072bb1`; engine receipt `sha256:01fb73df0849a188dfd668d8f8549cd1b64ed201e5fe706bcaa391967ab61f0f` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-SCATTERING-001-01fb73df0849a188.json`; empirical-validation hash `sha256:910bbc6b8cb2daca6e00dc9b9792851b70d7126484b3fa78a522d2928594fd60`; measurement receipt `sha256:9fce4306e8bd1dc729a61f34e3166eb388153ff8bd3fda93842ffd1258dd5f6e`.

90. Particle transition and decay

Claim identity: `SFT-PHYS-MATTER-DECAY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Decay is a recurrent constituent word transitioning into a lower-recurrence composite of output words while every conserved held label and total energy-momentum carrier is paired.

> *Particle decay is exact conserved transition from one recurrent word to a generated output composition.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-SCATTERING-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-constituent-transition-channel__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`closed-constituent-transition-channel`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:7c3d62e4e9f23f4c61a96a2f4a3c7fc7179dd71964b6da4dc7ba44e1b86d4f7f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:794db5bad533dc878ce7f5091dd8534c2f9b9037ae26c75f69e0107f804a4582`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c7a9c9cbc99ddf70f9a58b2de21043f733beb858f4329a69b6178a07b623d150`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a924a3d1c129b90a5c746a6d723c15627d3abe0de4c5e58cbb5c0e489706ef33`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d6c967315e23cdb063fdc12735b33154fa9cf0db3bcc1c548c6e25c3b7f57343`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:809d2705df4cf4ab7132c2df98d03b54a449771aa0de6420c9061b0900e27e3a`. Independent certificate: `sha256:7cc8dd2f74fdd1f9b4bce38797fef768190f35f07dc57353e11c917977d4847d`. Engine external-validation hash:

`sha256:14bb288aac3e051b88c97a368e17c106739741086958601638bdfe6689f262df`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-decay-1: predicted particle-decay-transition; observed particle-decay-transition; exact match True
- matter-decay-2: predicted particle-decay-transition; observed particle-decay-transition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:e38dcce46ece40beeadc86694d24f4621e7b92a5886b414d250b085528428231`. Isolation certificate: `sha256:92af8eeef129f8f13e711bb0fd2216e05afaa7ed77ad3938c508bff22e997ee0`. Custody certificate: `sha256:9b426857927415273d745cd432c067b61e5dfbcdffe6cbb352fc1a9ede77c1d7`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-decay-1 from PDG-2025-SUMMARY-TABLES
- matter-decay-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:c01f0e0c1c08c62ff3c0f887f1b65d8a94002e2e887cd018a30b2700ffd4c9ac`; engine receipt `sha256:67296b745325db1768a99db563e6a6966b0201b9fb107133746aaac222ae8291` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-DECAY-001-67296b745325db17.json`; empirical-validation hash `sha256:7081680de252cc8971336c5ab0e2296c5092c49e1dc9d4bba6cd0eaf4663fe3e`; measurement receipt `sha256:e38dcce46ece40beeadc86694d24f4621e7b92a5886b414d250b085528428231`.

91. State mixing and oscillation

Claim identity: `SFT-PHYS-MATTER-MIXING-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Mixing occurs when preparation and interaction observation partitions select different refinements of the same finite recurrent support; cyclic phase action then changes their exact overlap with recurrence count.

> *Particle mixing is cyclic recurrence of one finite support viewed through nonaligned preparation and interaction partitions.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-DECAY-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__nonaligned-partitions-with-cyclic-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`nonaligned-partitions-with-cyclic-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:146a2aa9a27a200c26d676c30ca782b2f8f302898f0512e0fca2ec46a910d9b5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8aad8060dbde458070e15cd15fa55313d0a0fb1efb7f79e6dab95208c93c6834`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9af76d3b28e78224c81f28b597e1a3b05c573059bf8de8f128315deb08295b87`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:763ea04baba4dffedd25ed238304c89f82e5f9d97a3ebf05d1e6842ed4f40dd8`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:65352a8468399e495fccb9ef4943f1e3c475b298ca8954d56f4a9c9d1567f479`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e2076b010706a7bcf0e70455f5f7701bc2b71fb07a36cabab6a08dab7e7d88f8`. Independent certificate: `sha256:c44e862daedd1331de9f2b39221975753d321a594023e4c0872c2e30a9205e8b`. Engine external-validation hash: `sha256:26022d6f191095471ef38e32bd069ab5a8b3a1ec98310673fc3841d0f5bb6ce1`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-mixing-1: predicted particle-state-mixing; observed particle-state-mixing; exact match True
- matter-mixing-2: predicted particle-state-mixing; observed particle-state-mixing; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:29c2514184ac55bea194bf4dd3a54c68dc30cb837dfc48626be37458aff7e7c4`. Isolation certificate: `sha256:6d7e34542666520a3ef0f24a90082fc7b5389d7e61f0eba3a7ade7abcdc20afa`. Custody certificate: `sha256:f8c15bb32625847abee7f91d00e1d52865182049cc4eda5ccef8defb4525de6d`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-mixing-1 from PDG-2025-SUMMARY-TABLES
- matter-mixing-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude

- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:1d735f19a4e97bfe8341691b055e6b7274fc5380833ef335cd4e314677047ee4`; engine receipt `sha256:6e42de870a32d7dd47c1d24d1410009498854662fc9b7b85ff2aae24642a3e2b` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-MIXING-001-6e42de870a32d7dd.json`; empirical-validation hash `sha256:462921308b12d9ad2785f45aac4e115da26e8e33f0cbad99fd658cc2809adb4e`; measurement receipt `sha256:29c2514184ac55bea194bf4dd3a54c68dc30cb837dfc48626be37458aff7e7c4`.

92. Nuclear binding and mass relation

Claim identity: `SFT-PHYS-NUCLEAR-BINDING-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Nuclear binding is the closed joint-word recurrence whose separated constituent inertial support exceeds the bound-word inertial support by exactly the energy released into named carriers during composition.

> *Nuclear binding pairs a positive released energy carrier with the exact reduction of separated constituent inertial support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-MIXING-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__composition-energy-inertial-support-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`composition-energy-inertial-support-pairing`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2be5949b1f34c8caf6ffecc99ee36229ff77bf51c669050b08f3c25282728b4f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:fd81a904c6c352b69b65f42e7480ce913ebcfba32e8ecf6697542de0587ecfe5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7319c9d7e19e03ea80eda886ea4656be56455b7390a319a4ce06e0f54790b57c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:195fca22a5ce2ebd1b964d027360a7ce9191601530fb05ea2c1b697f7ff04d37`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f357cf87333e729a5c24cb2024a070e081048c02ff807adc26935fa8e3e998df`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:13230a01de73f6c8dc99c99cc990cfe58483ed3d9c24013610fc2d941f3b834a`. Independent certificate: `sha256:ab40eb1ab93ae74c3d7c9804d11fdf239d29147c5926a59521295a02f0545202`. Engine external-validation hash: `sha256:f48f7d60b5964811413e74536897fc7e7553aac2fa6f8bcc1fc23d279f5207a8`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-ENSDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-binding-1: predicted nuclear-binding-mass-relation; observed nuclear-binding-mass-relation; exact match True
- nuclear-binding-2: predicted nuclear-binding-mass-relation; observed nuclear-binding-mass-relation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:d7cbe20b60e87671320f3a1d90549b701b9a751b5a00c01ad1bacf84a19ac624`.
Isolation certificate: `sha256:9bad8064d730e0acaa879175458f47f30b30e9a8b1485b58ddea5705666cf26d`.
Custody certificate: `sha256:7cceb30b648541c1045c1c6100a4bdc77259e601d2794df5dfadbba0a48d7ec8`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://nds.iaea.org/ensdf/ (sha256:c17d03b30d117a48b776fd143a2e4224bd4f53bc774d394c1343645329a0ce93)

Registered target identities:

- nuclear-binding-1 from IAEA-ENSDF-CURRENT-2026-07-23
- nuclear-binding-2 from IAEA-ENSDF-CURRENT-2026-07-23

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:2db6a24ffdb2047e02bea855ac44381848b991bd43fbb9e3ad460b5675c53552`; engine receipt `sha256:8ebc11747ec4bfd0552de732d26bbb60bcbbeb6ba273885ebec783ffd43dbb3f` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-BINDING-001-8ebc11747ec4bfd0.json`; empirical-validation hash `sha256:7f9bc2dd84794e9d4709ca05dc1e74efe1515768885a50211800d6daab707073`; measurement receipt `sha256:d7cbe20b60e87671320f3a1d90549b701b9a751b5a00c01ad1bacf84a19ac624`.

93. Nuclear levels and transitions

Claim identity: `SFT-PHYS-NUCLEAR-LEVELS-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Nuclear levels are the exact boundary-closed recurrence classes of a composite nuclear word; transitions are conserved carrier differences between named classes.

> *Nuclear levels are discrete recurrence classes of bound composite Fold support with source-bound transitions.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-NUCLEAR-BINDING-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__composite-boundary-recurrence-classes__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`composite-boundary-recurrence-classes`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:76f9d0a0965161c735fd68ccc9c32b70f95af9a608797eed5a7337de433bc2b3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:327a71cc8de73d2a1c78a174dc6970825c5522d443715a42c4395d6b1d8af7e7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:653c74bcd47a95a1311142e94053c696505967b25fd90664363116fbecdc26e4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:300df7b3eb317cfdfeb054c30f8bf07707fb63158e6be614b57d9e13805e0c2e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ebfadd76b8db50a577d6da9f56e084e6260e6f00893d29aa7afe61517b56a497`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:57005aa5b4ea43540c6c5c2b6c90a9d03f7ccabdf65f58e8b37041d3847fa204`. Independent certificate: `sha256:bba43a62fb1fccfdb86973be39bb8b5fca71241a3ca036d137ac74bf796514b2`. Engine external-validation hash:

`sha256:0fb69d726eb3ce0606c271f0b73ed48ed240477c14bea40d08d4f2058d398d08`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-ENSDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-levels-1: predicted nuclear-level-recurrence; observed nuclear-level-recurrence; exact match True
- nuclear-levels-2: predicted nuclear-level-recurrence; observed nuclear-level-recurrence; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:6a1a8e1613eb6cac74c66843c4553f3e5416c30fb09ab78eb6eb000b90ad0961`. Isolation certificate: `sha256:12e041dddac9d4f3dc13bdd464cf9365bc517bb906960253ecf9ec780a9f1d30`. Custody certificate: `sha256:5ff4164d03023279c6566e0888ace4022d1402e23245a54186231f3ab68fec70`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://nds.iaea.org/ensdf/ (sha256:c17d03b30d117a48b776fd143a2e4224bd4f53bc774d394c1343645329a0ce93)

Registered target identities:

- nuclear-levels-1 from IAEA-ENSDF-CURRENT-2026-07-23
- nuclear-levels-2 from IAEA-ENSDF-CURRENT-2026-07-23

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:37409d7dc276088da25835592fb355d85cbd857d5586f3161e147360d3e5c1b1`; engine receipt `sha256:8361256cd14917c077f36a6250d7706d6b97eaf4254633721107775112a94479` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-LEVELS-001-8361256cd14917c0.json`; empirical-validation hash `sha256:cba148bdbd2aedc9548d29c9b027d0467962567fab73bdbeaa016c59037b48fa`; measurement receipt `sha256:6a1a8e1613eb6cac74c66843c4553f3e5416c30fb09ab78eb6eb000b90ad0961`.

94. Radioactive transition support

Claim identity: `SFT-PHYS-NUCLEAR-RADIOACTIVITY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Radioactivity is a nuclear recurrence with lawful output channels but no stable self-return on the observed support; exact event weights arise from complete deterministic hidden-path census.

> *Radioactivity is deterministic complete-path transition from a nuclear word lacking stable observed recurrence.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-NUCLEAR-LEVELS-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__nonreturning-nuclear-transition-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`nonreturning-nuclear-transition-support`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3baaa11050ac054e4bc27061f6acafb7c8b1d20ae9a06ab2044e93c38fa43780`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:7e9ccd542cf7e94950e6e4aa1e519f1337c9368433868c2ca3703d920a9562e4.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:83f0826fdd6d0e6db144c50f30bdbcc9be7c25e0aa72c829226abc11ca3e6314`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:404a845a88efd25dbe794b73e078814ad4636aeff85114ecb1a00b642a80963f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:eaa65ad66b7acc1ad1ded6aa28a4ca7d3f4bda1c0265fe92bc6cadcfc21bfad7`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8bc09f9171ba803b1633192af125e6695a777ea38bf18ad5d9da6015d11f0924`. Independent certificate: `sha256:c8c4cf90ccf96270f8e98d74e35afb66cdb835387dfa783b2f8778240884f741`. Engine external-validation hash: `sha256:149cb62c5efcf8332acc63666d7cb9b7bb365f75008af5ffab5480b974373286`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-ENSDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-radioactivity-1: predicted radioactive-transition-support; observed radioactive-transition-support; exact match True
- nuclear-radioactivity-2: predicted radioactive-transition-support; observed radioactive-transition-support; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:2fb40b4979572faebf3f41b84ffe8676f1e7659c4d9a3524d72f8800d851e569`. Isolation certificate: `sha256:589b8da919a2ee61eeac0d08972ca0ff258eeef6a398dc2066a6d0252bb374e8`. Custody certificate: `sha256:fb23907b939fece460b56cf9a2907665b35c092380d6275aa6c3d44a69646668`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://nds.iaea.org/ensdf/ (sha256:c17d03b30d117a48b776fd143a2e4224bd4f53bc774d394c1343645329a0ce93)

Registered target identities:

- nuclear-radioactivity-1 from IAEA-ENSDF-CURRENT-2026-07-23
- nuclear-radioactivity-2 from IAEA-ENSDF-CURRENT-2026-07-23

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:273f3a543b43ca1522707cc521dc8ca7b294b36cffd379a51963940262ca28f0`; engine receipt `sha256:abfd4ba82c2ecf7fba7473fa029c2b3c06ad54d536c4fef8a7b354fbdb587576` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-RADIOACTIVITY-001-abfd4ba82c2ecf7f.json`; empirical-validation hash `sha256:2eb2570415ea91beb3310bbccedcaeb644075e14b7a193c71a9ebf7c57c9724c`; measurement receipt `sha256:2fb40b4979572faebf3f41b84ffe8676f1e7659c4d9a3524d72f8800d851e569`.

95. Nuclear reaction and channel accounting

Claim identity: `SFT-PHYS-NUCLEAR-REACTIONS-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A nuclear reaction is a complete map between incident and outgoing composite words in which every material, energy, momentum and held conserved label is paired across the boundary.

> *Nuclear reactions are exact closed maps across complete generated incident and product channels.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-NUCLEAR-RADIOACTIVITY-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__complete-nuclear-channel-accounting__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`complete-nuclear-channel-accounting`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:c4ca1115e7810dd0426286893e7508475510d5861221f6eb392b3dbfe9d0dace`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:65887188ea7c159786c06b3c69f0295b3bb31780ee1b2a068f1b91429f2cd077`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:51f7f0eac3a58bed76c33fa2c48b1790fd7134b7f80ccdb63877b6c421e655bd`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3201fac045e03496efadcc814789850c6a1dec7f2e47278b0ebfaef3ed7decb1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6bf41deed8cdec0fd0fde987f16271b96d3490911393a5366d923a4f6e52828a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5dae5b17cd7967171984f76fe3a5fc7931906d98c4d43ef699ca0bdc2213f099`. Independent certificate: `sha256:0222736e5d94d5fd9354aac097ea8eb3337b66bf92a6858f908a9e7918252573`. Engine external-validation hash: `sha256:4ce0e55b5061edd580437fa059d871c66d9c40475dc0f2d9633b0244767fbe6a`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-ENDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-reactions-1: predicted nuclear-reaction-channel-closure; observed nuclear-reaction-channel-closure; exact match True
- nuclear-reactions-2: predicted nuclear-reaction-channel-closure; observed nuclear-reaction-channel-closure; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:90df60f2143b04166e63a04b6073725d94456b6a4f9bbf2b614efdd72a911793`. Isolation certificate: `sha256:8f9cd6cfb1475a8d1976125e63a314024123bfe2366d61e368899143faf36c1f`. Custody certificate: `sha256:94a227c1a68856fbce05c9c808c686c569510cc4a6bdefd2f2d356d49eafb027`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://www-nds.iaea.org/endf/ (sha256:16b8187ef7ef9024914aa7b54b6af07928a2f1d3bb97a6999f7563fa676ccbd9)

Registered target identities:

- nuclear-reactions-1 from IAEA-ENDF-CURRENT-2026-07-23
- nuclear-reactions-2 from IAEA-ENDF-CURRENT-2026-07-23

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:39b1e4d4ec2f64e601b1c6724c2838f65797441cb1df32d4406b2c37bac7b8ac`; engine receipt `sha256:d619676a96d0ef76e876a31388e3ead4e581cf513277f87928f75ac490b62127` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-REACTIONS-001-d619676a96d0ef76.json`; empirical-validation hash `sha256:0d3686a4a22399ad626810c76085b19c7b8ac64edc30664bdece602a19e3b55c`; measurement receipt `sha256:90df60f2143b04166e63a04b6073725d94456b6a4f9bbf2b614efdd72a911793`.

96. Fission decomposition

Claim identity: `SFT-PHYS-NUCLEAR-FISSION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Fission is a nuclear transition whose output support contains multiple separately recurrent composite words and named released carriers, with all labels conserved across the decomposition.

> *Fission is conserved decomposition of one bound nuclear Fold word into multiple recurrent product words and released carriers.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-NUCLEAR-REACTIONS-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__bound-word-multiple-product-decomposition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`bound-word-multiple-product-decomposition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2fe7f5f41c727546bdfe8ccaa4acaca5e3fb753d05a8d8888fb8643a34a7ecb8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8945ba3b665b979784ca31d3ea3de9e420d5ec2ca8c234a5719fbb9bf9e64de5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b69d6af60d92b523ec3246adfb4b1039b12979d2e4b73f1128080af26d5d2008`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:446277cc78578caf8243313372fa8da0542b513e37fca900760b935dde0d3165`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1a1d0d29304c8b346c5eea4f464658c45618a20cb1cc5135b55cecdd87bb657c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ac12c4c5196542894a2ab1002ece360d2c5ad971332c46e66f77d4be44587307`. Independent certificate: `sha256:c378af85ed65ad3e447c5497f76e416380c3d70bfcb24eeb663135f589afa458`. Engine external-validation hash: `sha256:1b6ab3af2e44aba9eeb8cb9a2103eed55ca96ec0dfc141befdf42faf90ad602b`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-ENDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-fission-1: predicted nuclear-fission-decomposition; observed nuclear-fission-decomposition; exact match True
- nuclear-fission-2: predicted nuclear-fission-decomposition; observed nuclear-fission-decomposition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:08869f2c7c3b92aebb67f0c94c0e50f8271c407f7b088280ae658630c7b2533b`. Isolation certificate: `sha256:28d6f24280e39b7e18069a8103b9291716c1b6a2d12b096c3d47bb7a67d92a05`. Custody certificate: `sha256:3e771ca4008a532dadf4c32962e122088da4d99bb2a0331fa0e1112ddbe23dbb`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://www-nds.iaea.org/endf/ (sha256:16b8187ef7ef9024914aa7b54b6af07928a2f1d3bb97a6999f7563fa676ccbd9)

Registered target identities:

- nuclear-fission-1 from IAEA-ENDF-CURRENT-2026-07-23
- nuclear-fission-2 from IAEA-ENDF-CURRENT-2026-07-23

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:de201d487f30f1538d3129d5586fc1357329f6bb4b7cc60e1495711119d9246f`; engine receipt `sha256:37bc1d64923a33e5d6548f4701606b8566eb51f9d6871e6e7815a289926a3ceb` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-FISSION-001-37bc1d64923a33e5.json`; empirical-validation hash `sha256:3c4f86c17f05f95b240fa9ef4de61ca95114514b336de96606dc54275857ecba`; measurement receipt `sha256:08869f2c7c3b92aebb67f0c94c0e50f8271c407f7b088280ae658630c7b2533b`.

97. Fusion composition

Claim identity: `SFT-PHYS-NUCLEAR-FUSION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Fusion is a transition from multiple incident nuclear words to one more tightly boundary-closed recurrent product word plus the exact positive carriers released by the mass-energy relation.

> *Fusion is conserved composition of multiple nuclear Fold words into a bound recurrent product and exact released carriers.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-NUCLEAR-FISSION-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__multiple-word-bound-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`multiple-word-bound-composition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4614d77c25311196cc1ebc5f0ce5f47897195b3c0a060fcf755c59ba113efd96`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:dcf009d6b2b76f5c957ff2982be5c3854b417e1d327c14c98dfb98c4b66ee982`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1a789c350ea0afb3aee5c9f53654677f9a627b2f0c989b3a34eee08369dca1d0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:de889777d2509a339723f1f13e24af5fb765b008fb431795f7f6e06dc10b9161`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8f1439a35ff8a600cf9fde87e3351fedad3eacc9488c343d570f0fc1d2654c16`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:290be909c8c1da1aeb994b4ed9541ada13c006702379e79ed7c326f7032ce4dc`. Independent certificate: `sha256:aaa7e043eec1826489229193189f1c25aad22db59b2635e678317e3c9a58f54e`. Engine external-validation hash: `sha256:d6379dacb6640551cccc3dcc2e0e5b0cdf128b5c7bf0258a2ba5c560f68fcde8`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-ENDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-fusion-1: predicted nuclear-fusion-composition; observed nuclear-fusion-composition; exact match True
- nuclear-fusion-2: predicted nuclear-fusion-composition; observed nuclear-fusion-composition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:230a7c0b1b00ce89a6ae991f6ea31d37886fadf99dfe45272a755e2951f1e273`. Isolation certificate: `sha256:c1c5003ad034af463d8b70bdbc9b006d5080f77742f1232c852b1cf2b77b7658`. Custody certificate: `sha256:154c0d4014c7ab3ce2562efd98ea5aab06ca34f1a96e177f59f0efe86486b681`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://www-nds.iaea.org/endf/ (sha256:16b8187ef7ef9024914aa7b54b6af07928a2f1d3bb97a6999f7563fa676ccbd9)

Registered target identities:

- nuclear-fusion-1 from IAEA-ENDF-CURRENT-2026-07-23
- nuclear-fusion-2 from IAEA-ENDF-CURRENT-2026-07-23

Meaning. Matter is classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. Empirical tables determine observed membership only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:9f4db1843099be37632c5ceb2509b9c49a8b486d75fe5aa23f44b63887db43e2`; engine receipt `sha256:9277671be548ed512279ff99a5347ed91531056cecddc1f22ed96da39a0a7676` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-FUSION-001-9277671be548ed51.json`; empirical-validation hash `sha256:849afd978cbb0ca37c0a51c1143457fa6acdc09d55356b8edf5691d61334d4a4`; measurement receipt `sha256:230a7c0b1b00ce89a6ae991f6ea31d37886fadf99dfe45272a755e2951f1e273`.

16. Spacetime Gravitation

Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise.

98. Relational spacetime event

Claim identity: `SFT-PHYS-SPACETIME-EVENT-RELATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A spacetime event is a physical Fold transition located only by exact relations to generated recurrence clocks, spatial support and causal predecessor events.

> *A physical spacetime event is a source-bound transition with exact clock, spatial and causal relations.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__event-as-clock-space-causal-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`event-as-clock-space-causal-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:57158f6c5c035ddb048f810a643a6017ebbbce195a4c0588312c905e96a5bc3d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b7773e6b2ff8d06c5214ded33789524114c64d4484a5670e1a4f605059e96789`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8477eec669255c0e170818818a86a100ecda40bae844ba4019c1752c9e65db54`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b18bfc28a43fc2854e21066cbc6d1247cb14cecc546dc46ce09de064318cf37f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fecb336074e0c41f4d6ec1eec8923db6075bf2c4a69c0a19ed362d29e889d150`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:7fb975e8ff2ba01379b95f1223d62e94f244d303e7486926461154f0e3e40e29`. Independent certificate: `sha256:ceb24ff9b7ff53a8e2f1279b4659c17bfce039260279ca8494f93b65139f003b`. Engine external-validation hash: `sha256:dbbf84b3d2ced73f56340293db1f23759f9ccd165e88571f6fecd80d5221761e`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- spacetime-event-relation-1: predicted relational-spacetime-event; observed relational-spacetime-event; exact match True
- spacetime-event-relation-2: predicted relational-spacetime-event; observed relational-spacetime-event; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:8dada648a51868ea3569d4c92ba039fad71738b6191c83b0ab311e3aad6072e9`.
Isolation certificate: `sha256:e8564e388adb1acf9c6e6d9aceea33ba5e1d5126892fac89084435645445902a`.
Custody certificate: `sha256:67687003795e730cfc3d720b1145cffa6d1a03d271e45f7469a21e764cc9362a`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- spacetime-event-relation-1 from GWOSC-V2-CATALOGS-2026-07-23
- spacetime-event-relation-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:1432661c0f6d82366ac4f9242b50b0cb701690050a23fd3a14003aa89079fa30`; engine receipt `sha256:82fcaa4e269454cbc2b6c7308f0505bc092b436bd8a240c9ad6799d2bd5e3622` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-EVENT-RELATION-001-82fcaa4e269454cb.json`; empirical-validation hash `sha256:696e6ebacc729052b3b4ae64b4149ab84ea68d38cfac3f247505f0a79d33cf3e`; measurement receipt `sha256:8dada648a51868ea3569d4c92ba039fad71738b6191c83b0ab311e3aad6072e9`.

99. Invariant event interval

Claim identity: `SFT-PHYS-SPACETIME-INTERVAL-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The interval between events is the canonical equivalence class of their exact clock recurrence and oriented propagation-support separation under reference relabeling.

> *Spacetime interval is the reference-invariant Fold class of exact temporal and spatial event separation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-SPACETIME-EVENT-RELATION-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reference-invariant-event-separation-class__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`reference-invariant-event-separation-class`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:fec28317c8170661e758a9419a98c1debf2cb1ab1c4166cc27c45d6f6e79b004`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ce6b28bb82bc59bdd033240abe690be1d0975058ebd0eeea5a81cfef20c143a7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:743da5168c1ece2e75de4e5c4d4e98f8aa2a750be492e54da39ffffb955995af`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c8a03de0d32ff02986b3511d82d721222393e34eaa6192d6b70ee1c7bd2f5307`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c5068957099301afd9005fc768ab8997fcdb26b8b587fd1ece663c56d13b325d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:1c2f6139e6822a4f6143d4e727b590f756a442630bfb5b8fa0c6694f8072d72a`. Independent certificate: `sha256:50c3ed5aa9bee665adf665dabbef51f40cebd1394415c1e8d4a1843c91d6308d`. Engine external-validation hash:

`sha256:77aaa397de56081eb78d5310378433bd205de1868243de36e40d7ea95d1d2362`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- spacetime-interval-1: predicted invariant-spacetime-interval; observed invariant-spacetime-interval; exact match True
- spacetime-interval-2: predicted invariant-spacetime-interval; observed invariant-spacetime-interval; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:e2bde2298077d8e2c5fed178ea398024d00549e3d72d24a2c05a2c66432223b8`.
Isolation certificate: `sha256:cd342dd16ae11ba15020243d71220ed758ab08ebbb3eaefc4837e5c23613543e`.
Custody certificate: `sha256:54f90d96d76784160f28d4edd19d6410d0d8a7de43041662228c78b4ed6f75ce`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- spacetime-interval-1 from NIST-CODATA-2022-ALL-CONSTANTS
- spacetime-interval-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:e4d316c3500dbca3efc77b55638d28b6e9f6a12af7611643374a3cfd28a65ad0`; engine receipt `sha256:50d120a00b976766cf5679b950c5b17e11f3c7718e90f8350bfb0f4de166298f` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-INTERVAL-001-50d120a00b976766.json`; empirical-validation hash `sha256:2a2776461015bd10876a20724e3935b290f458bf6725d340951e4114c8b2ca6f`; measurement receipt `sha256:e2bde2298077d8e2c5fed178ea398024d00549e3d72d24a2c05a2c66432223b8`.

100. Causal order and accessibility cone

Claim identity: `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Event A can precede event B exactly when a generated adjacent propagation path from A reaches B within the limiting recurrence support; the complete reachable set is the causal cone.

> *Causal order is exact generated path reachability bounded by limiting propagation support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-SPACETIME-INTERVAL-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__limiting-path-reachability-order__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`limiting-path-reachability-order`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:c6f2c57dcbff3f4dcef21939a575a89514f5ee9dc17f9f6d9dc5a1229187e23a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

`sha256:e94442b5bb53ef2b5de6881020d27b9d07013f476e63ff55922e7c574c7b0285`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8be3895f058cdc5dbb68c13f8f5b74cf024aac7b5f19f6ca8c896d79208e9b5f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0ecf97d49c8e894ca1a702750e0dcf902e1287d8aa6a7907f2456279e0cc0fa1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d59f9e9634e3091169f87061ee34a4ebe45752d4dc067ddda43e8e0b0b483615`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d997d62bd528d094409a6febb7838577ed38e68e6594d69616cf0c8eebdf2768`. Independent certificate: `sha256:118cf1eb0c90c71598c96f9d3e8a2f8bfd9885ea5133bd149f50dad48a4ead8c`. Engine external-validation hash: `sha256:787a7a0b5cbddef61c57956bcb3bfc30031b79c4958a8ae40a5f9c4aeaee1865`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- spacetime-causal-order-1: predicted spacetime-causal-order; observed spacetime-causal-order; exact match True
- spacetime-causal-order-2: predicted spacetime-causal-order; observed spacetime-causal-order; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:25be46db3ad5277c12a1891ceae8822285bc4b86090bb022f136f7bfafb4aad6`. Isolation certificate: `sha256:8f958eeb9494faf3478fd50d2aed0176e84ca1c96ccf047846c9118807fd3895`. Custody certificate: `sha256:30982855826057f365e2cf2458626a3a2a18d43dcee90eb63dd6a451515cf6ce`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- spacetime-causal-order-1 from GWOSC-V2-CATALOGS-2026-07-23
- spacetime-causal-order-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:ebac04b824e98f9d993423d4e8eb91d0a11b27106eaa53b2f134cba581aba6ef`; engine receipt `sha256:1bfc8212981c9de42edb974516b15a92e3313182b0e61c7048f60ed8883d2678` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-CAUSAL-ORDER-001-1bfc8212981c9de4.json`; empirical-validation hash `sha256:d8c887f8b816e18cad223094fd3ab55579943d65e68a11e7ee4553a067807603`; measurement receipt `sha256:25be46db3ad5277c12a1891ceae8822285bc4b86090bb022f136f7bfafb4aad6`.

101. Inertial frame correspondence

Claim identity: `SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. An inertial-frame transformation is the unique bijective relabeling between uniform recurrence/translation records that preserves event coincidence, interval class and limiting causal paths.

> *Inertial frames correspond by exact interval- and causality-preserving Fold bijections.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__interval-preserving-frame-bijection__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`interval-preserving-frame-bijection`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:80ed23b843c0af6dd2a5dbf3cee8b049dfa37aadc382217d825be30d28a81e8d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f9297fdf7205dde4aa959fb5ff0c226f839fb3bf06aa40b29fac6e3a2a23cddd`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:18d929aa636d7afda9393834957574c0fbc60fc64c1c0ea4c0a122f011c08b56`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c44d24816d8e519c021582ddc7b5e6dc80a1d0118982f7005e1f0d6c0e5c317d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:879f262781f49174bae996767595daa69cf72f1d253b7cfa8f546f530ab1de8d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:6e481b4aa80234a618d5d8a306575f66818fc68bffd9e0a8f9ecb7a409db3e4b`. Independent certificate: `sha256:52baf3a0668d6cd44be3f6bf6380c3ef2ff6a548bb046038bd2bad31ffc7809e`. Engine external-validation hash: `sha256:14e5897d7826e6d4227103a3240f8b6e6c5f7e9531b4d16515baf775d9a0a763`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- spacetime-inertial-transformation-1: predicted inertial-frame-correspondence; observed inertial-frame-correspondence; exact match True
- spacetime-inertial-transformation-2: predicted inertial-frame-correspondence; observed inertial-frame-correspondence; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:ed541a487244cbea726608732cbff8123eba2f7e749162ab5d6f111e7cf1d863`. Isolation certificate: `sha256:f7816ed17d3e67f421348b6da5c0c9a2416324d4108724b97e0c3895ca562334`. Custody certificate: `sha256:298236f30403e5813aa34b62764a242f44691570f4c3f8d12f710132090b7260`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- spacetime-inertial-transformation-1 from NIST-CODATA-2022-ALL-CONSTANTS
- spacetime-inertial-transformation-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:bf6777d88a537ebb9ece0f7805aa0d6f0052ed940444b7f178d08469ef3148f0`; engine receipt `sha256:a9a1243a2a78ae05c7b16a006f1841a05abf263e00f35ca77ea68d5a5d2dcbb9` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001-a9a1243a2a78ae05.json`; empirical-validation hash `sha256:ca79b98d79e30fc96b7a7f154dd3908e663f5a9fd38543595403aa25a8b69169`; measurement receipt `sha256:ed541a487244cbea726608732cbff8123eba2f7e749162ab5d6f111e7cf1d863`.

102. Invariant limiting propagation speed

Claim identity: `SFT-PHYS-SPACETIME-LIMIT-SPEED-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The limiting speed is the exact spatial support advanced by one elementary causal recurrence; every inertial correspondence must preserve this boundary ratio.

> *Limiting propagation speed is the invariant exact ratio of elementary causal support to recurrence duration.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__elementary-causal-support-per-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	elementary-causal-support-per-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:8a38374d90b1ea52f160ef519a60f4312e2c73e7516cb051a542021a2645e4e8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3f24907ad869c851e94358c6d2e1589485d8dfc57a3e47429d9e67fec3b1b807`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f0065fbc2140dc09cb06db2d8f0fed9292f7e037bc3543987d690031ab0fc8f9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d403016dd04c1b0dbe76390cc38dcb92301e6b787b9c7d905ee0c6e4a5172e64`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:70e1003982b55a829dedd23bbd2d3c5a8b811ed7f9bc72eacc9d8f3309a6fc1b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f9f2b248a9aa95dd1cd4f693c9ec4d5bce43afe14d0559a70bed45bcd9308135`. Independent certificate: `sha256:9229fc9c2465f5e749ccf72df5c46a67dfe460021a0128ede2badd6dde62c28a`. Engine external-validation hash: `sha256:7dd92703c6f203aa8680cc03a6d43eb6d14d91abda7c941a8575db209bd9d7b5`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- speed of light in vacuum: exact predicted interval [1616237000000000/5391307, 1616273000000000/5391187]; external interval [299792458/1, 299792458/1]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact Planck-support/recurrence interval does not contain the released CODATA limiting speed or the displaced control is accepted.

Measurement receipt: `sha256:af13f413002d4934e1d6e502ec6563cc9afcbf087fe6443bed2bb4c863451b94`. Isolation certificate: `sha256:5feabf42262e853d3c7d56b84e0c6b407f56d88158b15ff9055aa3982fe5245d`. Custody certificate: `sha256:0ea675b46a394b6e93bff604617015845701dc158e6838ab84f0f6494c1105f4`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- spacetime-limit-speed-1 from NIST-CODATA-2022-ALL-CONSTANTS
- spacetime-limit-speed-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:fbe9031ec69bbcf574735a42d111b6098057e3ebd47aba0ba6b244143e4c7803`; engine receipt `sha256:9bef8332bbd2d865002cece22dfd62dfbbc9a9084876156ff04bb8d388551a3e` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-LIMIT-SPEED-001-9bef8332bbd2d865.json`; empirical-validation hash `sha256:97e83cbfc50402a837d53fcd83feb9ca236caf6a396f55dfc1fdf3a778c68433`; measurement receipt `sha256:af13f413002d4934e1d6e502ec6563cc9afcbf087fe6443bed2bb4c863451b94`.

103. Relative clock-rate relation

Claim identity: `SFT-PHYS-SPACETIME-CLOCK-RATE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A clock rate is the exact recurrence count along one causal path; between inertial paths it transforms by the unique positive ratio preserving the common event interval and limiting boundary.

Relative clock rate is exact recurrence-count correspondence between causal paths sharing invariant event separation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-SPACETIME-LIMIT-SPEED-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__path-recurrence-ratio-at-invariant-interval__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`path-recurrence-ratio-at-invariant-interval`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9e396ba01214b05e387f267920f4137e52df5920e3450b69fd99c35b7deb7b40`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:943cf51afadc3eea33e1917e2654588cd5ffb852a9b66d055f31edb06a920b7d`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cb4c45ed3892212961529a7859a527ca986a6ffb4e44f83ce60fc5016d7fac6b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:575773c48e443966b9c33c32c94c1ea43757f7839befd628e08b309b52f8278b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:49b8effffa4e055079087bdc448a43dd780c1cb38eae3e71224b8d58cee7a105`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:22904c14e9462cef8a843160336eb4f8b6cc423c4f567aa2155f81febeed99a5`. Independent certificate: `sha256:9eec8c27eeab2fd5cb74ae22d8d68cde4653e993b1ee0f1cb459951c561224db`. Engine external-validation hash: `sha256:76c050cfea5e77bb4ddb071e33092e3000691c2d531be6109bc1c2538a274130`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- spacetime-clock-rate-1: predicted relative-clock-rate; observed relative-clock-rate; exact match True
- spacetime-clock-rate-2: predicted relative-clock-rate; observed relative-clock-rate; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:6d5fbd1c9a090fe1f28fbecd908e63eafcdf60ebaf732e2d7ba7749b1bc29c1d`. Isolation certificate: `sha256:e68434c8ed886c0d69de56c0ddcb8173c09226de20ee0ba6365c8b36a25673c3`. Custody certificate: `sha256:7d99a6df31b443d9118a0dca1d0c77ebe2c6055e51c3d98099c0f58b8bb48c4c`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- spacetime-clock-rate-1 from GWOSC-V2-CATALOGS-2026-07-23
- spacetime-clock-rate-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant

- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:e2c7e10a97fa2126fb26c770b9c8e23423ca25e1a9412836eaa2cfb1db9dba47`; engine receipt `sha256:e7413efbd9b5a70f627078003097d132d54f3a872a7ffb58545a56e106487d34` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-CLOCK-RATE-001-e7413efbd9b5a70f.json`; empirical-validation hash `sha256:3795eb8d27b90586e27be5332749347c83245b4339a2898be1d50e1b3a0b0eba`; measurement receipt `sha256:6d5fbd1c9a090fe1f28fbecd908e63eafcdf60ebaf732e2d7ba7749b1bc29c1d`.

104. Relative spatial-extent relation

Claim identity: `SFT-PHYS-SPACETIME-LENGTH-RELATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Spatial extent is the simultaneous support count under one clock partition; changing inertial partition forces the exact extent ratio paired with its clock-rate ratio so the event interval is preserved.

> *Relative length is exact support-count correspondence under interval-preserving change of clock partition.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__clock-partitioned-support-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`clock-partitioned-support-ratio`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9233374abcde90b8197eb26613fc1d6dba24c389ab958255f29f42d68e14a127`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:01fbbc74d963f944aec1a95d254af9a06b05424fef643181c1e652078e61cffa`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9eebb4ebf15a0efc40bd52f8a03163195485f5c32e1d62e802380539f86a8619`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5c03ab3952ed6a4bc31c5fe759f384a92af1015b79645066ebfbe551a7f61e21`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ee13c0e3ae4a4b7073813fa16fd78284ce2aa8b1c28a2a48addbe4a1832475a1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a4a71b3973fbd61b62fc84b2bb2e009fc86edeada7c2bbebe2255f0ce7eb9817`. Independent certificate: `sha256:d035014e134e4cb6cecc72653f05c6bc1ca0d36bba21361fb670e65c303eded6`. Engine external-validation hash: `sha256:103c40127b1866cc738336a787db008e1c392506fa017acb5bddd6de53a5c8a7`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- spacetime-length-relation-1: predicted relative-spatial-extent; observed relative-spatial-extent; exact match True
- spacetime-length-relation-2: predicted relative-spatial-extent; observed relative-spatial-extent; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:a32487b92a5a6536054fc88923113da4d4464fe04bb45ac3f57ecb92a98578cc`. Isolation certificate: `sha256:407b00521280075f2b3bd4e34c79de3eef73ee4d229b713fee0ca5d30a81bc80`. Custody certificate: `sha256:fc11aea1cf19cb506ff35768487e465585527e2a40a049486fcc4ed57825351d`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- spacetime-length-relation-1 from NIST-CODATA-2022-ALL-CONSTANTS
- spacetime-length-relation-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:4e5f6f72c85b5622f17b4e8847f5657f6491f1d55346b47e2df5e9afd62442c0`; engine receipt `sha256:34de5f07fb814d9d840ebfb262cff11c2fa45736e26a95d94c26328edd162a7d` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-LENGTH-RELATION-001-34de5f07fb814d9d.json`; empirical-validation hash `sha256:adaee49ca1b23978a9d18c187e8ff3cfe0e5fa447d2ba63c8ce27fe41c52cf4d`; measurement receipt `sha256:a32487b92a5a6536054fc88923113da4d4464fe04bb45ac3f57ecb92a98578cc`.

105. Inertial-gravitational equivalence

Claim identity: `SFT-PHYS-GRAVITY-EQUIVALENCE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Local free response to a gravitational source is observationally equivalent to uniform change of the relational reference support when every constituent follows the same causal path class.

> *Inertial and gravitational response are locally equivalent complete Fold traces under universal free propagation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-SPACETIME-LENGTH-RELATION-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__universal-free-response-reference-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`universal-free-response-reference-equivalence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0d512e99dd1d9639c6df83f0b5ab9c2a70461b3c61d94c2c33ab68bdfdf90c7a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:73aa11a193c0b20cc2a035d9f42bc3740a8a3ce235bc3da7f1badbffe27147a5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:92df6588bcf888aaad40ca734ec326adc9765e2fc6bd382c5e746b6cf2368d0d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8862ade98a9553248fe679256e7b4d33b52f0dfaa585520ff00e310bd9e1adc9`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f041cf78472d82e37e466383945ec3bc74481b6b52ad05634a1a27f122ac85e3`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:3c06fca37a2b1be088f1a2c5652ccc01ab5951b29fd61569ec0e2ce49eea9b99`. Independent certificate: `sha256:bd958e9791dc9ae04d5aa573f9d94372281a1ae9b2f163f93e43d158bdab4d82`. Engine external-validation hash: `sha256:24a9c5fa2fe8d79fd19d060c22856a547d82aeda443f5d050e255c1af5ccff8e`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravity-equivalence-1: predicted gravity-inertia-equivalence; observed gravity-inertia-equivalence; exact match True
- gravity-equivalence-2: predicted gravity-inertia-equivalence; observed gravity-inertia-equivalence; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:8da3f466abdecd2171edccd7bc27493a6d8730c6ff2fde273ab946f273f0f07a`. Isolation certificate: `sha256:80f679c512c8dc48bf87c898f7e679da4e4ca5d7a73747e25b2925facbd26936`. Custody certificate: `sha256:464151ed3300d04050baac9f6a5b24b2a23a2ddd2c0db77d7411424529c0d3d9`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-equivalence-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravity-equivalence-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:debdf06bf6ab446c0e570406a49f5da060c602c5d45563b682df99144f160c05`; engine receipt `sha256:391143646a9cbdfec7c08931a1e1f662f1132224e4303f02c6ca1e050213e459` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-EQUIVALENCE-001-391143646a9cbdfe.json`; empirical-validation hash `sha256:e8a5760895c5fe0229b3d28c7081c53bb1a517c749f497002bbf880043c93703`; measurement receipt `sha256:8da3f466abdecd2171edccd7bc27493a6d8730c6ff2fde273ab946f273f0f07a`.

106. Source-linked relational curvature

Claim identity: `SFT-PHYS-GRAVITY-CURVATURE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Curvature is the failure of source-linked adjacent event relations to close to the same transported orientation around distinct generated paths.

Gravitational curvature is exact source-bound path dependence of transported Fold event relations.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-GRAVITY-EQUIVALENCE-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__path-dependent-orientation-closure__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`path-dependent-orientation-closure`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:36ecd3deb2f47612ccb41670d5b909d075daf9ad2e2c9ea6d0fd38df84723be3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:cc296bc892c14918ca793090c889169d6a420f86cda52974fc4454296a920104`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:296028aa42392a91f94031efd1eda98594633cd11e145432a956394e715d8d2f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:267e131843693985631 8206e4807adee62447f2541e0602f82d97a15cfd24771`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:5427c13bc35996d5dc625b40524d58f651b52064201989dca085bb5c20eff4e0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:60dc41684edf0c9909df0e2dfa1dd501118544d2b9d3e08bb2ee5f7aebb88b8f`. Independent certificate: `sha256:4338bb649a7ef62c7b507d93622192e8c473ba74270c4380cc37ff1a332ca050`. Engine external-validation hash: `sha256:0e36fafef0802525aa67f7fb1e142a635fe090fb619cf478bf0e5a7b698180fa`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravity-curvature-1: predicted source-linked-relational-curvature; observed source-linked-relational-curvature; exact match True
- gravity-curvature-2: predicted source-linked-relational-curvature; observed source-linked-relational-curvature; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:ad2de4c531e6cc4451e44954e488fcd55cdc5ed63a7fdd0d45605474943d8a46`. Isolation certificate: `sha256:f9358b2558fc21e9b92921152e1d4c9c4da49e9fc90f066fc15b7f0c84b06c1b`. Custody certificate: `sha256:518dcffea1b2ae15d09e30d72f2d573e7554d5b49dfa9a4af5865b576f0b130c`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-curvature-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravity-curvature-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:c882cbf1ee5ee7f660f46f000d82d7434a3174d8a745f73e60fdc255e42058ae`; engine receipt `sha256:4a4e70826c4fd7be1d00b9aa003261fb437402fb7c43ab61130e3f17852828b3` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-CURVATURE-001-4a4e70826c4fd7be.json`; empirical-validation hash `sha256:fef1f1331caa9c032a29ad4ce7977f4119571cf1ccb39f17b20a5f0d8b233438`; measurement receipt `sha256:ad2de4c531e6cc4451e44954e488fcd55cdc5ed63a7fdd0d45605474943d8a46`.

107. Free propagation on curved relation

Claim identity: `SFT-PHYS-GRAVITY-GEODESIC-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A geodesic is the locally unforced adjacent path that preserves the propagating word's inertial recurrence while following the complete source-curved event relation.

> *Free gravitational motion is locally inertial Fold propagation along the minimal complete curved-support path.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__locally-inertial-curved-support-path__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`locally-inertial-curved-support-path`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.

Axis	Forced form	Eliminated form	Elimination reason
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b7027658c2c3b9c24d408fc0ff0a242c8dce9cc00b77b8a3581a14f863e7faa7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c0dd995225e3068b28acc56c8e6f66418abd154c843e41d3148593bbea25d211`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9982d9cbdfb704442072dc38365d63f2a65efed5391a5ff4f9e2f9a2ef03d976`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6b9d7defb3d6154f7fffcf998c8750eca69274a2760a27dab5dbe6f299e88d36`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d779b05ac5f9cc3a8bfd895b3e03292d2761344d1f2d09b3cd6968798e61ae45`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a34d39b8e9a0e93721e04115fe1c670f9194546f60bd479e2aafe9a19f652181`. Independent certificate: `sha256:1aaae7eae0755b0584d825c67752d6059a5e66c1abbecc7bbffba773b19be9a6`. Engine external-validation hash: `sha256:5786c1d8194603af22c8d718b37d8e0bd1118cf0ee8f845b01de5d6e838a6ccf`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravity-geodesic-1: predicted gravity-geodesic-propagation; observed gravity-geodesic-propagation; exact match True
- gravity-geodesic-2: predicted gravity-geodesic-propagation; observed gravity-geodesic-propagation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:2e87a9a5e3e82e5521850fc66255d692f4bdced8d84830787b33cdb3ac7783ee`. Isolation certificate: `sha256:a5a13691d20b4468f53851defac36713913a05d25dcce1737d7922c65056c24c`. Custody certificate: `sha256:db317311a002ae98ae85890dfae6cb9296800ecea1e04ed36bb3be6765a437da`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-geodesic-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravity-geodesic-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:374514b1682e1464b36702b385c0fa0f5956f5d119eaf8a44230d2386388cbcf`; engine receipt `sha256:fcd6e73b219916aa38b32b7f2f781e0711935f16db94bae6359634106e177782` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-GEODESIC-001-fcd6e73b219916aa.json`; empirical-validation hash `sha256:abb3be49e4d9abc5657257c748d69b0d531896492951089c3eab598e745f11fe`; measurement receipt `sha256:2e87a9a5e3e82e5521850fc66255d692f4bdced8d84830787b33cdb3ac7783ee`.

108. Gravitational field-source closure

Claim identity: `SFT-PHYS-GRAVITY-FIELD-SOURCE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Every gravitational relational change is paired with an exact inertial-energy source carrier and its propagation trace; closed boundaries pair interior change with outward field transfer.

> *Gravitational field closure pairs inertial-energy source, relational curvature and boundary propagation on complete support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-GRAVITY-GEODESIC-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__inertial-energy-source-curvature-pairing__source-bound-proof-trace__capability-closed-predict` `ion__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`inertial-energy-source-curvature-pairing`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:75a8cdb60ed35199567225195e0c447cb2a94125726f1ac037b1f2763a71a236`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6de7b89644fb09a81e548b3a6503b3ae4903f29a50146da6a9213c0cb3701be8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ec1b9c32b20f0b63851cbaf3c076294a09139217c74e0b40d6211ceda7ad63c9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:31847463ec37ed5d0dcfc0131a741bf1cc77146584900bdb0b9ae432fec15b57`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:3f224f13c7965a287af13df8a33132cbe7f70ce8aca7796cc86109d47a7e3d31`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f0ed74962fa452e895824dceff5799f47a305d51bd3625919fb3d7dccd6ba72e`. Independent certificate: `sha256:8cb3c50c0d9bed56fa6c08931db1d4ad43098c2fa827d7973af9c6907313aa9e`. Engine external-validation hash: `sha256:402a48c0845de430f8dedfd4ee830568e19d46a4c4e001fd3bbcecfca80192e9`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravity-field-source-1: predicted gravity-field-source-closure; observed gravity-field-source-closure; exact match True
- gravity-field-source-2: predicted gravity-field-source-closure; observed gravity-field-source-closure; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:b7f53de6ed47b473111318347f5cdc8ab679312ee0cd53db38f0220ed397a1b9`. Isolation certificate: `sha256:9510864d6ee491b49b0f57f875d0de1b9cd2abb194b5b2c03c53dc76207eedb1`. Custody certificate: `sha256:6268fb115bc902a506d0f7afd136a664054c7da056cc1cbce1a57d570895db62`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-field-source-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravity-field-source-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:586326ad2962e36fb72e63267cf73f63a2b7a2769ef76b79936138c42547ffb3`; engine receipt `sha256:80e7d6df387d7f3fec02ef0deaf4a31f5838e858ed740f648e2f5950a4375140` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-FIELD-SOURCE-001-80e7d6df387d7f3f.json`; empirical-validation hash `sha256:e58b1b4ad9f490bce8fac6023b462c676cf34c47ae54f106d172a5e67d92c328`; measurement receipt `sha256:b7f53de6ed47b473111318347f5cdc8ab679312ee0cd53db38f0220ed397a1b9`.

109. Gravitational-wave propagation

Claim identity: `SFT-PHYS-GRAVITY-WAVE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A changing source-curvature relation that closes into transverse recurrence propagates at the limiting causal speed and transports energy-momentum with held polarization.

> *Gravitational waves are limiting-speed source-curvature recurrences carrying energy, momentum and polarization.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-GRAVITY-FIELD-SOURCE-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__self-propagating-curvature-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`self-propagating-curvature-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b7b136872d39bf5145b9cd5a7c9e144f069b09595ba417bec1954264fa4bf639`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a4668b977a7dc243e8d7cffb5281dd522f7e141c9d57a414c7fa4b9d7ff5c07b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:99c266006d03d674de429c1914a35d332b4e2fb65b4d77de084eb84f01ec84bc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

`sha256:5d26782234031d446a0c0401268a54c749a24761281beb8753a6d453d8e5b91f`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8b6850e209743b529c53b98b2167da4edab5bf543e2b22f1b57f3feb3e79deec`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:62dcb4de31ba440c9b80227a2b922bdb5b626eafabaa6041c4b368edeefb57fb`. Independent certificate: `sha256:de8127768a7fdeb63d0f4d9e9da3ab09f73ba114d18a96b60074a66d8d08e4be`. Engine external-validation hash: `sha256:efa538d7a76533cf7044f4245e7a706f77e667dfe9f63dff2541c02761bb7a82`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravity-wave-1: predicted gravitational-wave-propagation; observed gravitational-wave-propagation; exact match True
- gravity-wave-2: predicted gravitational-wave-propagation; observed gravitational-wave-propagation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:b294c31d9d0cccfdb72234bc8edd7fc77f2e8f2e81328ac54b3f94683b099b00`. Isolation certificate: `sha256:7194df1fb9a8b0f56df0e38270db161bbb4c9e5fed5c22d744b89c9f671b9162`. Custody certificate: `sha256:cf71825378cc4fcf95eed5616ad3196c2d5b5647c2e53932a9f7ab134eb7e2bb`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-wave-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravity-wave-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:d3c2e4ade0bcd74a1cb6731c8db906e8b9e9a53d4ce309301b39790715679923`; engine receipt `sha256:f2bcf8f3bec360481dadd724546028e3c47c633bd288adecbf650d92a8d0f1ba` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-WAVE-001-f2bcf8f3bec36048.json`;

empirical-validation hash `sha256:013c0d29ad9a8ad2f3850f481e9b0be54e0a291f3baea625227a050109934f05`; measurement receipt `sha256:b294c31d9d0cccfdb72234bc8edd7fc77f2e8f2e81328ac54b3f94683b099b00`.

110. Causal horizon boundary

Claim identity: `SFT-PHYS-GRAVITY-HORIZON-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A horizon is the generated boundary separating events with a limiting-speed outward path to the observer from events whose every outward path remains source-confined.

> *A gravitational horizon is the exact Fold boundary of outward causal accessibility on source-curved support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-GRAVITY-WAVE-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__outward-causal-path-accessibility-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`outward-causal-path-accessibility-boundary`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0436ccab8b956bf9b4ff8dd85553679f27286fdaa52750759d8164f019c258a5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:90c176ed08a43d27a61c034b0213577c6468523dd8cd8a190252fb874fdb3686`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:158073e681f1b8a87a714e63e93fef55959987dcadfc116ceee7a281f1c6eb0a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:aefc1806213b3b9c4642d09fc10e8ea8a2663ef634218e22bfcac424218d4d22`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1a3295b020e67bd329f48fcaedc43302734345fba5abfe1ead76a577d5b8e4e0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:04d5416a76e6cae6eed78368afc175b43ea8899624372c23524cd3ff369b6219`. Independent certificate: `sha256:3782e0a1753794145e8d5cc5e1b84d7e8e490a067d75dd3203db16660946f9ad`. Engine external-validation hash: `sha256:fee7d362ca3bf1c92f3c4b6446b3f575b487959c41e54737a5176ccfd4cbde4d`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravity-horizon-1: predicted gravitational-causal-horizon; observed gravitational-causal-horizon; exact match True
- gravity-horizon-2: predicted gravitational-causal-horizon; observed gravitational-causal-horizon; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:25a77549a95884a6c8b2f2f636f8d2521bdf5c406b96c298326e7b5fc772a502`. Isolation certificate: `sha256:3d97d1dabc2f89390e65e54273b762de3256ca4605bea78726cbe2479fbdfde8`. Custody certificate: `sha256:ffe0b5a8a931e75710308a7f166bbc63de7f2b82157da11a57c9c01e59155316`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-horizon-1 from GWOSC-V2-CATALOGS-2026-07-23

- gravity-horizon-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:dc739559de7afaf7c8fff7beb4ceb24afcf08cf87fe9832f66b9f235c694fd65`; engine receipt `sha256:7abc9b68655deb7986f0846887236bae69f14d4d6656d2ed71c388d0f889c52d` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-HORIZON-001-7abc9b68655deb79.json`; empirical-validation hash `sha256:8c7e559f057c9378e631a19da8f7ac4d999c223aee5e3673ab8260e1014bbcbd`; measurement receipt `sha256:25a77549a95884a6c8b2f2f636f8d2521bdf5c406b96c298326e7b5fc772a502`.

17. Continua Collective Matter

Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance.

111. Finite coarse-grained continuum correspondence

Claim identity: `SFT-PHYS-CONTINUUM-COARSE-GRAIN-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A continuum description is the observation quotient of a completely generated finite cell network when adjacent cell records lie in the same retained macro-class.

> *Physical continua are scale-declared observation quotients of complete finite Fold cell networks.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-cell-observation-quotient__source-bound-proof-trace__capability-closed-prediction__separate-`

measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	finite-cell-observation-quotient	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:1dec3b2c7252a9da60284fb03232de86118a42a89d245fa02b42340a7e075165. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:d6bce9c5b6034211c3297165bb209983ecfcda45e5061635706fa854895e9b27.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ce09b9ee426c6690cbbd588473c9e32e0298a7911dfbda3b4453219f8598860c.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:7d15bff6005e58bb09c1a112387b838e5228cf03409007959e7e994af98e04a9.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:be709c0e451aa29ed4ee66afac372db7eb3ee5c6f1eaf735d720c38950c4b44d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:1a51fc9553d35360f449393d395f3ea0d072eb3095fa63397d888131fb34cfcf. Independent certificate: sha256:1efded5d8390c6f7d639702f9f91759b6aacb5442b8ccee2e9fead7d5280239c. Engine external-validation hash: sha256:950b7dca6d924fb7d678d1b1ffe159f617caf5a76e6e03e926c4ad8b7fd9fe5d.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- continuum-coarse-grain-1: predicted finite-continuum-correspondence; observed finite-continuum-correspondence; exact match True
- continuum-coarse-grain-2: predicted finite-continuum-correspondence; observed finite-continuum-correspondence; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:54e4a46c2fbe6c04e91bc6b288e9125ced225bde8547a007b72529f9fc3ff051`. Isolation certificate: `sha256:e20576f597c4f605d5a0f188309be1861b259ebff1d4badbb100e1355ab5a860`. Custody certificate: `sha256:ad650f6a6076a3fe9d36f56a95d41bce05e0b33204e440451f33773de30f4047`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- continuum-coarse-grain-1 from NIST-SRD-INDEX-2026-07-23
- continuum-coarse-grain-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:7eed273621cf5466b155832334d6233f73b2969d81c206d2099fbf4eff584b37`; engine receipt `sha256:fca2926aba2b9fca7aaeb55fd3121f9a52d64846397776cbb26e30b884b1a509` at `receipts/engine/model_admitted/SFT-PHYS-CONTINUUM-COARSE-GRAIN-001-fca2926aba2b9fca.json`; empirical-validation hash `sha256:9350cfc2a37afabfe63d2c94540dea6bf21904ee8ba60b7b34f5988d94c95022`; measurement receipt `sha256:54e4a46c2fbe6c04e91bc6b288e9125ced225bde8547a007b72529f9fc3ff051`.

112. Density as exact content-to-support relation

Claim identity: `SFT-PHYS-FLUID-DENSITY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Density is the exact positive material-content count or inertial carrier divided by the generated spatial support count of the same observation region.

> *Fluid density is an exact content-to-support ratio on a complete generated region.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-CONTINUUM-COARSE-GRAIN-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__content-over-generated-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`content-over-generated-support`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:04ca869c52dc5303a5ee090b315b4fb0a44e2599e5abf5be6b9584c018790c84`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d6b53ecc44e62942f0af864b87f43a598c49def06da854548f868257c5bb21ee`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:878e57a9dda01e2d5c20bbdeb5eea018d2b2d3ae390cfd4cab2ca48fc9368008`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4acd658a8909be1728fc3579dc413b6181aeaf168f7519d34b98036850c37a43`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:3f265911c3f4a001248376002fe6c3ff2d0a07b17a94b7009487f9f7824043f9`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:39d3d3ca8f88165fea6399c5b3889e5a0708134a39f25135c8e21b1df4d138aa`. Independent certificate: `sha256:fc4ec7e77d06740835e907c86bdb712e8f4a5cc2d38f7af667a5a06fc0c66a05`. Engine external-validation hash: `sha256:bdff48145835a803f3c2873f3cf43d06b34c5f3b90e4a2a70b9b188ebbfa3ffd`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- molar mass constant: exact predicted interval [625000000461504249/625000000000000000000, 48828125066636203/48828125000000000000]; external interval [50000000037/50000000000000, 12500000017/12500000000000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact mass/count interval does not overlap the released CODATA molar-mass interval or the displaced control is accepted.

Measurement receipt: `sha256:6f1ec3ffdd790df3ad9700de6a7bcf02e133f22bdede346e418c3ba41dfba650`. Isolation certificate: `sha256:efba3e6e2eddac59ece37a8836a99f02a3a3b62dd50045d9ec331999c315e759`. Custody certificate: `sha256:7c3ff6327bf489bc25caa5550519abe874a599823ea361f62380818528533b8a`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-density-1 from IAPWS-RELEASES-2026-07-23
- fluid-density-2 from IAPWS-RELEASES-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:2174bd50493277533f2bcb3ca5327a31201118dc149504ea5603ec2bbe31deba`; engine receipt `sha256:042d8b74f0692ac0c11dac25c449690e74d8f5a8b1196641080e377627b9bbcd` at `receipts/engine/model_admitted/SFT-PHYS-FLUID-DENSITY-001-042d8b74f0692ac0.json`; empirical-validation hash `sha256:5f9caf1c3efb73312888b0181c4595a61ad9f10cd6ba1288e1a34ccdd179f078`; measurement receipt `sha256:6f1ec3ffdd790df3ad9700de6a7bcf02e133f22bdede346e418c3ba41dfba650`.

113. Pressure and stress transport

Claim identity: `SFT-PHYS-FLUID-PRESSURE-STRESS-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Stress is oriented momentum transfer per generated boundary support and recurrence; pressure is its orientation-closed normal observation class.

> *Stress is exact oriented momentum flux and pressure is its normal orientation-closed Fold observation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-FLUID-DENSITY-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__momentum-transfer-over-boundary-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`momentum-transfer-over-boundary-support`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ac4f7c24a99afa7c33f18e744fc5afbf29b6ba686c32fdc2cbd5c72c17e6576f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:fb7fa3ccfb18ebce02d1140a68e3555e7593365a1e5ca5c14ca402e044e7d6da`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:92447125cc883fb979c3cde345b03c998c687a9bfd04232867dae08b530ba12e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:429ba780f58f3ff79a79b892f1c98dfb54cbcf84380a1b291c26ab05f649ea56`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4be9f236a07b5d7b7e6a9996faf428ba56d911ade74f2569ab2a032a4b52896a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:dbd7d23c7160c57567f66f104c05c598d4c86b8e7d2096e1957bd5138b6f6204`. Independent certificate: `sha256:30d13302ce554ba52ecf0fe8e5cd7044a5f3a1850cf73ab3189034001eafa953`. Engine external-validation hash: `sha256:37c52185ba81e358e7625a85e7d4a0c062d3425fe2d4882c3c8a8b93d0f3f9ae`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- fluid-pressure-stress-1: predicted fluid-pressure-stress; observed fluid-pressure-stress; exact match True
- fluid-pressure-stress-2: predicted fluid-pressure-stress; observed fluid-pressure-stress; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:452d42fbae15faaae0856ab60e86b7c067f1e7d0e29a7d3e366e5fea07c856af`. Isolation certificate: `sha256:993024f8b77bc093509ed733a0de02aa6b13b8919c957c8cc4a5161ba787117c`. Custody certificate: `sha256:d86b1066ca74dc7a0879d484992d89103cb1a38c19d216a8af5c15df019597b9`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-pressure-stress-1 from IAPWS-RELEASES-2026-07-23
- fluid-pressure-stress-2 from IAPWS-RELEASES-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:fa1293b29e0e53ca46291f56f2a6d76cace71777596bb5d5c4008416b3a759ba`; engine receipt `sha256:94ae8c496c0ad93534009db375be075997529f8bb40292d7bff956f9dce7d853` at `receipts/engine/model_admitted/SFT-PHYS-FLUID-PRESSURE-STRESS-001-94ae8c496c0ad935.json`; empirical-validation hash `sha256:59b462aa2e41aa117ca7b45e6d676ddd7c60b09cd67e2b15299d72979379f5c7`; measurement receipt `sha256:452d42fbae15faaae0856ab60e86b7c067f1e7d0e29a7d3e366e5fea07c856af`.

114. Flow conservation

Claim identity: `SFT-PHYS-FLUID-CONSERVATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every complete cell region, material, momentum and energy changes pair exactly with named transfers across its generated boundary.

> *Fluid conservation is exact pairing of interior carrier change with oriented boundary flow.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-FLUID-PRESSURE-STRESS-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__interior-change-boundary-flux-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	interior-change-boundary-flux-pairing	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:221074ca4b603a432a39e472f4924f258ef2628d4f0ee8442dc47c05e977bc5c. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:6c2960986f4d55e887ca5bcba3b6ae11b3c5dbc032efec8e183b0a7ca2146e21.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:8905302ed286a1762db5fbb766b12964a3b6646cd3c4e09e7a9e46f466742f0b.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:02ec9fd11b8633c435c5a45764bc7e6dd4b02e881b1fdf86755bd78e360aa3f2.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:ac8b616e32bac9ad28c3cbe03c4ca63bde1df101ba74d479f5f6d06f957620b6.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:4436963e2bcb6dc07b5a59fde089e6de3c5263fde0fa90e6a6c19c3d8acd969b. Independent certificate: sha256:185dc33929ce6029c79c250a4e28bc2c1eb622501a7838fd233b6d446f906830. Engine external-validation hash: sha256:b429eff314d1343a5a79a5c80990c96fdd961c3aab03c72612c6523267815984.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- fluid-conservation-1: predicted fluid-flow-conservation; observed fluid-flow-conservation; exact match True
- fluid-conservation-2: predicted fluid-flow-conservation; observed fluid-flow-conservation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:c85a4e2ddd3f216213e75a4418da06e63cf1c171b339770422c9d9434fed12d9`. Isolation certificate: `sha256:09249dd3f9c13a897149dee7901f09c2f2da5993128f52f051f8f73eab8487d0`. Custody certificate: `sha256:7b9e0b55f0b2c11511fdeb9b5007bcdfb3aac3df2dee9c83decb8850e22f717e`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-conservation-1 from IAPWS-RELEASES-2026-07-23
- fluid-conservation-2 from IAPWS-RELEASES-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:17c21e0e61208e61b4b3a30ede51a89dd492c9720b01444d85b5b4cd8676f409`; engine receipt `sha256:f6882fdf667263556f24e891247deb5f60533155505e7dfa31bdee8fb68f23ce` at `receipts/engine/model_admitted/SFT-PHYS-FLUID-CONSERVATION-001-f6882fdf66726355.json`; empirical-validation hash `sha256:2f2c901861c6a9e382e381bc5f63e24f1b166f45c8c7a2217f0c435bba14d978`; measurement receipt `sha256:c85a4e2ddd3f216213e75a4418da06e63cf1c171b339770422c9d9434fed12d9`.

115. Inviscid flow correspondence

Claim identity: `SFT-PHYS-FLUID-INVISCID-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Inviscid flow is the boundary where adjacent cells exchange normal pressure carriers but retain their tangential motion labels without intercellular transfer.

> *Inviscid flow is the exact fluid observation with pressure transfer and no resolved tangential momentum exchange.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-FLUID-CONSERVATION-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__no-tangential-momentum-transfer-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`no-tangential-momentum-transfer-boundary`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:6ff158fdae68bc989a9aab5baaa2aa97d5c267297b855cca481e25f8bf76597e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2dfd81c580bb59efca01f992f8cc1573ced5ceeb378ec1cb2fc54c598d76c618`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:518face2fa90408cb1537676e5ae2dc65a40dafcd9dddcbfcf09675c18b61ed9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a996c280c1b2b3383a30b0b528487bb4789d5b1a04fb9c144bf8194547ec8fca`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a1b5cfa22dfeededb22eb4d43fddd0b01e467092976b426d90f7ee9ba26ef751`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4be4eec85be994c725ed2e8c41fe1f8b05b5fccfda1c37693de5235861561d7b`. Independent certificate: `sha256:274aeb210d2fc6e06167b3c816daeac66339cd3d4d3137099997026ca200d934`. Engine external-validation hash: `sha256:c039ca4c27a2d4e3e2871e7ce4cbbd371b4084243996a6d85d7d2b724c96332a`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- fluid-inviscid-1: predicted inviscid-flow-boundary; observed inviscid-flow-boundary; exact match True
- fluid-inviscid-2: predicted inviscid-flow-boundary; observed inviscid-flow-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:b859bde36d69a637bb96f9cbd9d011e09bd29ac37789d276326d3e0ae3406be5`. Isolation certificate: `sha256:612b7ca508a9c572df39d2b3db94b634e709e37def01f2c9d6b1938a091506b4`. Custody certificate: `sha256:a4cf17c1f96e86f3d6d81e9f87517697d7947693bb8e9224ce6c6a154b968389`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-inviscid-1 from IAPWS-RELEASES-2026-07-23
- fluid-inviscid-2 from IAPWS-RELEASES-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:77981b1d9a4897ac79c4b93ee40fe02ef653ce17a93c9ddbfdf7b4137a8e45d6`; engine receipt `sha256:f697547d491a454368e7a58afdc7b0558bd77fc4a4c07419bae1d30623e21025` at `receipts/engine/model_admitted/SFT-PHYS-FLUID-INVISCID-001-f697547d491a4543.json`; empirical-validation hash `sha256:240df644cb228a10a0388173cd15ee75473126fa655c7fc8d8295b93c8666cf7`; measurement receipt `sha256:b859bde36d69a637bb96f9cbd9d011e09bd29ac37789d276326d3e0ae3406be5`.

116. Viscous momentum transport

Claim identity: `SFT-PHYS-FLUID-VISCOSITY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Viscosity is the exact tangential momentum transferred between adjacent velocity-labelled cell classes per interface support and recurrence difference.

> *Viscosity records exact adjacent tangential momentum transport conditioned on velocity recurrence difference.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-FLUID-INVISCID-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__tangential-adjacent-momentum-transfer__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`tangential-adjacent-momentum-transfer`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1992dc71f303b604c938e0e775750e54770e9f9b7e55124a58b7be68b74cee30`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:243b176f433d91fd498e68242b70e2bb66b2c1db852a059a126084da67db8fef`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:dfb3417d8cdf99d0bd693593953db666c9a0a1b443675f187fffc9a50a4360ab`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b0bda1d0ed11869cb1296026168adba986e70db0635133673b73975fc618acad`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e65a0f8e56f9657a81b05bd6cfbc99dcafaad256167c24be0dd6d3def944c411`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:dd2ea2c2404d6fb46987bb7c28819da623e8b20a855d6fe34ab68fc4294d8570`. Independent certificate: `sha256:88c8c29b5b3ad197093011b38958cbdba4cc0e1bad0af766083087ebd9723dcd`. Engine external-validation hash: `sha256:8246faa833391984580bdf87419a607300e82dac64d1f8446551dd97b2a67142`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- fluid-viscosity-1: predicted viscous-momentum-transport; observed viscous-momentum-transport; exact match True
- fluid-viscosity-2: predicted viscous-momentum-transport; observed viscous-momentum-transport; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:0dd046ca2d7e3d78f77c993ead65e6947505c7971b1647557a49f35914f8990b`. Isolation certificate: `sha256:cba153e98c6f7d6301365a14864a9b0fb1244a2adcf23514a8f6f93eeb2fa0a1`. Custody certificate: `sha256:90c414b23a8a13ecc14409cb3a656aaede90dc601c787ececbeefa0db0166fa8`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-viscosity-1 from IAPWS-RELEASES-2026-07-23
- fluid-viscosity-2 from IAPWS-RELEASES-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:7ef2a3786fb504b69aa64fd2dc22720096cad991236c9713b9310c8827270ac5`; engine receipt `sha256:b395ce239e1415b4afbd98cc99ab5f8a91dce5fb8a4914d2bdffe7e9b2108c71` at `receipts/engine/model_admitted/SFT-PHYS-FLUID-VISCOSITY-001-b395ce239e1415b4.json`; empirical-validation hash `sha256:708e3c3f3fe96ccc69bc563b2145f46739256206c6ea6e68f6f125e055704f93`; measurement receipt `sha256:0dd046ca2d7e3d78f77c993ead65e6947505c7971b1647557a49f35914f8990b`.

117. Finite turbulence and cascade boundary

Claim identity: `SFT-PHYS-FLUID-TURBULENCE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Finite turbulence is nonperiodic multi-scale recurrence on a complete flow network, with conserved carriers transferred among generated support refinements; cascade claims close only at the enumerated scale boundary.

> *Turbulence is deterministic multiscale Fold flow with exact carrier transfer and an explicit finite census boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-FLUID-VISCOSITY-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__multiscale-flow-recurrence-transfer__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`multiscale-flow-recurrence-transfer`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:aea1f033f8c07f5d9a9aca67733f05b355be5a92f81322e1ecee504cb720b81c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a73887a8fc98a863be96b9f31736681c2e9fcb1954acc76cd4f49a6e03a526b2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f9f3711bbcac605f36d171f20e96d6353e8dd48c3b973fb7a8cd91b9d1e43a6e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:533dac907b77fc5c47d01f99e297d5f36749f922d27a4ab684160748ba86528a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:33b889a2994f57a1e38443abc4b3013b70f1d44f495c9345f75daf08293cc931`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:15da7684842244e814ad50b42670a8cf90fe83b67039dde3ebbaf3f65b5060bc`. Independent certificate: `sha256:97481c4173ee2b5fa96182afa033951fdcf78787204793058f1343605430479a`. Engine external-validation hash: `sha256:de0b9864fa2f3c8b50e3c88fd5fd64c08378ccb1dccb301ee47c8c967641960d`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- fluid-turbulence-1: predicted finite-turbulence-cascade; observed finite-turbulence-cascade; exact match True
- fluid-turbulence-2: predicted finite-turbulence-cascade; observed finite-turbulence-cascade; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:39abd31200a9319915f19bbbff0d68a4084b5e3520ddf1035af29c7ab7085135`. Isolation certificate: `sha256:f4170b99db1d143cb46049cb050d793266638dfc4a8d52c8b11b45a236691f0c`. Custody certificate: `sha256:7c1dfd264cddd8f35e2cf6453bdff83c9addcd247c449d04d410942aed8b1bf0`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- fluid-turbulence-1 from NIST-SRD-INDEX-2026-07-23
- fluid-turbulence-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:1d7f62f73bd0650e4705ba612c4d86f3640108c8d91e45efcd8db87d7fb3e369`; engine receipt `sha256:d8a59dabe554a169c1f5d8f4c7c70feb2a0d3e1ebaa0d26186d1e2316a2e406a` at `receipts/engine/model_admitted/SFT-PHYS-FLUID-TURBULENCE-001-d8a59dabe554a169.json`; empirical-validation hash `sha256:35c6d95008dbd73d5f4b12364f48f9d2e20127a35bd2d15eae7c18d989aae7c3`; measurement receipt `sha256:39abd31200a9319915f19bbbff0d68a4084b5e3520ddf1035af29c7ab7085135`.

118. Plasma collective response

Claim identity: `SFT-PHYS-PLASMA-COLLECTIVE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A plasma is a mobile two-orientation charge-carrier network whose long-range source-response support couples local constituent motion into collective field recurrence.

> *Plasma collective behavior is source-bound electromagnetic coupling across a mobile held-charge Fold network.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-FLUID-TURBULENCE-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__mobile-charge-network-field-coupling__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`mobile-charge-network-field-coupling`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0b7b164840d922e51eb41ccc2cfc303980fcd520bc605d0d716cbef879f8166b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:a89f9ffa8ee5ae6d2a673c5e62fd60a6c313076e86184c2417c1ca9b3216d22e.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:d9ab01dd2e6633d7a172025c5f01c04d1247d6177636d4d81f3085c9c84871f8.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:c9f04994806a3462214b469cce636a358c665007a36ab75140155e59e7cfbbbf.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:321294b1ba2e87e4b07d90a4815ddffc2bb23729469f21499c9e8d9fe5a7971d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:2926da1c4cc5aa86a14edf2ae9353aa3c87a2e7b05fde8ea1fd505bb63162e1d. Independent certificate: sha256:ee358c5d98fe74f5a041dfc0a4cb8794420c2d0880172a452adf402ed78c9365. Engine external-validation hash: sha256:1c55941937e1b1b72cbc062a2b9692791e18aefad9768e0aa9a001fa754bdc6a.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- Faraday constant: exact predicted interval [120606665154137523/1250000000000, 120606665154137523/1250000000000]; external interval [2412133303/25000, 9648533213/100000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact charge/count interval does not overlap the released CODATA Faraday interval or the displaced control is accepted.

Measurement receipt: sha256:9b1424fffe6a51cf3dd08684b0aad17f3157ec20a53bc0cff2aaf25ed4655e50. Isolation certificate: sha256:805ec634b9172e2c8f7289a0c37170c288544ae116a8c5e921baaa7ecc5df77d. Custody certificate: sha256:79c010fef9d99afb8be2b2f6a7a01eb9405e1b29014abc0f4fd93833994ea92b.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- plasma-collective-1 from NIST-ASD-5.12
- plasma-collective-2 from NIST-ASD-5.12

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:1a17a43a87ddbe90c7ffff4645ea017347a3955e923efcc63bd5280c4c9e8186`; engine receipt `sha256:31da2d601e05d224850b377d80396438c38447c40e994dd8d51f613238a07499` at `receipts/engine/model_admitted/SFT-PHYS-PLASMA-COLLECTIVE-001-31da2d601e05d224.json`; empirical-validation hash `sha256:71356ff0a424d73c918de32c5bc719462f1fa7619fa68772d092921c212690ed`; measurement receipt `sha256:9b1424fffe6a51cf3dd08684b0aad17f3157ec20a53bc0cff2aaf25ed4655e50`.

119. Plasma oscillation and screening

Claim identity: `SFT-PHYS-PLASMA-OSCILLATION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Displacing held charge orientations creates a restoring field recurrence; phase-mixed mobile responses redistribute until distant observation classes close the source distinction, producing screening.

> *Plasma oscillation is charge-separation recurrence and screening is closure of its source distinction by complete mobile response.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-PLASMA-COLLECTIVE-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__charge-separation-restoring-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`charge-separation-restoring-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:64603a48c22c3e60a1451bd13367f889763a402e2bb95d7bde325192862515bd`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:4262741298915584680848 30d17dbe35712d20f0c3a86ff0a1710366f2664387`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:711474c269625d4f16c0a6dc355a0879c88aaf11cd9469298fa8234a20bc6fed`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:19a9f5fe7f3a6a1f91a35c7b3a4f4dc0c6c9abc2eca2ded59c1ff60bcd415aa4`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c2581546a9b49c3c450d381bfdbc49f43a1c3919f89e39649c06e1a7e4f632f2`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2acf9e656c728c6d6ba48f454d426f61e8b538b29873bf9b356571ce2b19416d`. Independent certificate: `sha256:24c27df396a58d082fb49a4efb2043962553983784ee0eb71dc0b21f6918c958`. Engine external-validation hash: `sha256:a7d8b427e4486a43fb6e368b4398da334392b319358c4901394ea96458831f9b`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- plasma-oscillation-1: predicted plasma-oscillation-screening; observed plasma-oscillation-screening; exact match True
- plasma-oscillation-2: predicted plasma-oscillation-screening; observed plasma-oscillation-screening; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:d7876b6ca9718f6b2e4efd047847aee1442c7c0c7da6464c9e8083d715ce7bc2`. Isolation certificate: `sha256:3776290cd8a4fe826c3c40a69415cea190673f87d0a4c3f5d1f8199de40e8fa9`. Custody certificate: `sha256:c566c606b03bfebc676080627bbb6291fbd463364ef52553e7c03e6241ec0301`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- plasma-oscillation-1 from NIST-ASD-5.12
- plasma-oscillation-2 from NIST-ASD-5.12

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:32e35ac55e76c5661d80eeaf2b1cb3ab1f8caaac3e3a1c29bcaea6e6798d39d7`; engine receipt `sha256:14cdadbe5421c2cb5f6f231d20b0456b8dfda9d76e01c4c4ab815a2d915efc82` at `receipts/engine/model_admitted/SFT-PHYS-PLASMA-OSCILLATION-001-14cdadbe5421c2cb.json`; empirical-validation hash `sha256:084f84514358c4efb0b9b9a64b5dfe4a310fa571c5b95ea31751036bbb13601b`; measurement receipt `sha256:d7876b6ca9718f6b2e4efd047847aee1442c7c0c7da6464c9e8083d715ce7bc2`.

120. Magnetofluid composition

Claim identity: `SFT-PHYS-PLASMA-MHD-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Magnetofluid behavior is the compositional closure of conserved fluid flow with oriented charge transport and magnetic source-response on the same generated cell network.

> *Magnetohydrodynamics is the exact joint Fold process of fluid conservation and electromagnetic charge flow.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-PLASMA-OSCILLATION-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__fluid-flow-electromagnetic-joint-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`fluid-flow-electromagnetic-joint-composition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9d5231006c1b977b269ed159e74da0db1dc1813fec47ffa9ade235c0562c76f6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7ac127a6edcf37c11b9d4123d9d0dd9158cab0aa3a8fd744eed289494f624b03`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:731f7b6746cb28d3129ee5047a211addaaa884961de1792249a10515532126b2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5c6de5e550bd5e51ca3b2d461671df02ec347b65c2a831034beeda1eb2f967b1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dfeb70b8f6190c73fe7c021d94bda529f597d4f32a1d2f699ac928cf3097c1a1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9b9af7f6ff1f479de604d080ede292c2367aca22e4a244c29bf4f0ec70872582`. Independent certificate: `sha256:048987901a7d00652dd28d6b307ed8caac174301fdd75a358afc547be3376929`. Engine external-validation hash: `sha256:7912fe7b4ecb3d7e042f26663ce6f1d1eac8bb7350f4fdcdda8950e92b45e7b8`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- plasma-mhd-1: predicted magnetofluid-composition; observed magnetofluid-composition; exact match True
- plasma-mhd-2: predicted magnetofluid-composition; observed magnetofluid-composition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:2849ed848c310b0815fcc89987abf99503720c5a4c5dcdd8394e982d2a5df506`. Isolation certificate: `sha256:2394e9378ee62d0d856c0c44e818cc3191a4e368b8d39821063581f832f38f93`. Custody certificate: `sha256:4c57aa904d3c44057f14d5ca6a561d281a6a905524b819e44f063b950e03e77a`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- plasma-mhd-1 from NIST-SRD-INDEX-2026-07-23
- plasma-mhd-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:bc7e3dd69af9792887aac4aa519554faed5ecce78cbbd933307323f97348d3b2`; engine receipt `sha256:758229c88e39f5606a215e2bdd8ed689d2f31b0cb739eacdda0d8c53ecad0e5c` at `receipts/engine/model_admitted/SFT-PHYS-PLASMA-MHD-001-758229c88e39f560.json`; empirical-validation hash `sha256:5e92998418418aeb0f51f1c25eba8bd0411a2a2839db5fd39a2f7866a6e37a33`; measurement receipt `sha256:2849ed848c310b0815fcc89987abf99503720c5a4c5dcdd8394e982d2a5df506`.

121. Lattice order and excitation

Claim identity: `SFT-PHYS-CONDENSED-LATTICE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A lattice is a finite adjacency network whose local constituent word and neighbor relation recur under generated translations; excitation is a source-bound departure propagated through that recurrence.

> *Condensed lattice order is translation recurrence of a finite Fold adjacency network and excitation is its propagated labelled deviation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-PLASMA-MHD-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__translation-recurrent-adjacency-network__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`translation-recurrent-adjacency-network`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a35d7000c96d02c087b673175ad732a45a7ed2abfd58d2f16ea7a77f0ec25dd0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b1cc8064129d3557b22b833afa9a83a704ab693dea418f8b40835f3df2f3d8ba`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8ed9ef2406b9bd6616e7f7cdd01a97adc21e8c5ed9bf668b5b67fa9f4f4e4017`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:07bab574e37e3d080d2079c1f26a7ab081f926d4f752ade2a505022ad9528967`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4013bffdbbb665a8dabe7e065bb971764b0c102b235cc83180695b542cc35eae`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e8d201f82909c9dc391dcc47256dc3fda9b22754bf0d69c9f55e8cf5242d3eed`. Independent certificate: `sha256:90a98406eb4fbb2f21d6615ea1ace1c4bcc5c80290eabba33b2cc4b92dda55b4`. Engine external-validation hash: `sha256:74ecf5daa0e778da0b65afed9c2871671aae334209ead28be74e052b62faf51e`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- condensed-lattice-1: predicted condensed-lattice-excitation; observed condensed-lattice-excitation; exact match True
- condensed-lattice-2: predicted condensed-lattice-excitation; observed condensed-lattice-excitation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:183d206a75d8e56ee46c5727c0ecb1de78cffd0de89bb0a640b98d9f8dfc6aca`. Isolation certificate: `sha256:5d0cfa6a495c1b77c2881f00333bb6451e13fda86048d2980be9b9c7c86d9b1b`. Custody certificate: `sha256:792f1815913b2689ddd15e12133d5fd0407762750fd464a20d17edbcd58d76a0`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- condensed-lattice-1 from NIST-SRD-INDEX-2026-07-23
- condensed-lattice-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude

- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:fdacf91e04fd5ab26a7fdd1cb2db5b3e0efb319be91b292307cfe73e2c175963`; engine receipt `sha256:46d5a4142f4a53f2665b0f45f89c00e0afe2b5cc8989e0f5169644fc100d6ea3` at `receipts/engine/model_admitted/SFT-PHYS-CONDENSED-LATTICE-001-46d5a4142f4a53f2.json`; empirical-validation hash `sha256:5c3b3b0f9927d524776ca6218313f669ed527bd513d6f85e0f7076cc5600b3ce`; measurement receipt `sha256:183d206a75d8e56ee46c5727c0ecb1de78cffd0de89bb0a640b98d9f8dfc6aca`.

122. Band support and transport

Claim identity: `SFT-PHYS-CONDENSED-BAND-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Band support is the complete set of phase-compatible constituent recurrences on a translation-recurrent network; gaps are observation classes with no compatible generated path.

> *Bands are phase-compatible Fold recurrence classes of a lattice and gaps are complete-support absences at the declared boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__lattice-phase-compatible-recurrence-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`lattice-phase-compatible-recurrence-support`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ccc73313cee745d4be6a4ef2c7e14cb669ebc2be5007249574e88135bf082c85`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:85b0b149042eefbc0921b56113028b1ec5a08a7cbf25076eaf8a71115bfa4c1e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:038dd4e7c0ee189026496752a144279c36795917b9005b100802ed42d08e9b27`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1e383fb722a2a19061bf3948397e286dbfbfb8768aa42018acd8c5eb07a310bb`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:13b3e4761ad16cd3e803b3254843b144ed07e75271da670f911db4bd18f516de`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:6fe2d12b549ad0d66bc826b8346857be50e5047b7e8696df922a4049cdcfc31d`. Independent certificate: `sha256:e52c462dbedfce2764ab700ba086e59a0cf92943d3d9b817f85ebf71ba8c2367`. Engine external-validation hash: `sha256:f0586d281a640ccda8ca73c962992d707e3246bf5cd2907b22444ccf0a38aa64`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- condensed-band-1: predicted condensed-band-support; observed condensed-band-support; exact match True
- condensed-band-2: predicted condensed-band-support; observed condensed-band-support; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:f27b2b1dbc982d5f064420c6fb2637d57ea98952c119c87ee199885bdd3c2987`.
Isolation certificate: `sha256:1dbf6766a8fdf443a61ff95f469ee3c2e74002e9e53f6044330cf808257bf909`.
Custody certificate: `sha256:a9b2ee7b8db7720351f93e4cef9a9ba625c09a73c8dc43b79e9f21397b76f93d`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- condensed-band-1 from NIST-SRD-INDEX-2026-07-23
- condensed-band-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:b3c1b1b6488a433372bf26faa0a8c351bc843da6ca2fda996ffc739298e4e684`; engine receipt `sha256:c9b70ce35823bff4a59e68486c4b7ebfe9820c03ec913c33c313e786bfc3dea4` at `receipts/engine/model_admitted/SFT-PHYS-CONDENSED-BAND-001-c9b70ce35823bff4.json`; empirical-validation hash `sha256:48dfc545c3f094e43ce2e88aa88e63aca96856868be3678e531b86fcc3cbedea`; measurement receipt `sha256:f27b2b1dbc982d5f064420c6fb2637d57ea98952c119c87ee199885bdd3c2987`.

123. Collective order and phase transition

Claim identity: `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Collective order is a shared recurrence label retained across a connected support fraction; a phase transition changes the stable macro-observation fibre and its adjacency symmetry class.

> *Condensed phase order is shared Fold recurrence over connected support; transition is provenance-preserving change of macro-fibre and symmetry.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`

- SFT-PHYS-CONDENSED-BAND-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__connected-shared-recurrence-order__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`connected-shared-recurrence-order`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:51b477ad9854d3e3390f794aade6d2fd828da8fd3a67a9a0dfc508a1feb83d23`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e1921214fd5a796742099527107d08bc253142690a0a512c1fc37c6d2ab883cb`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:151cbb565925f8890c997ccad95fcc1527c320948f80cf34a1b7701021a1433d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b562b5f1cf958f42a347f9408cc549b2ee44d2b8324c40590f7166d452f8af7f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ffe3538d4e78cb676aeee96d370bd15b9cd0814d8e838e92bfa11f0083bab4ba`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:125652ffac712d8fcd5648b50691367d6bda8c6041d39b69c72f68dc4e2d7393`. Independent certificate: `sha256:43d1226773c7d47f0390a4fbe3152c93d5fc630c6e8a0c3b44bdb3c168e4a3d6`. Engine

external-validation hash:
`sha256:8fb64442ce196f7e53f91ab1aa9e6aeba315823a535821c6385754d78607e01e`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- condensed-phase-order-1: predicted collective-phase-order; observed collective-phase-order; exact match True
- condensed-phase-order-2: predicted collective-phase-order; observed collective-phase-order; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:2ad3641024604c67411e28aeb1308239b87f5f6e3b6467348f81a1416797f22f`. Isolation certificate: `sha256:02c2b432f186698e83742819604388a9f56cc674d580b2186d288c77158c9eb5`. Custody certificate: `sha256:24d419f2f6a151ed9f5e7f86c6ff7e1dfd68dc4ecdde97a515ffbee863881245`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- condensed-phase-order-1 from NIST-SRD-INDEX-2026-07-23
- condensed-phase-order-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:5ad5edea1584254b0a73286c2af6fdafbbfc5ec40352dee35cc2da4a1ab9ac51`; engine receipt `sha256:7f20f2a8757ec6d79c32a77dd07c3fca5c25e58b1ae233c68d9daa30e5c86ceb` at `receipts/engine/model_admitted/SFT-PHYS-CONDENSED-PHASE-ORDER-001-7f20f2a8757ec6d7.json`; empirical-validation hash `sha256:c00fc15ba1b4e90f8fc8522f35c3423d6e3666e7d4330b195d60b7a552056fa0`; measurement receipt `sha256:2ad3641024604c67411e28aeb1308239b87f5f6e3b6467348f81a1416797f22f`.

124. Superconducting coherent transport

Claim identity: `SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Superconducting transport is a phase-locked joint carrier recurrence whose accessible support contains no momentum-randomizing transition compatible with the retained collective label.

> *Superconductivity is phase-coherent collective Fold transport with structurally closed dissipative transition support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__phase-locked-scattering-closed-transport__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`phase-locked-scattering-closed-transport`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:06e4517ff4a130b52a47147c077429df7e15957c2a80e5c5b067afa9b4297fd8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d223d8cd28aabe2b9f36107ddde2e6e8c811a34a78fb47914256490babf5e5dc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:11c0231899eca9549f63ab1f75820adc479df5559982abe9bee90b00249037e9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fe78fcc493a1e048e85a7cbfa55e5bd3e71f74518243bc3fdca8f318fe01cc4f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fcd2a75b76e9935e1034b9e0a2477ce7849f7ff90320cdada985cdd1aa306dea`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d8554ea60284f22785ca2689d890bae4e78117970a0f1d45ebf2876bf94f4a14`. Independent certificate: `sha256:c18546fc09850ef3db9f7ba1f96d7c0996ab108991fb44df787f7306f9d6ec71`. Engine external-validation hash: `sha256:77d8527a95cce3821e1c646a21720aa374cd5365ceff4eddd7e275b946be1cc8`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- condensed-superconductivity-1: predicted superconducting-coherent-transport; observed superconducting-coherent-transport; exact match True
- condensed-superconductivity-2: predicted superconducting-coherent-transport; observed superconducting-coherent-transport; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:7d7f1311995147699ba306a7a965ed3cb550ae1d6ee4cf61ba47e1ed373d43d8`. Isolation certificate: `sha256:b64d457d47167925766ee7ae92a8bbfa2ceb71a8da3cc6417880dc694c7dfc3e`. Custody certificate: `sha256:4b504c98077ddaf0d463f06b614ed078fa64b0ce6281ed81e1b7f039e5d3ce64`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- condensed-superconductivity-1 from NIST-CODATA-2022-ALL-CONSTANTS
- condensed-superconductivity-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:8edc42cffa4a9b38f5ce59196ef66c6289fd4bc2ddd5acf024b4619d167f542b`; engine receipt `sha256:ee86d5d893add989390c63fcc6f11bc7d95e0dc8c9c7dabd724c3233931deb04` at `receipts/engine/model_admitted/SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001-ee86d5d893add989.json`; empirical-validation hash `sha256:6251be4d6b6ee93d576f77e76672172a19d45db415dd21fee237071438ad1258`; measurement receipt `sha256:7d7f1311995147699ba306a7a965ed3cb550ae1d6ee4cf61ba47e1ed373d43d8`.

125. Topological collective-state correspondence

Claim identity: `SFT-PHYS-CONDENSED-TOPOLOGICAL-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A collective state is topological when its complete global path/adjacency class is invariant under every generated local deformation that preserves connectivity and boundary records.

> *Topological collective order is invariance of the global Fold path class under all connectivity-preserving local transformations.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__global-path-class-local-deformation-invariance__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	global-path-class-local-deformation-invariance	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:efe1f42aa2165b92aeab1e52d6f02cd7289d590cccc86d437bfedda7765054d0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bb8a4e31e8197f486fdb8e593f2109bc41d28c1f3614edf3ff948fabcb3c07c8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:380060ab86ed9165d21af7087a574b93e6ba01d9c9e469c87f9afc77f9ee9f06`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5493ae6329eac437a63678ee19a7a4ca8371a4a133f85ecdfec1e2069c5bffad`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:70337bca18cd7f4f94b5b20c543b1cd7de6747f07a0534b809af74e2d10f6db6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:07694800478ee3eaef088816a80ceccd07cd8fc888af6a1fa111a4229664372e`. Independent certificate: `sha256:ed10d3766585d02702ad385d9fd6cd5a9c2bcd7251ede206f9c03b9fe5453657`. Engine external-validation hash: `sha256:7a76a4c5d25b7f18533f18bc6e290966484057fc6300043b55d2a86bc00712d1`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- condensed-topological-1: predicted topological-collective-state; observed topological-collective-state; exact match True
- condensed-topological-2: predicted topological-collective-state; observed topological-collective-state; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:e57b9563687b05ec272c5eaf1f9cf40387eb01335d284c07d15ab0197ac4e087`. Isolation certificate: `sha256:5fad2cf81b8fd81a32d3f45de9c2c552ed7ff624e90b1626ed14d61a20b7de0f`. Custody certificate: `sha256:e17f9ec64247559f7e5b834de5c97e3915c27cf1705776b0ea24aee4f96c0e56`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- condensed-topological-1 from NIST-SRD-INDEX-2026-07-23
- condensed-topological-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:f4dd6ad6fc8f7a583b5dbe8e244bb6ff41376c3394d8b72442aab461ef1f4d3f`; engine receipt `sha256:817047e583a84e4443cabe268f2de478eae52cc8db2d84a6248c743102ae9d6e` at `receipts/engine/model_admitted/SFT-PHYS-CONDENSED-TOPOLOGICAL-001-817047e583a84e44.json`; empirical-validation hash `sha256:d6751b11a96db285f6bfdf7c753ae7112a35fb5ee93b9d393cdf0ab457111a7c`; measurement receipt `sha256:e57b9563687b05ec272c5eaf1f9cf40387eb01335d284c07d15ab0197ac4e087`.

18. Cosmological Boundary

The final Physics group closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to the later Astronomy/Cosmology branch.

126. Propagation redshift relation

Claim identity: `SFT-PHYS-COSMO-REDSHIFT-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Redshift is the exact ratio change between emitted and observed recurrence carriers after source, propagation and observer clock relations are retained.

> *Cosmological redshift is an exact source-to-observer Fold recurrence ratio with complete path provenance.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`

- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`
- `SFT-PHYS-GRAVITY-HORIZON-001`

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__emission-observation-recurrence-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`emission-observation-recurrence-ratio`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:60131c2b825f16bb397f042a42440edb9c14ad3c4085e97db361d03983166874`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9597ef38113a9a2375194761d7288a9920dbafa12a79e4212ba7efeecea59c44`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:bd4d7ace5d03153879e4d4f67b95c4ac791180d31b65e2b9c58fc1e56a13d9d5`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:56f45be90500b8eee597a38e7b2eb11e7cd8e61296df3da3fafe7be046d3d90c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d09e058c6a6886e403b4fd29935aaa5cf253af5b5c402dcacc06049310a41d1f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d09a1391267a174beda0f5d62f8adb0b7ebda82318099ff975f233d451899d7e`. Independent certificate: `sha256:0dbb95ae8ca617e091662cc252d9aeb4b9c6efdbf49b3e104cdf12a0f8408369`. Engine external-validation hash: `sha256:d4a1137be42db9bb12af28fe9e67c29146078476f22e1935ffa866a3f876ccd9`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-GW231123-135430-V3-2026-07-23

Observed comparison records:

- redshift-1: predicted cosmological-redshift-relation; observed cosmological-redshift-relation; exact match True
- redshift-2: predicted cosmological-redshift-relation; observed cosmological-redshift-relation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:7371810eb276fce1cc21e2901b8055a3c0e48f0833584758695a0ea1cc33d296`. Isolation certificate: `sha256:6fedf4242f76a740ce399eb929ed0d161280c0a474ee5b6e175fadce7f09a418`. Custody certificate: `sha256:930f8df1cee6fff2559c714f23b4e11d6ce026920dd1e5140611801573e79674`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/eventapi/json/GWTC-4.1/GW231123_135430/v3/ (sha256:67266a5406cbee1dc98a9853beb42d771595fa474f5ddd33ab4b368d5b71c45d)

Registered target identities:

- redshift-1 from GWOSC-GW231123-135430-V3-2026-07-23
- redshift-2 from GWOSC-GW231123-135430-V3-2026-07-23

Meaning. The final Physics group closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to the later Astronomy/Cosmology branch. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:4eb9d9e702e73970639f63cb01084df7b21f51888583fa65d5a82780fa20b5b9`; engine receipt `sha256:0843f2c5ea47964fd54287546abaac627a210b4bfabd6c0ba87c415f665ccf8e` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-REDSHIFT-001-0843f2c5ea47964f.json`; empirical-validation hash `sha256:cfed03b376b686cfc7cadd95821b7cae29ab05266f6964728dff79a22a0de2b3`; measurement receipt sha256:7371810eb276fce1cc21e2901b8055a3c0e48f0833584758695a0ea1cc33d296.

127. Relational expansion observation

Claim identity: `SFT-PHYS-COSMO-EXPANSION-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Relational expansion is coherent change of source-separation support relative to observation-clock recurrence across multiple source-bound paths; it is not motion into external space.

> *Expansion is coherent Fold change of relational source support per observation recurrence over a declared finite census.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`
- `SFT-PHYS-GRAVITY-HORIZON-001`
- `SFT-PHYS-COSMO-REDSHIFT-001`

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__coherent-source-separation-clock-change__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`coherent-source-separation-clock-change`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e1c9e7035c96b1ad0f7584b3db1b9eaaf5ca97bb38bdd1bea037ea9f4fde09fe`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a2dbb0b406192e90d75444c33a1a0cc7bb0f88d0e13b4e41145f977438ba145f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b5ae118031c6058ace28ac8f9026ffb3493dfe178c532404c893156cac44ebce`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ee430e29142cb3594511f207700c87ea69687f6058b67f30d75c8df741a03357`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ff266063a5b0223e3d2a5cd0c274b948cd09fc341c0e0cbd98ddfc094ae379e7`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e22e8687206a89a280e1b93bac0efe44993a1487ad0ac1de544a4f4779bc672d`. Independent certificate: `sha256:d1d7a1544b2aebaa6b78c3af0af668553ff3b955fff404639b1e8b73e7f0aba1`. Engine external-validation hash: `sha256:ebceaa28e01641c304eb29cc3202db88fbc9350587c23383c5edb1ee8e8e47d5`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-LAMBDA-2026-07-23

Observed comparison records:

- expansion-1: predicted relational-expansion-observation; observed relational-expansion-observation; exact match True
- expansion-2: predicted relational-expansion-observation; observed relational-expansion-observation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:93fd59729996521dcfe3e36997dcbe6d913f4bffb4231c4558e9e6f85d15a34b`. Isolation certificate: `sha256:634e1764554cecd564b0a02fef77b8d1aced1a35e88895659db9b3a23c79b8d6`. Custody certificate: `sha256:d64cc948ec714c9a3715aa64eeea9d65fd107fc00f6795454834b5e4cb3376b9`.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center: https://lambda.gsfc.nasa.gov/ (sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddedffe94e50b2d4275915)

Registered target identities:

- expansion-1 from NASA-LAMBDA-2026-07-23
- expansion-2 from NASA-LAMBDA-2026-07-23

Meaning. The final Physics group closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to the later Astronomy/Cosmology branch. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:227354f67ed81a6825724f89cc89a803b4f8952d65e0ca19945f89ca4bf3a965`; engine receipt `sha256:70b48d5fa5822a719fde0ab0073d965387cc0c4548ef14fda2138061b3966f53` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-EXPANSION-001-70b48d5fa5822a71.json`; empirical-validation hash `sha256:ca1f3946976b4526c76109d23042f0b93b6e28caba7f1cfa19edd7f8daba7f72`; measurement receipt `sha256:93fd59729996521dcfe3e36997dcbe6d913f4bffb4231c4558e9e6f85d15a34b`.

128. Background-radiation field relation

Claim identity: `SFT-PHYS-COSMO-BACKGROUND-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Background radiation is a near-omnidirectional recurrence field whose thermodynamic observation class is invariant across registered frequency support after local foreground records are separated.

> *Cosmic background radiation is a source-history-bound, frequency-consistent thermal Fold recurrence field.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`
- `SFT-PHYS-GRAVITY-HORIZON-001`
- `SFT-PHYS-COSMO-EXPANSION-001`

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__omnidirectional-frequency-invariant-thermal-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	omnidirectional-frequency-invariant-thermal-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2b6cb1e58f97060778f59ea7a5b1935a2635333f35edc74b9800342f046d1a5e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8cc7e810ff2e58c660c84249ac346725d8b2e9d7a99b29cebf699bd8266b3ea8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:932c0513c1bb93eb7e383ca9c0efd027c242eebe1796f73e3f3ab4a35518d326`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d58a5c5c3ad1b7c2116b9a85bdcf7458049dc99b65136d6901944d3eeacd9669`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:651bcce56f70730f691d13534b72f1ad7dedb56692f9948c5401e78a6396ac4d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:64dd57d72ca9cf7c24e56b6c7e3aaf3caaf3ad3c0b77317ac71a66a3fb93b717`. Independent certificate: `sha256:cac3fc2eef8135972ca5755eac53ecebfab08ef65bbd56ca971fb89948f389a4`. Engine external-validation hash: `sha256:5c93c3c2d3fea1d178c5b5d5e22d800ccb9ff8348e5e4cd8371b8220613cc56f`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-LAMBDA-COBE-FIRAS-2026-07-23
- NASA-LAMBDA-COBE-FIRAS-MONOPOLE-2026-07-23

Observed comparison records:

- FIRAS background interval predicted (Fraction(2723, 1000), Fraction(2727, 1000)); independently released monopole interval (Fraction(681, 250), Fraction(1363, 500)); overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:5c3983615cf9824416f01352c48a21d4eeb5d3b05c0d5db1d210b7b20bcf2085`.
Isolation certificate: `sha256:ea6140797f74add12df811f0492e2399fb4f6c9943d05cd726ddf661e02afba5`.
Custody certificate: `sha256:fc67914925bc1270f3ac2a722d756ac6f0320c56405c838eef8be43aaae2ba5f`.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center / COBE FIRAS: https://lambda.gsfc.nasa.gov/product/cobe/about_firas.html (sha256:5eae175f5ea06f7a5cb863866eeee9d87a38d870937d9a93ebf28629af0d6ef9)
- National Aeronautics and Space Administration, Goddard Space Flight Center / COBE FIRAS: https://lambda.gsfc.nasa.gov/product/cobe/firas_monopole_spect.html (sha256:e42bcf7e3e2eccac3ac8c210d4287e2477c394ad03d76ac7f719515d29753f2a)

Registered target identities:

- firas_monopole_temperature from NASA-LAMBDA-COBE-FIRAS-MONOPOLE-2026-07-23

Meaning. The final Physics group closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to the later Astronomy/Cosmology branch. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:552759b9f4be9557b1196984bffcfc001d6d13c69e75b831b71999fb46070c65`; engine receipt `sha256:2e4f7d3e02b8d087c3cce4a29dd5ee6b13f28758400c6da93090007f6e2d7e29` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-BACKGROUND-001-2e4f7d3e02b8d087.json`; empirical-validation hash `sha256:7d88b311c29f98a6dd326a9d4944fafb5827c056276e1ed688bc555348b76b42`; measurement receipt `sha256:5c3983615cf9824416f01352c48a21d4eeb5d3b05c0d5db1d210b7b20bcf2085`.

129. Gravitational lensing relation

Claim identity: `SFT-PHYS-COSMO-LENSING-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Lensing occurs when source-curved event support generates multiple causal propagation paths between one emitter and observer, changing direction, arrival recurrence and observed support while preserving source labels.

> *Gravitational lensing is multiple source-bound causal Fold paths through relational curvature to one observation support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`
- `SFT-PHYS-GRAVITY-HORIZON-001`
- `SFT-PHYS-COSMO-BACKGROUND-001`

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__multiple-curved-source-observer-paths__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`multiple-curved-source-observer-paths`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4fd7675d343fb9d994d603a695eb72260228dba9375314c3532c6cda7cbcb4cf`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3cc2a9e0f5dfd8bf2a27996c4f376e3a512b0c3c176108f1b10a868461af918a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f4e6b91bd814d859f13b3adafad665231ba76827cb01a8b25abfa05843dc5606`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1293463a7a30cb51089c59d24177fdd106c31613e039b12ed6871cc7c9e2f89c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e74c60c8dd88ff2f5a79298eeafec74420d3d4a42b0c9589b9354bba36906f0c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f0125e16e59c6103aa719aff85e8c32e19455717c544a0cab29cf1a92362c292`. Independent certificate: `sha256:0f2974648713cade0dea1dc5e85c4fe4130a680fe87a1319f1a72a7ec58ba150`. Engine external-validation hash: `sha256:5639b145f3657d0684e93825141f4665d3f89b8805be09b6e4dfba48191298ad`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-LAMBDA-2026-07-23

Observed comparison records:

- lensing-1: predicted gravitational-lensing-relation; observed gravitational-lensing-relation; exact match True
- lensing-2: predicted gravitational-lensing-relation; observed gravitational-lensing-relation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:53e32cc58a397f1475b947f7f25db97c5eab1c1f964992f7dd5e1e582da7f0b5`. Isolation certificate: `sha256:be2f4bd026393f9cc4673cb386f1088ffaea6e38a6a05cb38d84525db1712e6b`. Custody certificate: `sha256:099f8e5d84a2e620b5e9357fc612071c6ba22bd388c4ba9c301ec4ca5983b131`.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center: https://lambda.gsfc.nasa.gov/ (sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddedffe94e50b2d4275915)

Registered target identities:

- lensing-1 from NASA-LAMBDA-2026-07-23
- lensing-2 from NASA-LAMBDA-2026-07-23

Meaning. The final Physics group closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to the later Astronomy/Cosmology branch. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:2ff9cfb087ab2d14e71a157108c18c09818244662fc701edfa132b7ce816a5d5`; engine receipt `sha256:463d7041874e0b45a8ccafb32f1ee9fba54c8e733caa8446f52cce2549a07d73` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-LENSING-001-463d7041874e0b45.json`; empirical-validation hash `sha256:e8dedb014783ed584746eef77f30aecb9246ff3004ebfd791c247a51910da85c`; measurement receipt `sha256:53e32cc58a397f1475b947f7f25db97c5eab1c1f964992f7dd5e1e582da7f0b5`.

130. Causal distance and observation-time relation

Claim identity: `SFT-PHYS-COSMO-DISTANCE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Cosmological distance is the exact equivalence class of source-to-observer causal path support conditioned on emission and observation recurrence records; different distance notions are distinct retained projections.

> *Cosmological distance is a declared Fold projection of exact causal support and observation-time relations.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`
- `SFT-PHYS-GRAVITY-HORIZON-001`
- `SFT-PHYS-COSMO-LENSING-001`

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__causal-path-support-conditioned-on-clock-records__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`causal-path-support-conditioned-on-clock-records`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.

Axis	Forced form	Eliminated form	Elimination reason
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b81da4313a2a535edac98d00f96fd5c2d3df8d3b2c91b004eb6876774027c760`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:aa383062c59596ea69991db0272859b5ac291a76ee550f5438a54416dcbf5522`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f03495752b1f8a3690542261a0db6ca12a1130d48a1a1ca86b52ece949c40932`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ead8c15e0123454492674e2991a83303cd7b9c50188c91d33ccf00d539c543b5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:72c6f70adb91d4a3030d8086e4c3b58234b48e54d4c05613afedeafb023fd6e2`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d92bab1acf1fc9757e600c3d188a5747dcb24385dc817a1cd7d948d5ecd3392d`. Independent certificate: `sha256:05c01482d13acea0e21a1464be758598c2f44d3d5dd3ce03654bbfeaac5a1981`. Engine external-validation hash: `sha256:cbc661549ac3ba23474490b164ca9700f31afcee7e1f8ab6b32e2cf8325b78bc`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-GW231123-135430-V3-2026-07-23

Observed comparison records:

- distance-1: predicted cosmological-causal-distance; observed cosmological-causal-distance; exact match True
- distance-2: predicted cosmological-causal-distance; observed cosmological-causal-distance; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:788ac7ad1dca6ccc1703fd4c4121fc4c17eb6a02d9e96b42dcf479aa2c1b6d79`. Isolation certificate: `sha256:ff57831a77dd9a7adefe215a5a1aad5f71cba62b55311bc363354fd26474fc27`. Custody certificate: `sha256:0ff4efd6aa10640c3b25a78199d855a0ff5f0dbecac776014d45a14a5c6b51d2`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/eventapi/json/GWTC-4.1/GW231123_135430/v3/ (sha256:67266a5406cbee1dc98a9853beb42d771595fa474f5ddd33ab4b368d5b71c45d)

Registered target identities:

- distance-1 from GWOSC-GW231123-135430-V3-2026-07-23
- distance-2 from GWOSC-GW231123-135430-V3-2026-07-23

Meaning. The final Physics group closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to the later Astronomy/Cosmology branch. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:8d540e77a05ef62593ddf4542ffb1fc546723823ef6fc086372c60ccfe3cf52b`; engine receipt `sha256:3652a0696753f255405d64eb358c82e5099ce743763a43a2b693946040f163a3` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-DISTANCE-001-3652a0696753f255.json`; empirical-validation hash `sha256:0298a6976febe7b847faf2e661c41e5c67f2f31f60af258af6f337b76255eaa3`; measurement receipt `sha256:788ac7ad1dca6ccc1703fd4c4121fc4c17eb6a02d9e96b42dcf479aa2c1b6d79`.

131. Finite structure-growth relation

Claim identity: `SFT-PHYS-COSMO-STRUCTURE-GROWTH-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Structure growth is deterministic evolution of a finite source-density distinction network under locally forced interaction and expansion relations; weights describe unresolved initial/history fibres rather than ontic randomness.

> *Cosmic structure growth is finite deterministic Fold network evolution under registered gravitational and expansion relations.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`
- `SFT-PHYS-GRAVITY-HORIZON-001`
- `SFT-PHYS-COSMO-DISTANCE-001`

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-density-network-dynamical-growth__source-bound-proof-trace__capability-closed-predicti` on__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	finite-density-network-dynamical-growth	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:2c10da892c1aded3ed2cad6743b2eef5425cfd153f79ee0c6b2fd5ee587fc6db. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:4e6a66e9753cfc7df280705717c0fcf90fabcc3c968ab2d95191d72dba295567.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:6294b6a7219d87acbd7ac9cd3fd06eeef95f57487d3408af2cafcad072092a45.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:6533e6345df8c7b31418838b2fd1c63bfe822ddd7f37fb0307e841748f779a62.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:7e741f8ba12c27a8e4a76258e9fb1f384f4c067b450ea022c48bad8b8d21bdeb.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:eb896632cf27d04f3c63a7581367588cebcbf9d67de990119b9522392de880c2. Independent certificate: sha256:2e2581a18ed8db4826faf50042bf61a441363d25c9ba0c02ebf744d886b53ff4. Engine external-validation hash: sha256:5d49103d5af99bed683de928900c63b91da202f7a6f0bf0b8506da517178294b.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-LAMBDA-2026-07-23

Observed comparison records:

- structure-growth-1: predicted finite-cosmic-structure-growth; observed finite-cosmic-structure-growth; exact match True
- structure-growth-2: predicted finite-cosmic-structure-growth; observed finite-cosmic-structure-growth; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:e4d0d65ebfcb3ae49d1835e3cffc745e67e2a981ffbb3125196a576e2cbca395`. Isolation certificate: `sha256:59e4c6eeab740920eb407ba2581338255007e51de8d809dab09da981bff5d400`. Custody certificate: `sha256:f472907947520fb1b48dcded45b5dfd911e819ba009269e185488ea60ee4cbd4`.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center: https://lambda.gsfc.nasa.gov/ (sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddedffe94e50b2d4275915)

Registered target identities:

- structure-growth-1 from NASA-LAMBDA-2026-07-23
- structure-growth-2 from NASA-LAMBDA-2026-07-23

Meaning. The final Physics group closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to the later Astronomy/Cosmology branch. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:7e09ebcf55d86d177b601ec853ea64167d5c7ac6dd6f573d8caed18256da3437`; engine receipt `sha256:b150f2ddf89d4a0ea2352ebc7ae1ac954835eb6ac1856f38c6ddd8f8d5961744` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-STRUCTURE-GROWTH-001-b150f2ddf89d4a0e.json`; empirical-validation hash `sha256:b9423cbed498d5c7190ff4f9823de2a3944613d4acd5d2c9b4f071f5be563f6c`; measurement receipt `sha256:e4d0d65ebfcb3ae49d1835e3cffc745e67e2a981ffbb3125196a576e2cbca395`.

132. Physics-to-astronomy/cosmology handoff boundary

Claim identity: `SFT-PHYS-COSMO-BRANCH-BOUNDARY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Physics closes the universal source, propagation, observation and interaction relations; object census, historical initial state and inferred cosmic chronology pass as explicit measured inputs to the later Astronomy/Cosmology branch.

> *The branch handoff preserves closed Physics laws and exports only source-bound finite observations for later astronomical reconstruction.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`
- `SFT-PHYS-GRAVITY-HORIZON-001`
- `SFT-PHYS-COSMO-STRUCTURE-GROWTH-001`

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__law-versus-historical-state-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`law-versus-historical-state-boundary`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:09b1913ced9baccd4ad6da10830944772c9628b7538a430663d9a19ccecacb1d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:43993065c8994e8e6caf618318ed6ae948b22df1bbd82a4678c2865d80ba89b6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9579c3b9ecba7097e4caae2712a7b0aeea983ae70b5d83fb8bb1b1a4290856a1`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9b6a0c84cbb54a1ac283a2806ecbf11c23551aa2ca7639fecb9fed1bdf4d349b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fff8bff53425a262dc269b5a84aba203bc7d05d2b4d16d384c5a3e830d104e52`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e34d327c6aef4d6ee556576812a387fe8696bd13e948e144e11c271e682b73c3`. Independent certificate: `sha256:d3c90ff6e0e964eff72260f53e9e2ff541241215ecfb3f945e51bb9286b525e0`. Engine external-validation hash: `sha256:787b275a646d773bd9baa3ed2d87d7e950771b93ae276ee74972812d12ff64ac`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-LAMBDA-2026-07-23

Observed comparison records:

- branch-boundary-1: predicted physics-cosmology-handoff; observed physics-cosmology-handoff; exact match True
- branch-boundary-2: predicted physics-cosmology-handoff; observed physics-cosmology-handoff; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:0b9e11696be7f741781838ec270ac7e215e09d009fc3c12042e6675586b2fdf8`. Isolation certificate: `sha256:ed1a02b134a440f9e2c5dd4945a44ae50240e798a31979b14b8034e3f8a80b45`. Custody certificate: `sha256:34122d9c28743e35a51d7e39977c8c8b10d24641b4cbcf195ca28e2508e7fb4e`.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center: https://lambda.gsfc.nasa.gov/ (sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddedffe94e50b2d4275915)

Registered target identities:

- branch-boundary-1 from NASA-LAMBDA-2026-07-23
- branch-boundary-2 from NASA-LAMBDA-2026-07-23

Meaning. The final Physics group closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to the later Astronomy/Cosmology branch. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:c5e9469ae63514fb6102751a96b8e9102244e843b8a7c3a959c9bade858cb3fb`; engine receipt `sha256:86e32f007bb9c6304edc82ae7e4f01db1f0391b5fbbef56d5ddedc12dd178160` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-BRANCH-BOUNDARY-001-86e32f007bb9c630.json`; empirical-validation hash `sha256:fa5c646653b938d6c7de35cadac6259a51aa2d0fdccab5e76f8482da14f4b618`; measurement receipt `sha256:0b9e11696be7f741781838ec270ac7e215e09d009fc3c12042e6675586b2fdf8`.

19. Supplemental exact measured-value admissions

These eight claims do not replace the required laws or their receipts. They depend on those laws and add exact post-seal numerical correspondence where the first structural validations predated the measured-value interpreter.

1. Exact molar Planck measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-METROLOGY-MOLAR-PLANCK-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The molar Planck carrier is predicted by exact composition of the Planck carrier with the counted molar carrier.

> *Planck constant composed with Avogadro count predicts the CODATA molar Planck interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MEAS-UNIT-COMPARISON-001`

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2dfda8fbde6076b09842497abdc5a00d44c02e05164440bd1f0d379f969a5731`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8dba5666f65856a7a2f6d46a59317ff158e90067b1d68739639625fbe1ff4c48`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b50f43ff8ea8b8b28920191600b58f1084d395bbb64a7c281b26769d5fd5e8cb`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:01234e1a45ed97261d6e6aca6fa6a09c82d85cc82a44e3c8668947aff0f6ed5e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9bd844cf923a2862db055f9e90a89436b1ceb8493ccc58ea2b1a9c50e1c5b732`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5c176bbcc9c5e3304bf168c8624612b897b8c0e0d982f5c4d493520690bc404e`. Independent certificate: `sha256:7e2cc5601355e962f71607ad460a5e96a5cd116059af12db77640845bd7d3877`. Engine external-validation hash: `sha256:7a1a8c04d1caba9681bd270dabe42b44d5c345725b3abb1e686394fded2b11ce`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- molar Planck constant: exact predicted interval [19951563564467157/50000000000000000000000000, 19951563564467157/50000000000000000000000000]; external interval [498789089/125000000000000000, 3990312713/1000000000000000000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: `sha256:36b3c9aa786510f0f47fd90fdb05d61bcf377891525319a794afa48d7e053939`. Isolation certificate: `sha256:9f92f36602cc0fb91dc9e8454a62563f74168e90f3d4277f25c735a7bce0a4b8`. Custody certificate: `sha256:e18166182e4381ff3211a687b7d140bde54dd6a9d1845024f320c5be75b4a84d`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- metrology-molar-planck-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. This supplemental claim tests a previously forced relation against an exact external value interval. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:9225e8b389f0d1611722e0ee9082aed36211d7cad352f164b9784d71c7e0bf81`; engine receipt `sha256:e1f0dfe8b6d4a57d7b408516d2e09ae677fabefc201c273f3f2a62182f9bea73` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-METROLOGY-MOLAR-PLANCK-001-e1f0dfe8b6d4a57d.json`; empirical-validation hash `sha256:6a8f75e99001bab6986b19ecb01b097c11c6765dfee5b6dfe76a5573ee66f334`; measurement receipt `sha256:36b3c9aa786510f0f47fd90fdb05d61bcf377891525319a794afa48d7e053939`.

2. Exact momentum measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-MECHANICS-MOMENTUM-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The atomic momentum carrier is predicted by exact composition of inertial content with propagation speed.

> *Atomic mass composed with atomic velocity predicts the CODATA atomic momentum interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MECH-MOMENTUM-001`

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:48db928fa462f39db0d3897bad3bca0b2ca40b7cd526065fad1afb314ce1b128`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:86f5eb2fbf987c6a15c200773f00f2697c12ae8673795cbcfa43408304a1c8e1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:32ddd7bd51c73ff395027bcbf5673d17534a4a83309101de6f9eb8697a2ee79b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e52819448d4eb62b03f6356b792ad24afbb305bbf741b588bfb21c460591077a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e93bf332fc53a176bfd807a7582b045628da9fbff31feaee0de3950334533b94`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:96232cb6e8a85b63a038a0dbb0c587d0c8a77c3dfc0570cbbfbab64dca1a6d89`. Independent certificate: `sha256:3e5bba783e37e7e0781b1b92e395e66f8d8791ca74db8886c7d0457e22cac2cd`. Engine external-validation hash: `sha256:b0c73c45da9d914e52907adbd52103c61d262347e522382e33fd52f6117bc967`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of momentum: exact predicted interval [9964259572669456670101/5000000000000000000000000000000000000000000000, 15942815331027492267/800000000000000000000000000000000000000000000]; external interval [99642595757/50000000000000000000000000000000000, 24910648947/12500000000000000000000000000000000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: `sha256:56ee927b7a59252157c870d7b617361dbccb8149618a1312b23aff974eabaf91`. Isolation certificate: `sha256:2c7010c8cfdb5340f59517f0da156c24aea2c8d702b7cabca21d6ff0d17a36b9`. Custody certificate: `sha256:6b8b74bdc78db7accc67131fb786add7b9f7235e7052985541f4200d6ba3148b`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- mechanics-momentum-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. This supplemental claim tests a previously forced relation against an exact external value interval. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:5cbeade6d136aa75bd5230ed8c319ca39ef239a071b3a340252e8b120e9b6ea2`; engine receipt `sha256:4a7e754fdd04ef6ffe92456b90c10648d0c4b64194fd53774beaae63d98c62af` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-MECHANICS-MOMENTUM-001-4a7e754fdd04ef6f.json`; empirical-validation hash `sha256:747a63254d71bc932e64a52bc276ac83ff79026765de82e9730c03b3c074abb4`; measurement receipt `sha256:56ee927b7a59252157c870d7b617361dbccb8149618a1312b23aff974eabaf91`.

3. Exact force measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-MECHANICS-FORCE-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The atomic force carrier is predicted by exact work-energy transfer per spatial support interval.

> *Atomic energy divided by atomic length predicts the CODATA atomic-force interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-FORCE-001

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:531b071bf736461b1a957cc542ca52ccdbb1931bee56372e43f83a97175e4b85`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7d0e248a615bf7fa85e0c4e63bb059d811484b30daa1e6562655d5a2d9e50005`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e05966041153e32d868b967f176746cd8bbf7ade55fadf948076da6c65fcc6a7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:154a326728312835d420ee7a8f72418b24f9eac1b385b3fbe5bcd514ea6f13cd`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dfd96e21b8455542399ab2e45dac987d56016ced10065bbd4a854c2cb45e6164`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:305c4b0bfef2f469c7b94f691536ca5c5bc9ef5e4d2b21edc8afe150c342f609`. Independent certificate: `sha256:6accffe40e2a38bcd1e59f58794b8201a2756f1d626bd4e6bed722d321b4ead2`. Engine external-validation hash: `sha256:69ffd95f7625f105e65a4439bdd5218dab1568d70b90fe4f195c961e16cb1019`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of force: exact predicted interval [10899361805503/132294302656500000000, 10899361805527/132294302615500000000]; external interval [3295489401/40000000000000000, 82387235051/1000000000000000000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: `sha256:31429d5b44a542dc11e26006c28ab86fd8d4f766eda307b47f1215069204902c`. Isolation certificate: `sha256:7c68fc1380bc71369cdbb7852e389d07819634c7221e3ce66f634c861aa95261`. Custody certificate: `sha256:eb5472c0209196d6ed943d556e449ebe0013bd3b360dc619bd26793043a890f7`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- mechanics-force-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. This supplemental claim tests a previously forced relation against an exact external value interval. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:542c07dc5b33cd59956904767eda58be58b9338a4f0df6fa6208904864ff4e6e`; engine receipt `sha256:e394dd3d87bf8f712e39291a4f8dcc0fe2d2bea75d3d1fd5b88a67645fb691b1` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-MECHANICS-FORCE-001-e394dd3d87bf8f71.json`; empirical-validation hash `sha256:a3fd66e3f20289357da1a8679e0fba1574529cee162cca2c5067d03d2b50af04`; measurement receipt `sha256:31429d5b44a542dc11e26006c28ab86fd8d4f766eda307b47f1215069204902c`.

4. Exact electric-potential measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-FIELD-ELECTRIC-POTENTIAL-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The atomic electric-potential carrier is predicted by exact energy transfer per elementary held-charge carrier.

> *Atomic energy divided by elementary charge predicts the CODATA atomic electric-potential interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:90794ad2d32fc15db13bf3e1b9212babd6cabaab3e2629a2af951a18f9e72610`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c5d5bc3b55e252b97377e395fd24d91276e5901912f446c273d050a5815e3452`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:60be210b04698f43634a547027bf2d8018948689ca5f24ffaf24138e79792bcf`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:465eda6c270bb44dc6c32e821e599098a9618ae570ddfffe2b3101603ed172f7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:86f4aea99ba7612cf9889c79870777cf6878cec798c0bb551416fdf9ed9a07bb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:eb299544fa0a2cb21cc9c01a748f9da3bcaeb47b0ea1b7ee954a9bbb3db44137`. Independent certificate: `sha256:10ed41b8443c1db52bc830ad1f60d83f261b7995f2ceb65587025a1056685460`. Engine external-validation hash: `sha256:59c4603aa1c7e15048f81d0cda1d2bcd823299418d17b9fd8f6b45d09b5cf019`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of electric potential: exact predicted interval [10899361805503/400544158500, 10899361805527/400544158500]; external interval [27211386245951/1000000000000, 27211386246011/1000000000000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: `sha256:4edaa6daf75bf922b1b265d0c71158cf9c4cbf46fba764a099d9abfeb8190114`. Isolation certificate: `sha256:86114e2038b630b41cf649ccebed03767644c0d0f3562cf9b9eefe02bfa9e534`. Custody certificate: `sha256:5034883424092c1d8f700dcb27b09cc1f63c7db2f5cfb817f1c9c7827904b131`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- field-electric-potential-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. This supplemental claim tests a previously forced relation against an exact external value interval. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal

- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:b4905946f588c0b676abca54102aea83324c31bb32bcf614ff4d1376a916e6c7`; engine receipt `sha256:34a88c7435b67d0c73792b2eadc0ff77c03291db4280c5b806c286b78ddbb00f` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-FIELD-ELECTRIC-POTENTIAL-001-34a88c7435b67d0c.json`; empirical-validation hash `sha256:490ac035799a28d5636017d5332003ba8a8c8242399bb2f73c8ad1549035608d`; measurement receipt `sha256:4edaa6daf75bf922b1b265d0c71158cf9c4cbf46fba764a099d9abfeb8190114`.

5. Exact electric-field measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-FIELD-ELECTRIC-STRENGTH-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The atomic electric-field response is predicted by exact potential change per atomic spatial support interval.

> *Atomic electric potential divided by atomic length predicts the CODATA atomic electric-field interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:68ad39b63f242c2bcab3cbbe275198cbbc364a97569665ad8ccc14b164fe4a02`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9935fe5faaef70de04a0fb1096d94b7102ff87ca5c74a8c907f9c2b86e268442`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c94feca21e55eed4366a19704f6e7d9d64ade592f25c5c765c4df02d3984a99b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e699961a2ae9dfb96d886fabba5d3137d5cb6ee283a558e308aaafd0dd70e3ca`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:450854a196136d47100d2bfc73504c78f85dde507440fc34cb3b9a950ac9b2bd`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f0f3fa02e9e303bb6a657ca72723370b1476bb9017f6bc0f60ef840ee05d0e8c`. Independent certificate: `sha256:b563e6fd9fedf5ba12fec207079a96b863ec69bc82d5b8f09497c229a7238ee8`. Engine external-validation hash: `sha256:1b5e48598ab764f78667fdb515e5e36c0c77e550d8740c21f2ec1430069f06ee`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of electric field: exact predicted interval [136056931229755000000000/264588605313, 136056931230055000000000/264588605231]; external interval [514220675032/1, 514220675192/1]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: `sha256:828c7e4fbb00764d4e9df5ddc3582e7aee6bca095db57f37bbb8deaf67c469a0`. Isolation certificate: `sha256:2e6da0f632329f5add36793dff84a23aa8ec7111ba4e5564985e39a81c939941`. Custody certificate: `sha256:435baa2ece8efe40fa8ae6ff70c0d08e55074f04f19e7377714855db66243d74`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- field-electric-strength-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. This supplemental claim tests a previously forced relation against an exact external value interval. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:bdb2a10e1b1f70da8cdf3c6f285dd547459ebf99e75ed39a00c3a1094e05ab72`; engine receipt `sha256:b5b06d49508b35b38f352d022c8edcf0f2e593225086d53472aa679a8e64dbc8` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-FIELD-ELECTRIC-STRENGTH-001-b5b06d49508b35b3.json`; empirical-validation hash `sha256:a1d5c9e8da6710ad18d84368335db10cf121930ebf59504a863157d7ff31baea`; measurement receipt `sha256:828c7e4fbb00764d4e9df5ddc3582e7aee6bca095db57f37bbb8deaf67c469a0`.

6. Exact wave frequency measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-WAVE-FREQUENCY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A registered inverse-length recurrence transported at the limiting propagation speed predicts its recurrence frequency.

> *Rydberg inverse length composed with limiting speed predicts the CODATA Rydberg-frequency interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001`

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3af1893b417c04c95f0233f2ed9a2047c13492d12b5761f7cf2eb35560d714c8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d1b1b26ccb8e0f088ad709031fb472de420836598102c0ceec7d13bcb415e688`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:657967992b8f1c2e12d67611da1293b2fb5cde33f726e9b486a9eefa47701483`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:213e3480e3dad607392a3f96d7656c451e251f3e53fcb53a77e67ecea56d5db7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4a60e6dd8a7cffe9b795bdb5dd2806f1a7f7c5d5118a1cb55f26bc43e7cb7d29`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f12b815d77906521b2090244dd8dd4e3afd91e97d51cb9ccbdfb1e7962a32aab`. Independent certificate: `sha256:1c07dd896d1c7074c9cb1e7e3aab7471662f3829120d5536b0534ed959b72048`. Engine external-validation hash: `sha256:e2ea7686a0d5b823e751e2a44c265fb17a66afd7d61bf1e3b1667f95b25388c3`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- Rydberg constant times c in Hz: exact predicted interval [328984196024638405041/100000, 1644920980126789534701/500000]; external interval [3289841960246400/1, 3289841960253600/1]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: `sha256:c273d3186b7d7c1f07493ee7b303e28d6299fa252e4abd65c72eaa968ecd3723`. Isolation certificate: `sha256:581833bcd6fc4a5fac836ce0d6e309ad5c780137389e98caab9a1e609a6ca126`. Custody certificate: `sha256:153a70cc8e477b12a568d1ee4719192061fe23a2d31ee2517fd7152e354b8bdc`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- wave-frequency-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. This supplemental claim tests a previously forced relation against an exact external value interval. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:0fc7adec00d3a5e509382fc951f24c3dad4a1836c18770735393de6818453713`; engine receipt `sha256:d75c1886968251dd42429c73c3c8303ae4d22eededbe87ae37e24951e7e938fb` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-WAVE-FREQUENCY-001-d75c1886968251dd.json`; empirical-validation hash `sha256:75b2bdcab27442712f9b8272975533b3121f04fe4ddb633a879e2ecf201f2111`; measurement receipt `sha256:c273d3186b7d7c1f07493ee7b303e28d6299fa252e4abd65c72eaa968ecd3723`.

7. Exact wave energy measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-WAVE-ENERGY-001`

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A recurrence frequency composed with the Planck transfer carrier predicts its transported energy.

> *Rydberg inverse length, Planck carrier and limiting speed predict the CODATA Rydberg-energy interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`

- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-positive-interval-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:52ef549cae6c9a11b223baff93ae4578bcdf3aa5d1d37132cb5d26c2b7cca9f2. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:80fababc6aacb0f4cc9169e4d261edb055e531385429c8d6d0712e082e2b8c8f.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:dcbdff65034657a9d951b337630fda7c01e70642ab9a1746257825b9f4b3fd42.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:4d7cb5e64989a5c0d5d54a87e838a6d7717f0352e221a6ad80c6a7a5f9c963c8.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:a5d04f8c4cd2f3b0152f661e985a1d9d0839aece8a9dc6f8319999702392e77a.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:6dd06d5f612f4003d26dafa3da17e7fd27e58f1dba04cf89bd205dbf7a056e7e. Independent certificate:

`sha256:2623704f4a5b8842046e10ea7adc04f8be5fbd323819cd0bda228158f507b9ab`. Engine external-validation hash: `sha256:22447689c894ce0c699c76930493d3e039386abb0aadfa9cceef616c52c8407b`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- Rydberg constant times hc in J: exact predicted interval [43597447222012104003715592523/20000000000000000000000000000000000000000000000, 217987236110537267024293705503/100000000000000000000000000000000000000000000000]; external interval [10899361805503/5000000000000000000000000000000, 10899361805527/5000000000000000000000000000000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: `sha256:bfcee7ac76277f2eca8503e567c67cb5661ad12a342cad3a531c126ae66c676a`. Isolation certificate: `sha256:9d5fa3a07cf45a376ac75853053ba25a5459fa4059c08a6c5d36df0a30927f5b`. Custody certificate: `sha256:78364198ef5c6f4b73962a2cb5151a3fd4f9e9050cca8c4ba96af9db7d5a75d0`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- wave-energy-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. This supplemental claim tests a previously forced relation against an exact external value interval. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:ee42680f05f96265966953bbaf31df5c983b1ea1ecf841ab7288e3826137da33`; engine receipt `sha256:fd222d0ccab49086ac4daa8928f6c9f0cd65ccf15dc94dfdc56562cec0c22173` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-WAVE-ENERGY-001-fd222d0ccab49086.json`; empirical-validation hash `sha256:ebd6ff2986706f73adeb320421bc15a51030bc97df03321697f53af5930a04d3`; measurement receipt `sha256:bfcee7ac76277f2eca8503e567c67cb5661ad12a342cad3a531c126ae66c676a`.

8. Exact molar thermal measured-value correspondence

Claim identity: SFT-PHYS-VALIDATION-THERMO-MOLAR-ENERGY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The molar thermal-energy carrier is predicted by exact composition of the per-constituent thermal carrier with the counted molar carrier.

> *Boltzmann carrier composed with Avogadro count predicts the CODATA molar gas-constant interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-THERMO-TEMPERATURE-001

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Complete axis elimination. The 256-form census is the literal product of eight binary choices. Each row below compares the survivor coordinate with the otherwise-surviving form changed on that coordinate alone.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-positive-interval-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:cda72729383c0dd8395a446dda7fafbf2343e3452ca591e2d56187f117d0e610. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2d76619ba4fffb95f44490faecc85c9503181235a3b97795c0cd00e91dba3d27`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:189f4aefc0e0d84013394e7c93774b570c9d30738b5bf0eca7850d92fa3f48c6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:18fc3e77d08c08e7911ab7bbecea77efb7cee6a6defc6de717db77287244a504`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1dd0e1d54a27ff5a0f5ebf3731e5842ce31f34eb8975dd1f742ee0d3862503da`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:7c74bfeab70b66c1a68ab28d93377be8592b949cde42658538a08e24dac8c6b5`. Independent certificate: `sha256:94c1f683599f41d967651a055f6c72c4d311a58710642443377dfa8144ce603d`. Engine external-validation hash: `sha256:6c6eaf91b7f7c35ea6d030b8f0b82887a5e00cf7eb097aa653935e7c6eb45f4e`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- molar gas constant: exact predicted interval [207861565453831/25000000000000, 207861565453831/25000000000000]; external interval [4157231309/500000000, 8314462619/1000000000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: `sha256:72a70301b05b1847cb68d81b26ec8b12ba9fd6ed0b7633034f7c4c062a5439c8`. Isolation certificate: `sha256:49e41a57f9c684a88cb3548296db3c9b373c50191e40b4d92cd9a0c0dad5f437`. Custody certificate: `sha256:8eb921aa0495f19a26d475ad77f38874da872edd0238379aa1adbd0167b0f181`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- thermo-molar-energy-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. This supplemental claim tests a previously forced relation against an exact external value interval. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest `sha256:55cdab99ac1372af5d769f493866a3dc73d31ed5fee6c0e24a30667d32590e50`; engine receipt `sha256:4c675313be3e4006a248563bd452cb64e04425dbef90838ba7625a59beb8b277` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-THERMO-MOLAR-ENERGY-001-4c675313be3e4006.json`; empirical-validation hash `sha256:2e08d99977b76a21e09ae04a2080633f83f89d247304f1b54df868ca5e9b42cf`; measurement receipt `sha256:72a70301b05b1847cb68d81b26ec8b12ba9fd6ed0b7633034f7c4c062a5439c8`.

20. Cross-branch synthesis

The completed branch has one compositional arc. Measurement defines what an observation can retain. Mechanics defines relational change and transfer. Fields extend source-response over local support. Waves close recurrence and propagation. Thermodynamics identifies macro-observation fibres and information-retention costs. Physical quantum theory retains complete support, phase/action and joint composition. Matter identifies recurrent constituent and channel classes. Spacetime orders events by limiting causal paths; gravitation changes their source-linked relational closure. Collective matter takes finite cell networks to scale-declared macro-observations. The cosmological boundary exports universal propagation laws while refusing to disguise historical state as a law.

This unification is operational: the same Fold objects - exact counts, ratios, words, held labels, tables, transition traces and observation fibres - appear in every subbranch. The claim is not that established terminology selected those objects. Correspondence is registered only after each Fold form survives its generated alternatives.

21. Limitations and falsification

The paper makes no claim of an actually completed physical infinity, exact continuum substrate, derivation of contingent initial conditions, unbounded astronomical census or replacement of later Chemistry, Materials, Astronomy and Cosmology branches. It does not treat a simulation as empirical proof, an exact SI definition as an independent natural observation, or portal availability as numerical agreement. Categorical external rows test structural correspondence; the dedicated value adapters test numerical consequences. A new observation outside the registered boundary requires a new preregistration and cannot silently broaden an old receipt.

Any admitted empirical claim is falsified within its boundary by a preserved external row that does not match the sealed prediction, by target access before sealing, by omitted adverse evidence, by a changed source snapshot, by failure of its independent reconstruction or by an engine-constitution violation. The correct response is a halted new claim and investigation of the derivation or adapter. Existing immutable receipts remain identities of the artifacts they certify; new evidence cannot be retroactively written into them.

22. Reproducibility

The repository uses Python's standard library for the core engine and supports macOS, Windows and Linux without Docker. Researchers may inspect every claim package directly. The one-command repository verifier remains available for release validation, but this branch build deliberately used proportionate admission-time checks and a read-only evidence census rather than repeatedly replaying historical derivations. The Physics inventory is `publications/inventories/physics.json`; this manuscript's exact evidence map is `publications/current/physics/evidence_map.json`.

23. Conclusion

At the declared boundary, the Physics programme is complete: 132 required derivations are forced, enumerated, independently reconstructed, externally tested and admitted through one engine. The measured-value layer preserves the project's strict arithmetic constitution while allowing official observations to invalidate a sealed relation. The result is an open, inspectable tree of physical laws whose authority rests on reproducible traces rather than credentials, opaque prediction or consensus selection.

References and official data bodies

- Bureau International des Poids et Mesures, *The International System of Units (SI), 9th edition*.
- National Institute of Standards and Technology and CODATA, *2022 Fundamental Physical Constants - Complete Listing*.
- National Institute of Standards and Technology, Atomic Spectra Database, version 5.12.
- Particle Data Group, *2025 Review and Summary Tables*.
- International Atomic Energy Agency, ENSDF and ENDF evaluated nuclear-data systems.
- International Association for the Properties of Water and Steam, official releases.
- Gravitational Wave Open Science Center, GWTC catalog and event APIs.
- CERN Open Data Portal, public dataset API.
- NASA LAMBDA and COBE FIRAS public data products.
- Smith, Maria, *There Is No Nothing*, Ernos Labs methods and foundation paper.
- Smith, Maria, *From One to Fold*, Foundation branch evidence corpus.
- Smith, Maria, *From Fold to Mathematics*, Mathematics branch paper.
- Smith, Maria, *From Distinction to Information*, Information Science branch paper.
- Smith, Maria, *After Turing: The Fold Machine*, Classical Computation branch paper.
- Smith, Maria, *The Quantum Fold Machine*, Reversible and Quantum Computation branch paper.